Conceptual Design of Buildings

(Course unit code 1C2)

Module E

Integrated structural design and execution of buildings

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European Erasmus Mundus Master Course

Sustainable Constructions under Natural Hazards and Catastrophic Events

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- Concluding remarks

- General
  - Multistory frame buildings
  - Present developments
- Structural systems for multistory frame buildings
  - Gravity load-resisting systems
  - Lateral load-resisting systems
  - Beams and column elements
  - Joints
  - Floor systems
  - Facade systems
  - Fire protection
- Response of frames to lateral loads
- New structural systems for seismic applications

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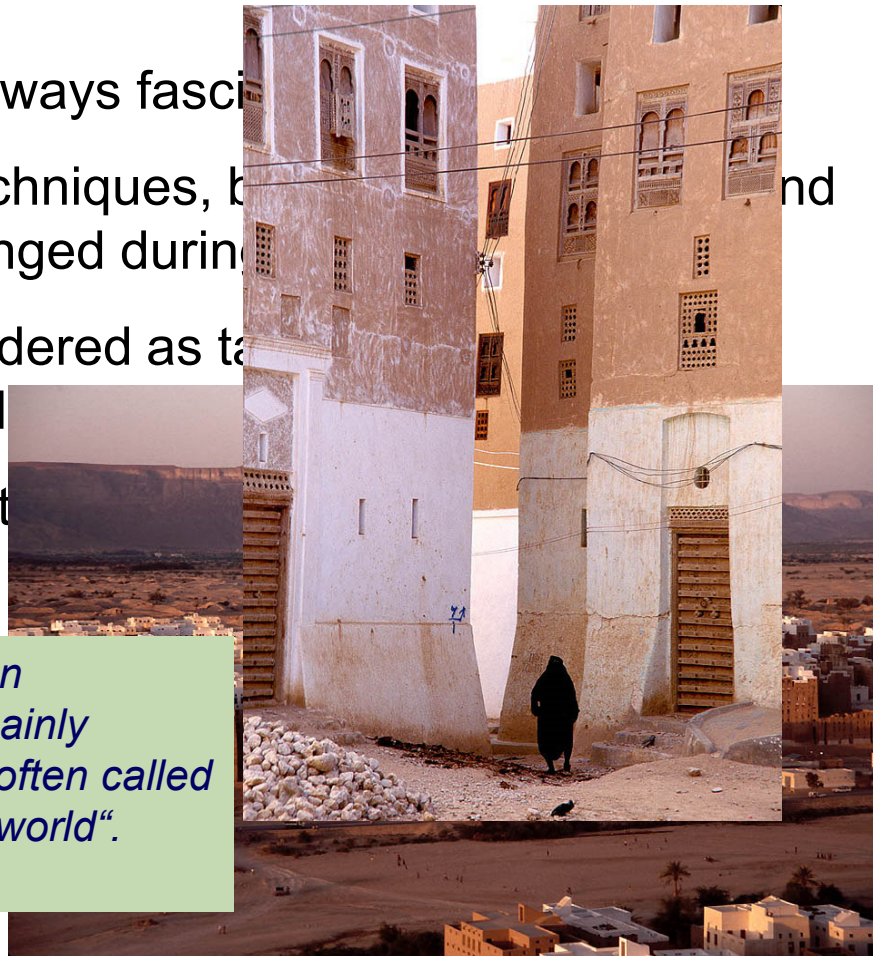
Multistory frame buildings

- Multistory buildings
  - Tall buildings has always fasci
  - The construction techniques, b nd suprastructure, changed durin
- A building can be considered as ta lateral loads is reflected
- It is important to take int wind as well as seismic



Old Walled City of Shibam, Yemen
Most of the city's houses come mainly from the 16th century. Shibam is often called "the oldest skyscraper-city in the world".
Buildings reach 40 m height.

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Evolution of multistory frame buildings

- The first skyscraper was the ten-story Home Insurance Building in Chicago, built in 1884–1885. The architect, Major William Le Baron Jenney, created the first load-bearing structural frame.
- Most early skyscrapers emerged in the areas of Chicago, London, and New York toward the end of the 19th century.
- After an early competition between New York City and Chicago, New York took a firm lead, culminating with the completion of the Chrysler Building in 1930 and the Empire State Building in 1931, the world's tallest building for forty years.

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Evolution of multistory frame buildings

The skyline of New York City

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Evolution of multistory frame buildings

- Multi-storey frame buildings
  - Skyscrapers also began to appear in other parts of the world (Mexico City, Tokyo, Shanghai, Hong Kong, Singapore, Kuala Lumpur, Taipei, Jakarta, etc.).
  - Modern multistory buildings use steel for the main structural members (or in combination with concrete – composite structure)
  - Despite the recent events that threatened the construction of very tall buildings, their developments have been continuously increasing worldwide.
  - Many tall buildings were recently completed or are going to be completed in the near future – Dubai has 18 completed buildings that rise at least 300 metres !!!!!! This includes the tallest man made structure Burj Khalifa

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Present development

Burj Dubai: 818m

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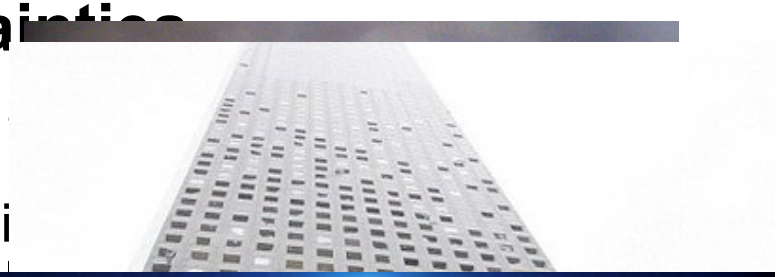
Challenges and uncertainties

- There are many challenges uncertainties related to the development of high rise bui



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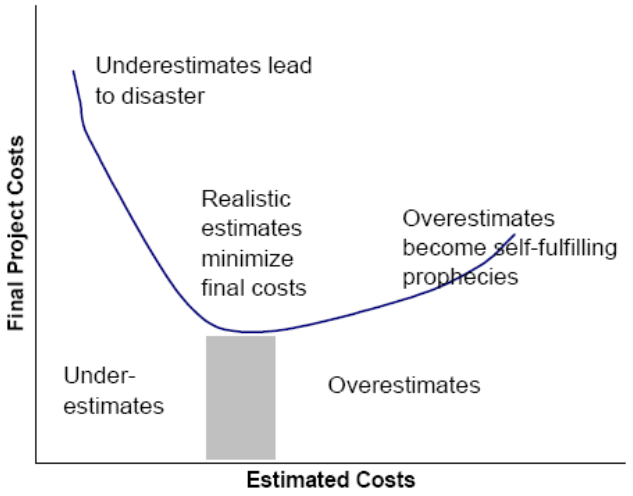
Challenges and uncertainties

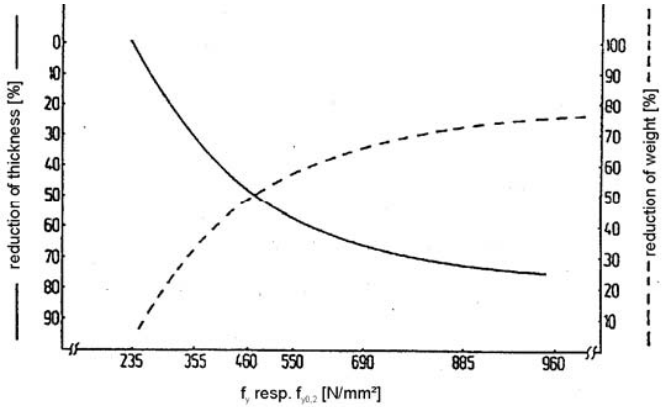
- Cost
  - initial costs
  - operational costs
  - dismantling
- Design and erection
  - new design methodologies (PBD)
  - new systems, materials, technologies

Sustainability (“Green” or “sustainable” buildings)

- Life Cycle Assessment
- Energy use
- Emissions from energy
- Water use
- Waste reduction
- Productivity and health

The wind turbines engineered into the Bahrain World Trade Center, Manama





Reduction of wall thickness and weight with increasing strength of steel

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Structural systems for multistory frame buildings

- Gravity load-resisting systems
- Lateral load-resisting systems
- Beam and column elements
- Joints
- Floor systems
- Facade systems
- Fire protection

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Gravity load-resisting system in multistory buildings

- Tall building floor structures generally do not differ substantially from those in low-rise buildings; however, there are certain aspects and properties that need to be considered in design:
  - Floor weight need to be minimized
  - Floor should be able to resist construction loads during the erection process
  - Integration of mechanical services (such as ducts and pipes) in the floor zone
  - Fire resistance of the floor system
  - Buildability of structures
  - Long spanning capability
  - Necessity to reduce the weight of the floors so as to reduce the size of columns and foundations and thus permit the use of larger space. Floors are required to resist vertical loads and they are usually supported by secondary beams. The spacing of the supporting beams must be compatible with the resistance of the floor slabs.

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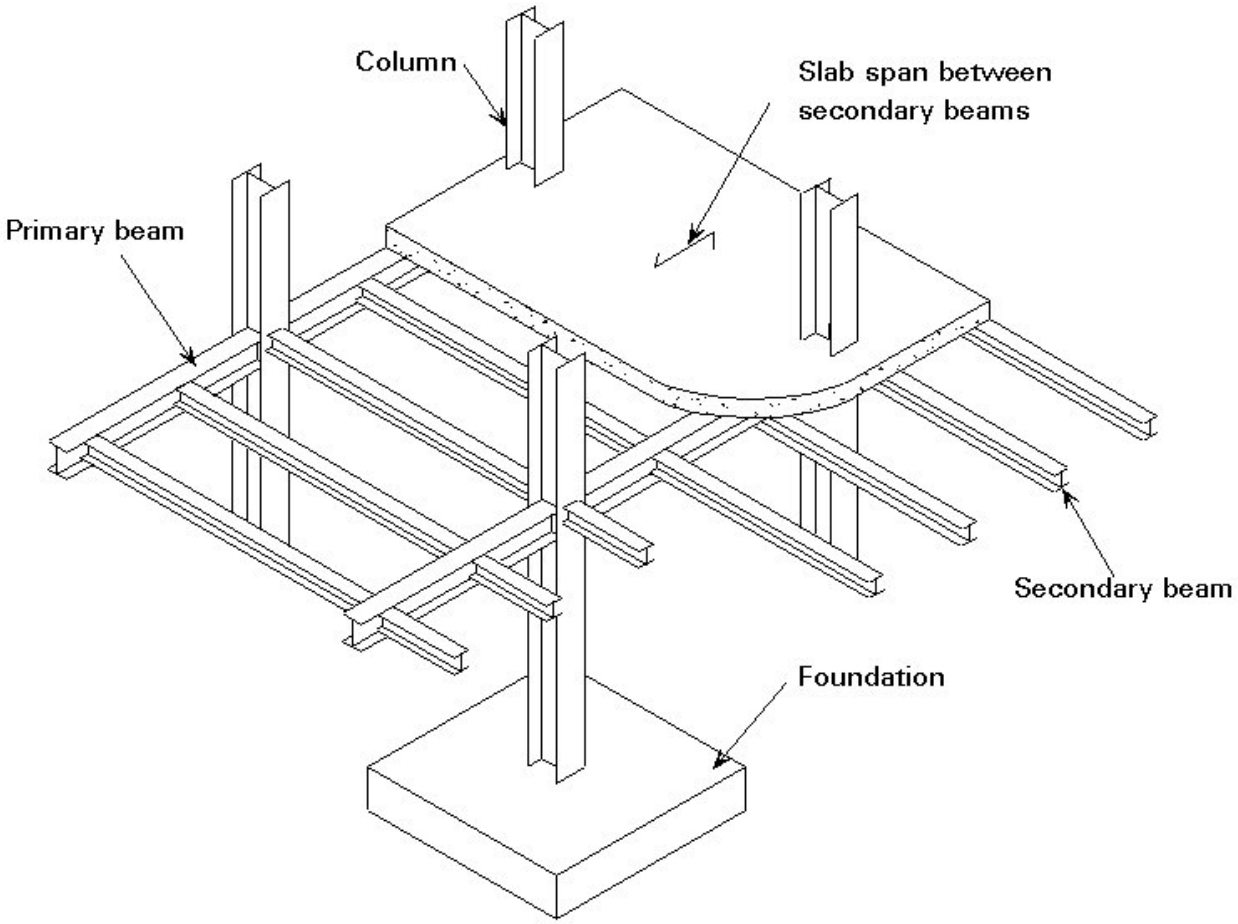
Floor Structure in Multistory Buildings



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Floor systems in Multistory Buildings

- Different types of slabs can be used, in either composite or non composite form.
- Most slab types can be designed to act as composite with the supporting beams if adequate shear connection is provided
- Faster construction if prefabricated or precasted elements of steel and reinforced concrete are used
- Openings in the webs of beams are required to permit passage of horizontal services, such as pipes (for water and gas), cables (for electricity and electronic communication), ducts (air-conditioning), etc.
- Floor spanning systems must provide adequate stiffness to avoid large deflections due to live load which could lead to damage of plaster and slab finishers.
- Where the deflection limit is too severe, pre-cambering with an appropriate initial deformation equal and opposite to that due to the permanent loads can be employed to offset part of the deflection.

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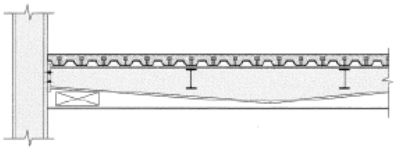
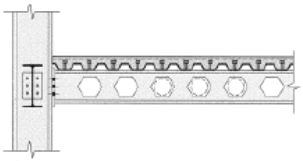
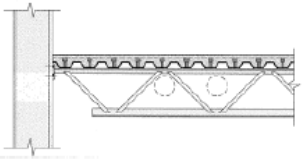
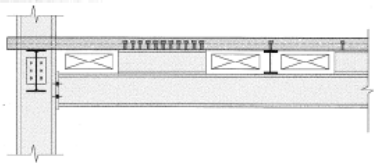
Structural systems overview

Type	Application	Example
Rolled beam: IPE, HEA, HEB, HEM	Columns / Beams w/o composite for spans **≤ 18m**	
Welded sections	Beams w/o composite action for spans **≤ 30m**	
Slim-floor beams	Beam embedded in the slab for economic spans **≤ 7m**	
Cellular beams	Sections with web openings for spans **≤ 24m**	
Divided structural members	e.g. trusses economic for spans **≥ 30m**	
Stub girders	US/UK system to easify passage of installations for spans **≤ 20m**	

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Why is composite system better?

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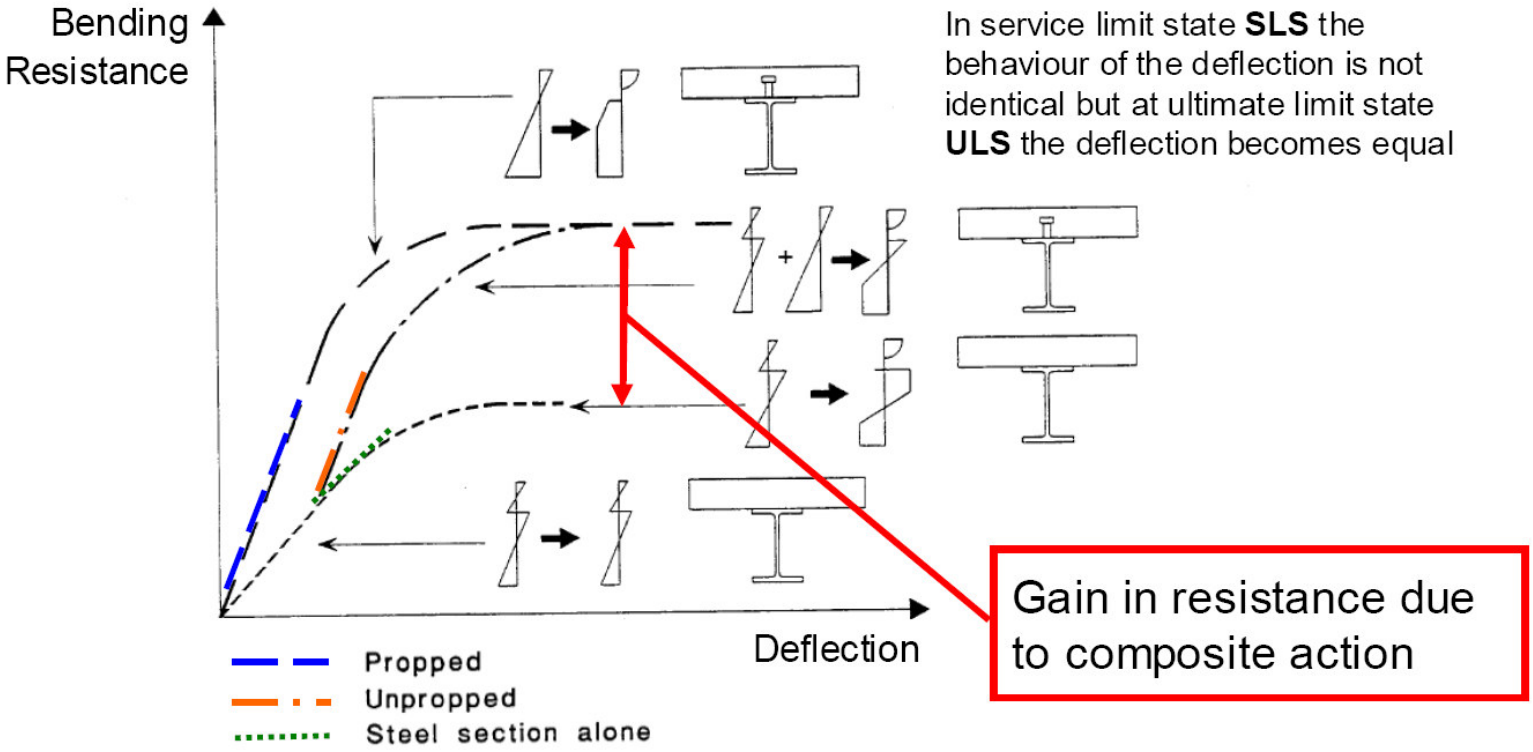
Why is composite system better?

Comparison between different construction types



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Why is composite system better?

Comparison between different construction types

- **Economic benefit** compared to traditional construction: with equivalent loads significant reduction of section possible (up to 30 %*, ⇒2 to 3 sections)
- **Decrease of deflection** (2 to 4 times) **and Increase in bearing resistance** (2 to 3 times) compared to pure steel beam
- **Reduction of construction height**: The overall height of the building may be reduced significantly (up to 30%* of the steel beam height / storey), thus an additional storey may be added with the same building height
- **Reduction in surface protection** (corrosion or fire) (ca. 15 %)
- **Increase in global stiffness** of building and optimised behaviour in case of fire or earthquake: The connection between steel beam and concrete chord is uniform in all directions; Damping behaviour improved as well (vibration)

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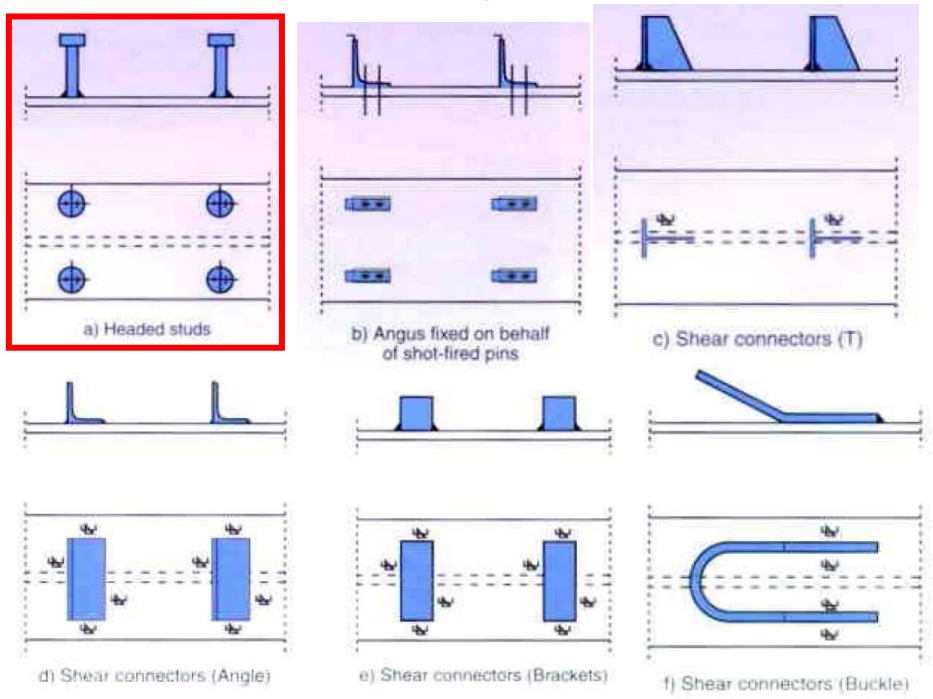
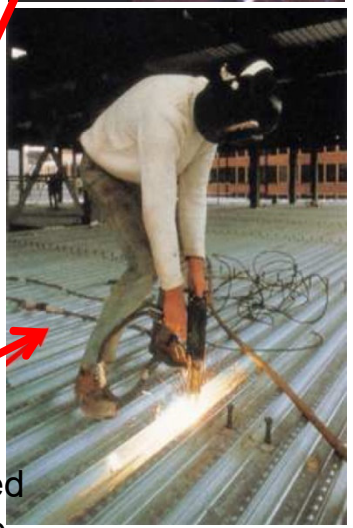
Different types of shear connectors

Ductile connectors according to EC4





Headed shear studs welded either in the shop or on the jobside

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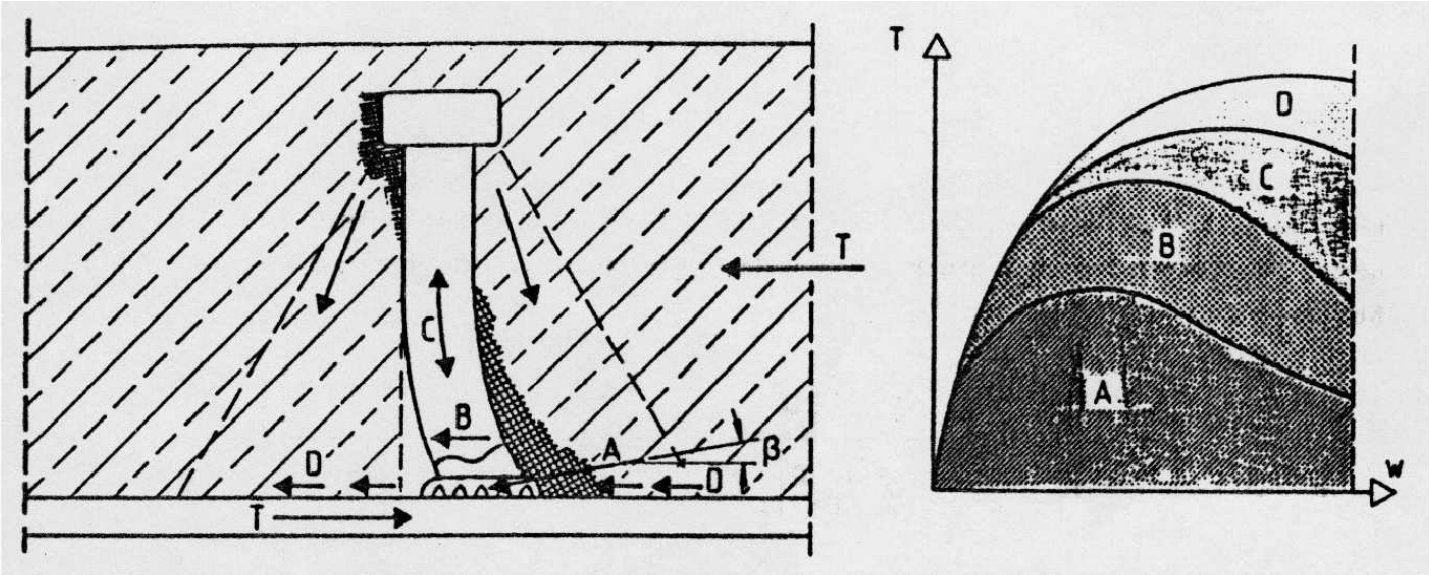
Bearing capacity: C20/25 – C50/60



Concrete failure

$$P_{Rd} = \alpha \cdot 0{,}29 \cdot d^2 \cdot \sqrt{f_{ck} \cdot E_{cm}} \frac{1}{\gamma_v}$$

Steel failure

$$P_{Rd} = 0{,}8 \cdot \frac{\pi \cdot d^2}{4} \cdot f_u \cdot \frac{1}{\gamma_v}$$

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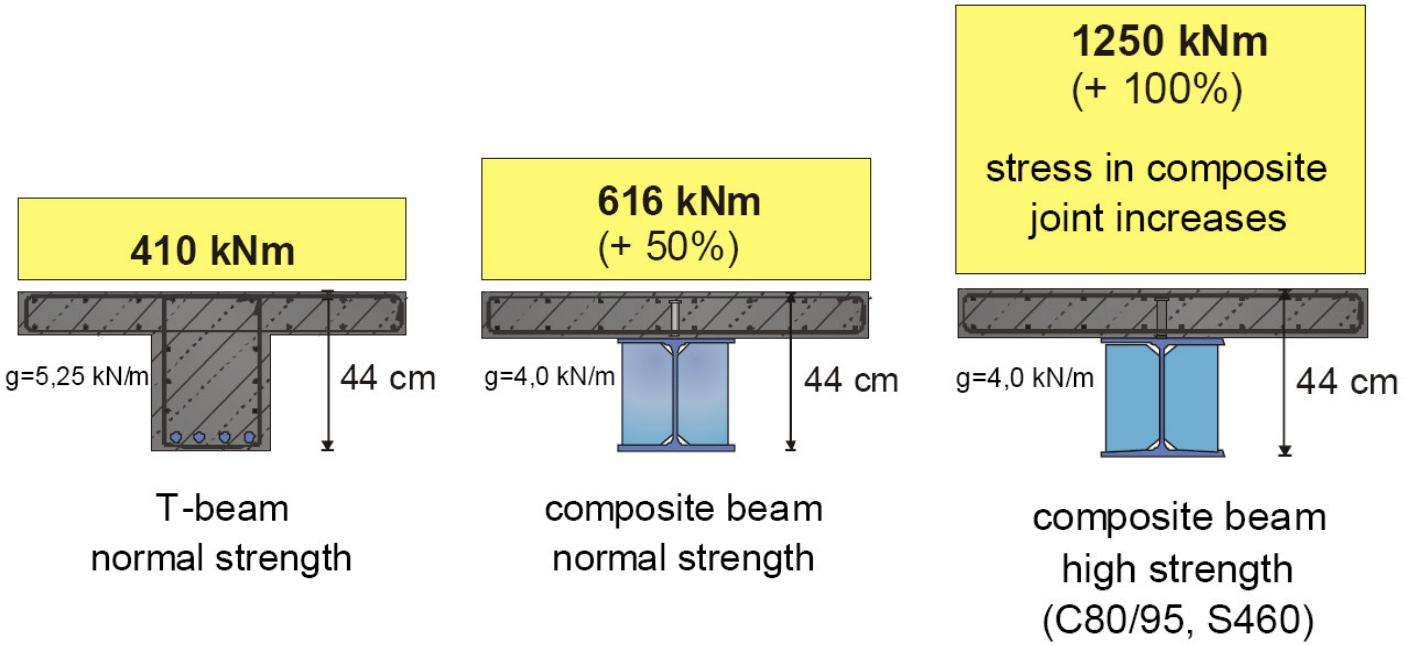
Shear studs in high strength concrete > C50/60

•Further increase in bearing capacity



- Requirements in construction
- Increased stress on shear joint

⇒ Higher demand on shear connector

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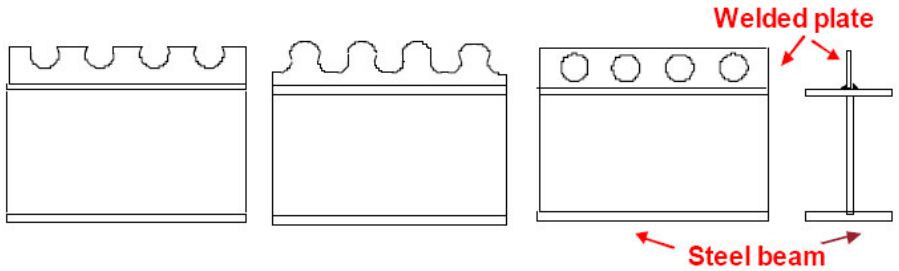
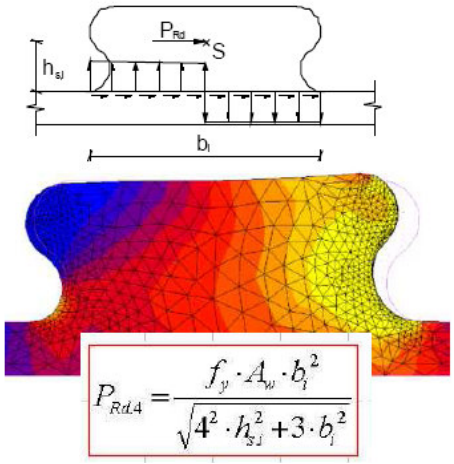
Shear connectors for special applications

•In composite structures using high strength materials, standard shear connectors show as not suitable as their ductility is not high enough



Crashing of Concrete



perforated shear connector (perfobond shear connector)

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Composite construction - Application

•In composite structures using high strength materials, standard shear connectors show as not suitable as their ductility is not high enough

Composite steel sections with camber before mounting of the metal decks

Parking Aix en Provence, France

Studs welded before or through the trays on site

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Composite construction - Application

Parking Aix en Provence, France

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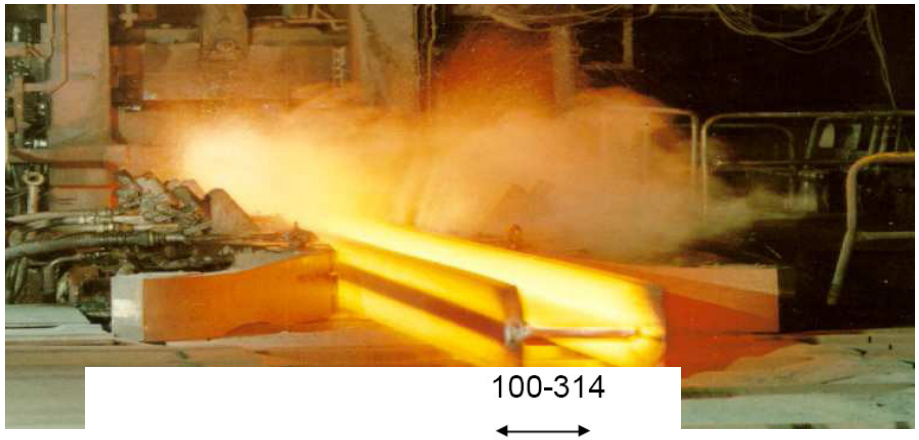
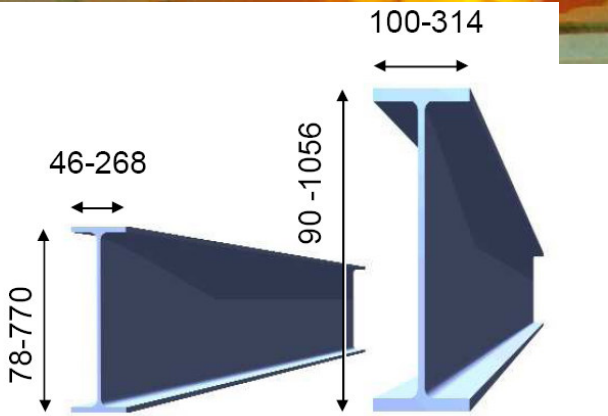
Type	Application	Example
Rolled beam: IPE, HEA, HEB, HEM	Columns / Beams w/o composite for spans **≤ 18m**	



Shanghai Financial Center



IPE 80 - 750 HE 100 - 1000

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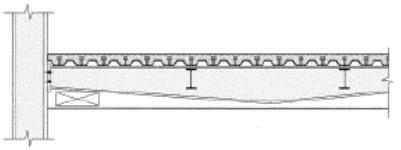
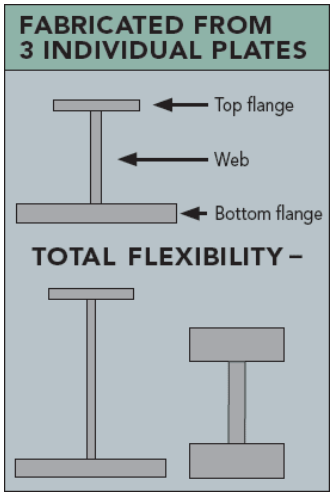
Type	Application	Example
Welded sections	Beams w/o composite action for spans ≤ **30m**	

- Welded section beams can be made with parallel flanges or tapered
- The economic advantage of welded beams is that they can be designed to provide the required moment and shear resistance
- When tapered, their characteristics can be varied along the beam span in accordance with the loading pattern
- Several forms of tapered beams are possible



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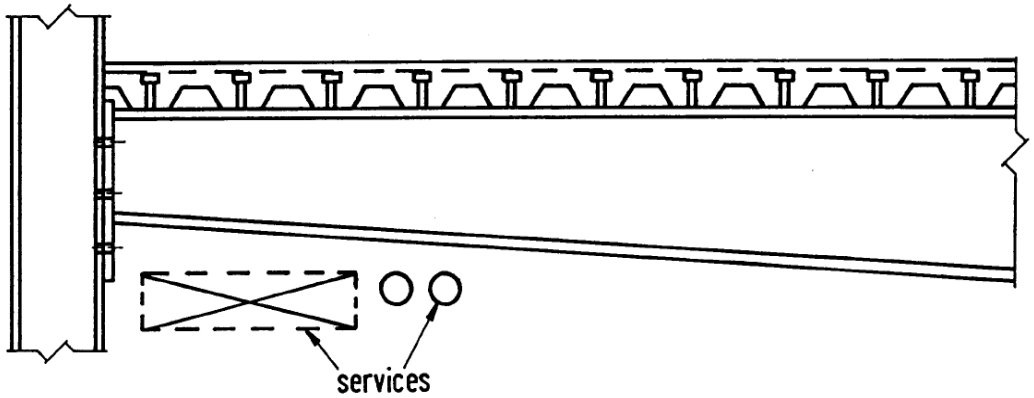
Tapered beam: simply supported beam

•A simply supported beam design with a maximum bending moment at the mid-span would require that they all effectively taper to a minimum at both ends, whereas a rigidly connected beam would have a minimum depth towards the mid-span.
•To make best use of this system, services should be placed towards the smaller depth of the beam cross-sections.



Tapered beam is found to be economical for spans of 13 to 20 m.

Tapered composite beams: simply supported beam

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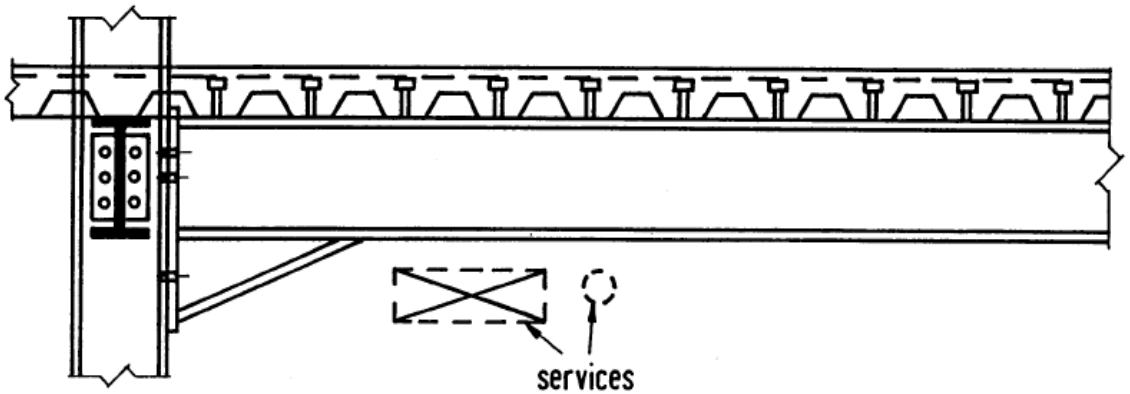
Tapered beam: haunched beam

- The span length of a composite beam can be increased by providing haunches or local stiffening of the beam-to-column connections
- The length of haunch is typically 5 to 7% the span length for non-sway frames or 7 to 15% for sway frames.
- Service ducts can pass below the beams as in conventional construction



Haunched composite beam

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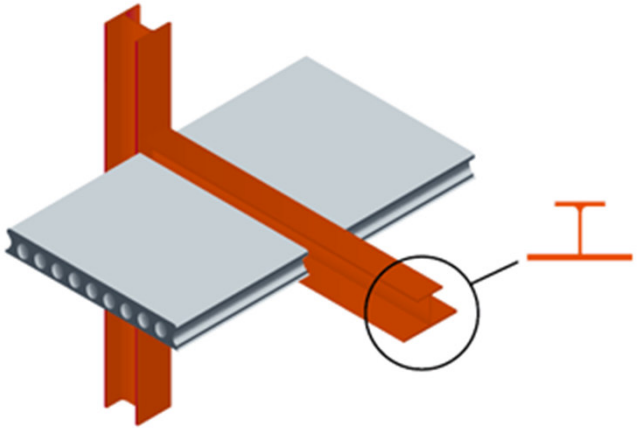
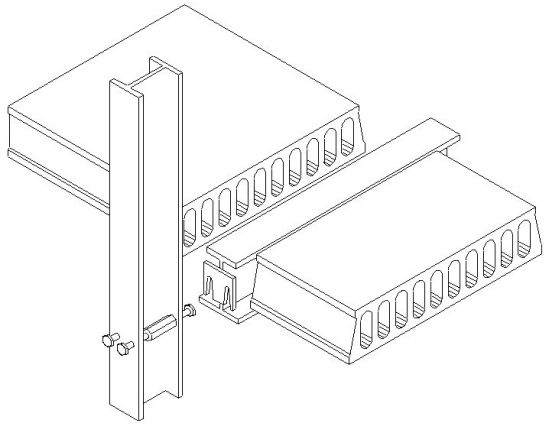
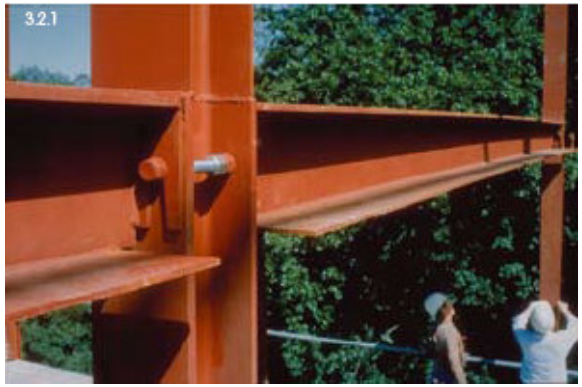
Type	Application	Example
Slim-floor beams	Beam embedded in the slab for economic spans **≤ 7m**	







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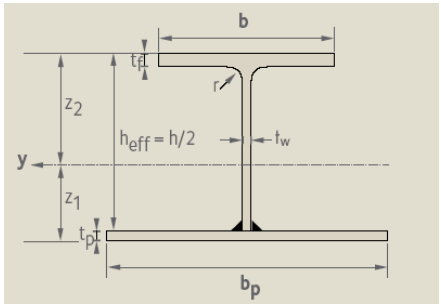
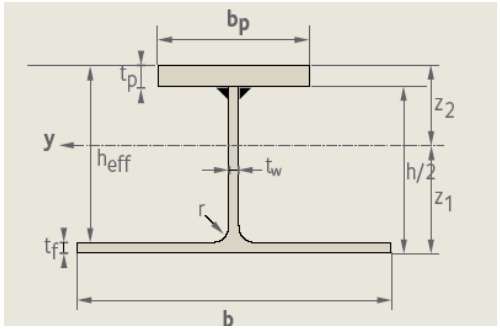
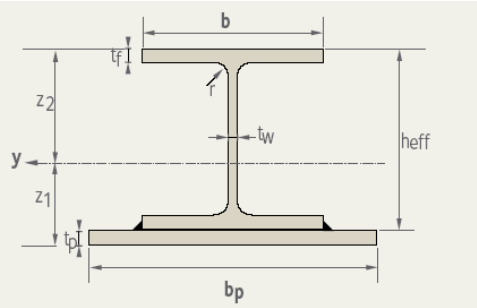
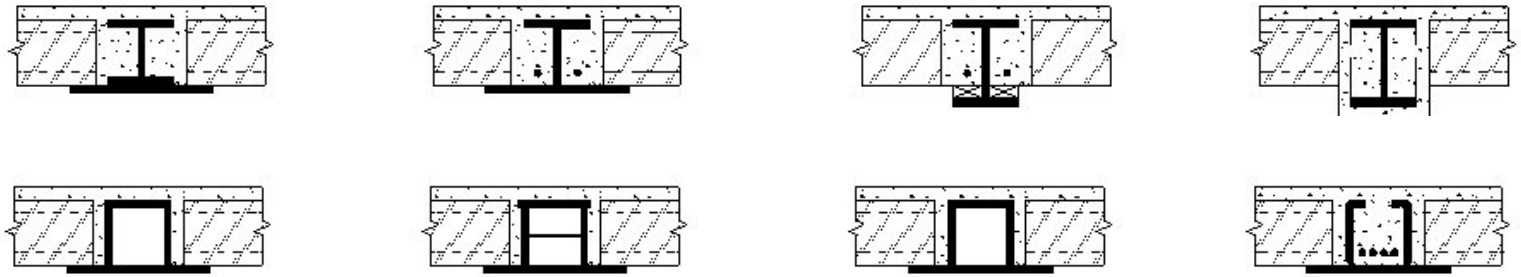
Slim-floor system









Integrated steel beams for Slim-floor systems

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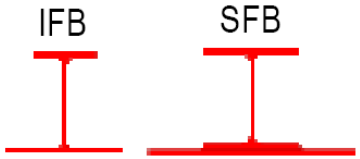
- Request of modern architecture:
  - Transparent structural envelopes with column free ground floor design
  - Flexibility for sustainable conversion of use
  - Possibility to upgrade services for multifunctional living
- Slim-floor construction (IFB / SFB):
  - Combines advantages of prefabricated slab elements with steel framed construction

⇒ Economic solutions fulfilling the above specified demands



- The “slim floor” system cannot be used as part of the seismic load-resistance system

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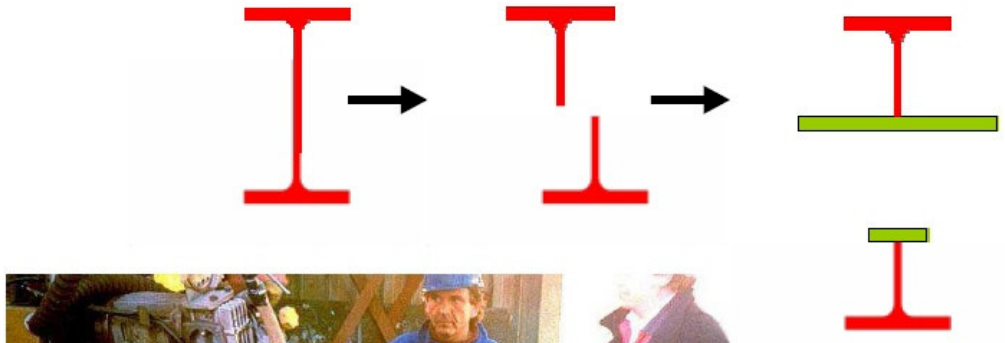
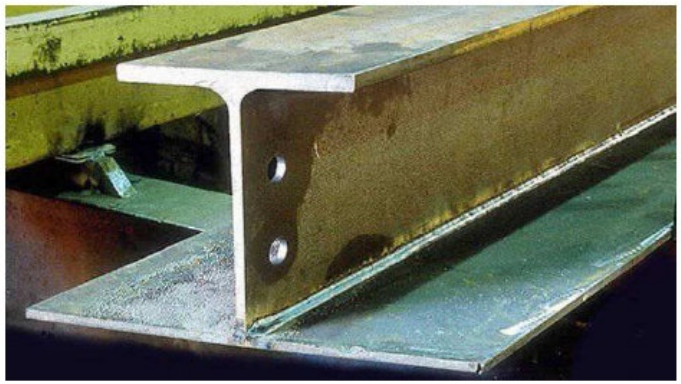
Slim-Floor beams - Fabrication



IFB Typ A

or

IFB Typ B



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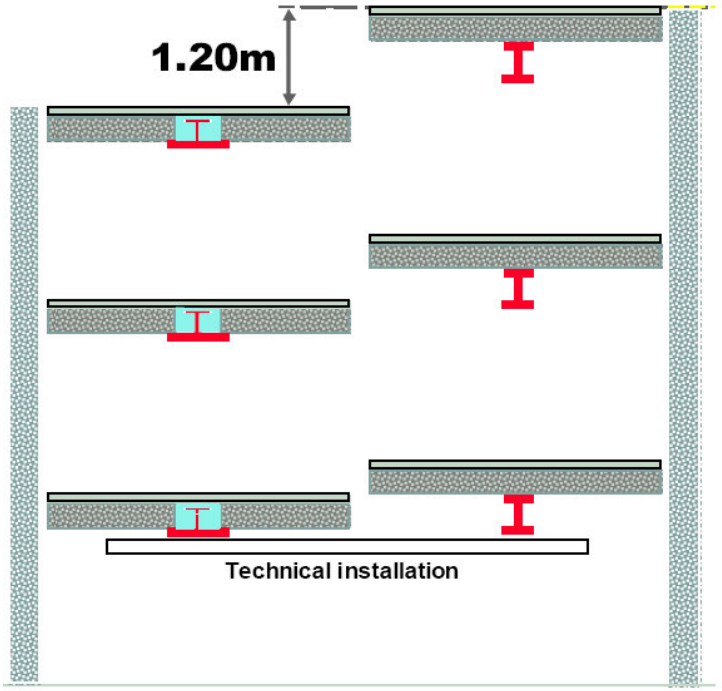
Slim-Floor beams - Advantages

⇒ Floor thickness reduction
⇒ Constructing floors of variable thicknesses
⇒ Incorporating under-floor technical equipment
⇒ Freeing-up working space
⇒ Built-in fire resistance
⇒ Competitive pricing
⇒ Easy to build
⇒ Sustainable construction
⇒ Lighter structures
⇒ Less façade surface
⇒ Heating / cooling cost reduction
⇒ Lower building height
⇒ More natural light



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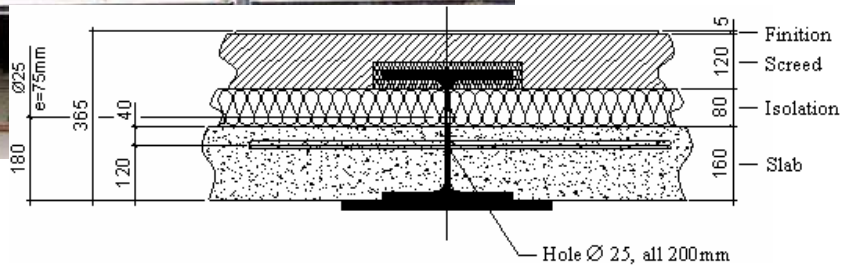
Slim-Floor beams - Applications



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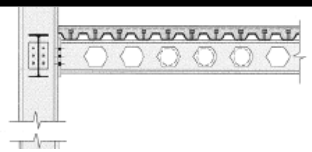
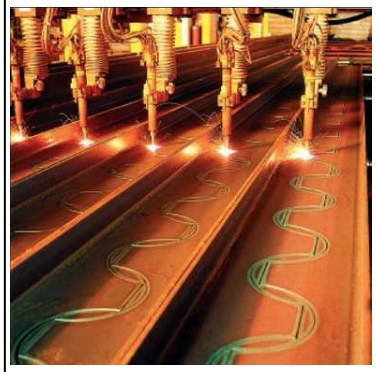
Type	Application	Example
Cellular beams	Sections with web openings for spans ≤ **24m**	



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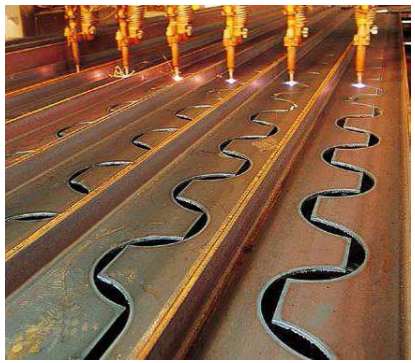
II

III

IV

• Cellular beams can be made with parallel flanges, tapered or curved
• Castellated beams have limited shear capacity and are best used as long span secondary beams where concentrated loads can be avoided.

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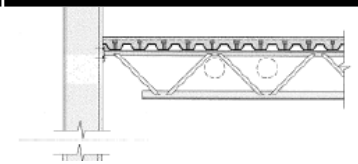
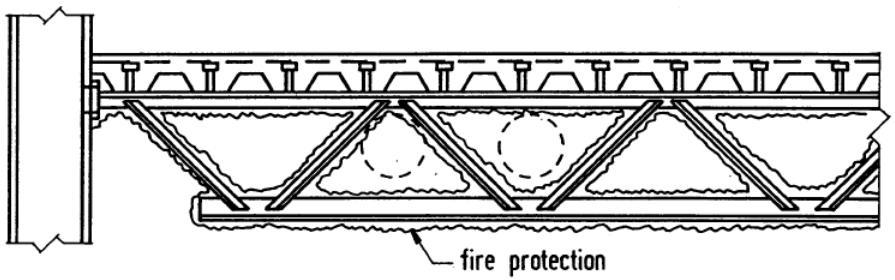
Type	Application	Example
Divided structural members	e.g. trusses economic for spans ≥ **30m**	

- Trusses are frequently used in multistory buildings for very long span supports
- The openings created in the truss braces can be used to accommodate large services
- Truss configuration creates difficulty for fire protection. Fire protection wrapping is labor intensive and sprayed-protection systems cause a substantial mess to the services that pass through the web opening
- From a structural point of view, the benefit of using a composite truss is due to the increase in stiffness rather than strength



Composite truss

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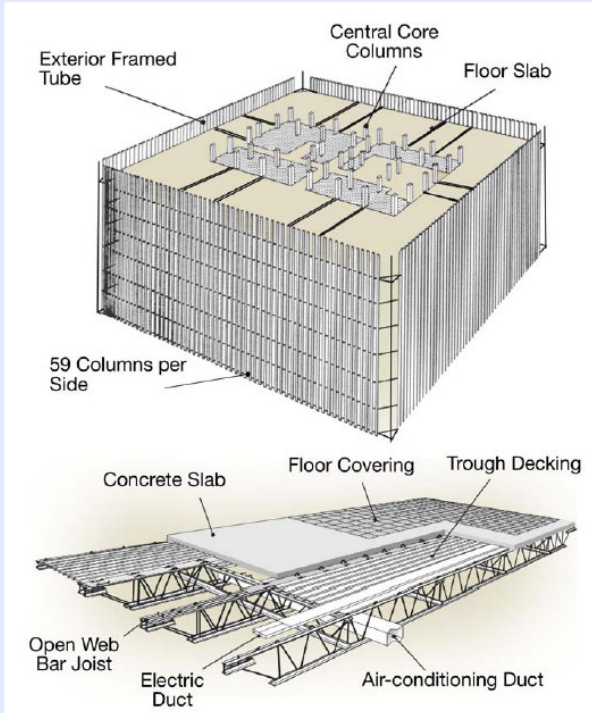
Applications

Composite Trusses – WTC application

Innovative WTC Tower Structural System



- Innovative structural system when built; incorporated many new and unusual features
- Two features require additional consideration:
  - Composite floor truss system using long span open-web bar joists and spray-applied fireproofing
  - Design for wind loads and control of wind-induced vibrations

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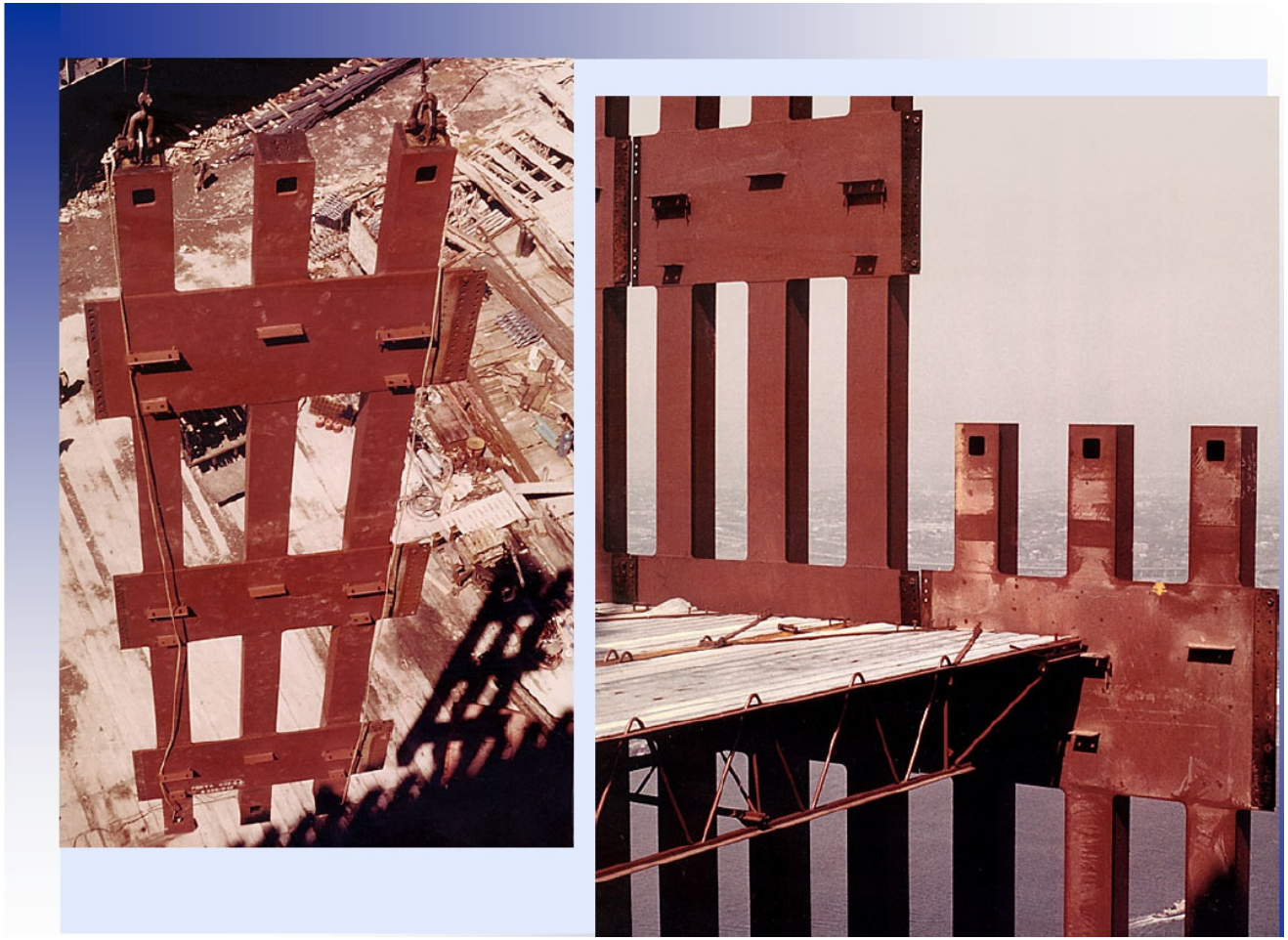
WTC application – views during construction



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WTC application – fire performance

Fire Performance of Composite Floor System

- Fire-protection of a truss-supported floor system with spray-on fireproofing was innovative and not consistent with then-prevailing practice.
- **No evidence found of technical basis in the selection of fireproofing thickness** to meet 2 h fire rating:
  - 1/2 in. specified when WTC towers were built to maintain Class 1-A (not 1-B) fire rating requirement of the NYC Building Code
  - 1-1/2 in. specified for upgrades some years prior to 2001
  - 2 in. for similar floor system in an unrestrained test (model code evaluation service recommendation in June 2001, unrelated to WTC buildings)
- **No evidence that full-scale fire resistance test of the WTC floor system was conducted** to determine the required fireproofing thickness; in 1966, the Architect of Record and, in 1975, the Structural Engineer of Record stated that the fire rating of the WTC floor system could not be determined without testing.

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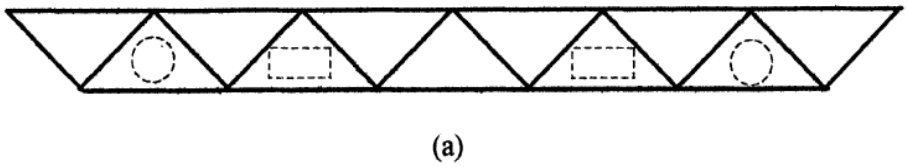
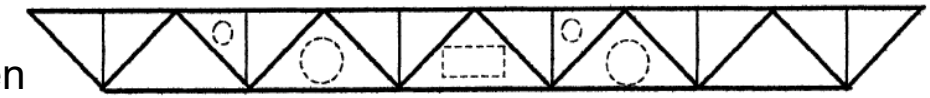
Truss types

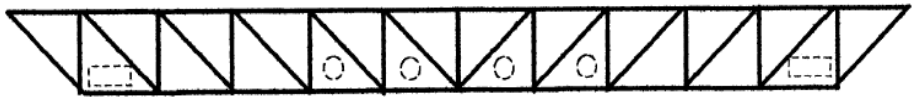
- Several forms of truss arrangement are possible. The three most common web framing configurations in floor truss and joist designs are:
  - Warren Truss
  - Modified Warren Truss
  - Pratt Truss



(a)



(b)



(c)

Truss configuration: (a) Warren truss, (b) Modified Warren truss, and (c) Pratt truss.

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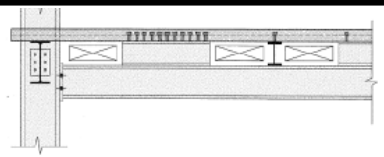
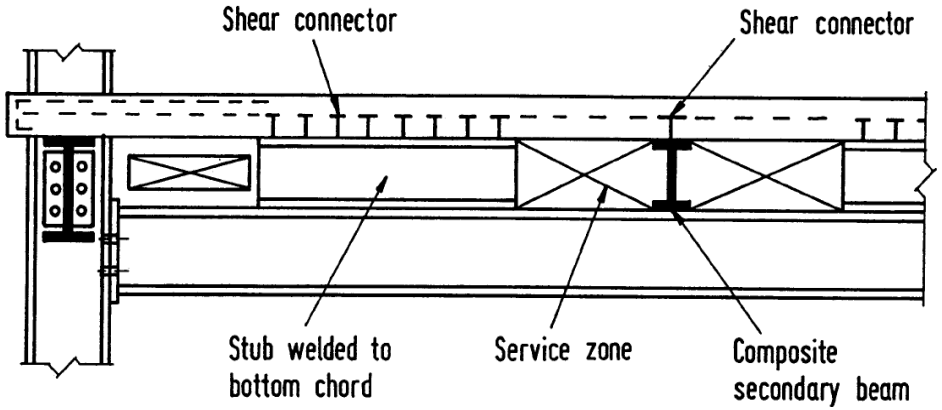
Type	Application	Example
Stub girders	US/UK system to easify passage of installations for spans **≤ 20m**	



•The stub girder system involves the use of short beam stubs that are welded to the top flange of a continuous, heavier bottom girder member, and connected to the concrete slab through the use of shear studs.
•Continuous transverse secondary beams and ducts can pass through the openings formed by the beam stub. The natural openings in the stub girder system allow the integration of structural and service zones in two directions, permitting story-height reduction when compared with some other structural framing systems.

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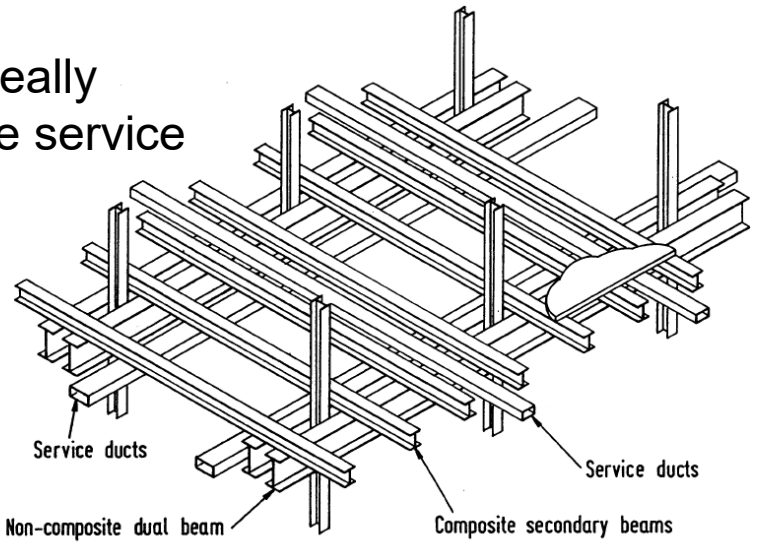
Special types

- Parallel Beam System
  - The system consists of two main beams with secondary beams run over the top of the main beams. The main beams are connected to either side of the column.
  - This will help in reducing the construction depth
  - The secondary beams are designed to act compositely with the slab and may also be made to span continuously over the main beams. The need to cut the secondary beams at every junction is thus avoided.
  - The parallel beam system is ideally suited for accommodating large service ducts in orthogonal directions



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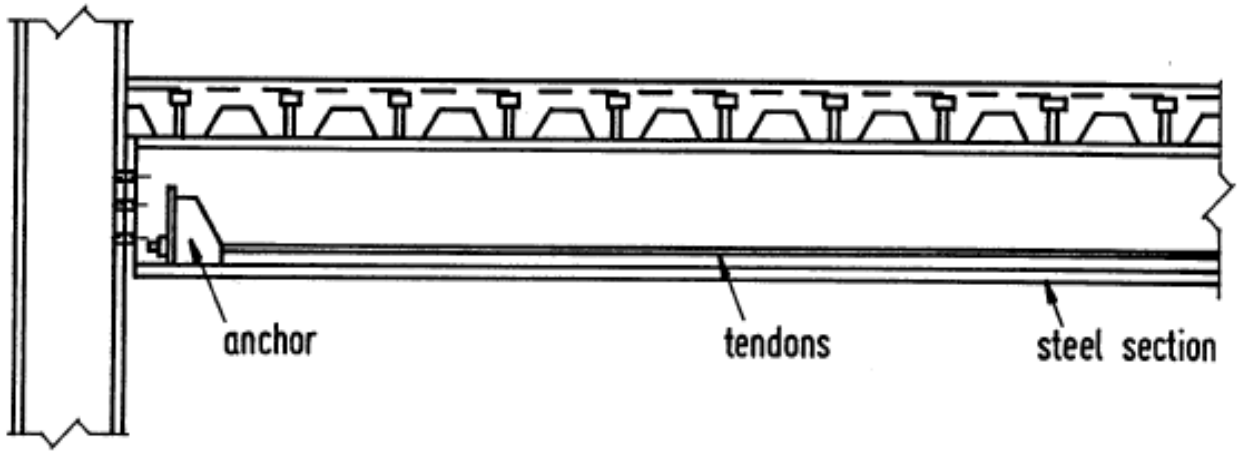
Special types

- Prestressed Composite Beams
  - Prestressing of the steel girders is carried out such that the concrete slab remains uncracked under the working loads and the steel is utilized fully in terms of stress in the tension zone of the girder.
  - It enhances the load-carrying capacity and stiffness of long-span structures



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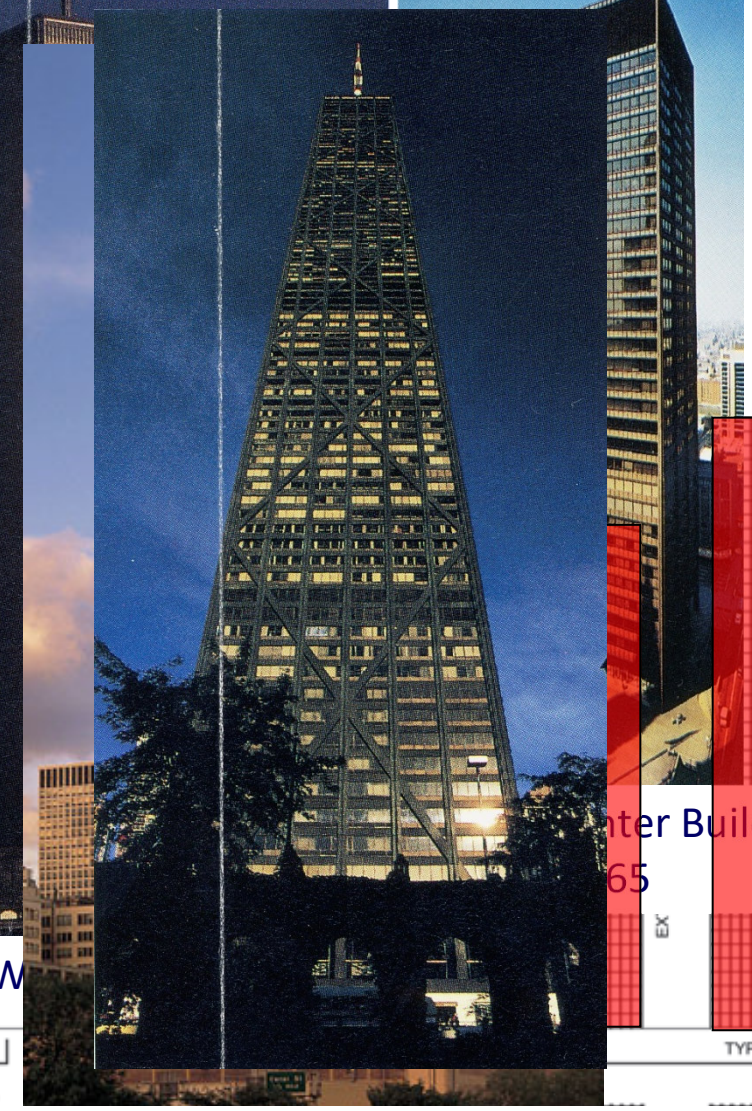
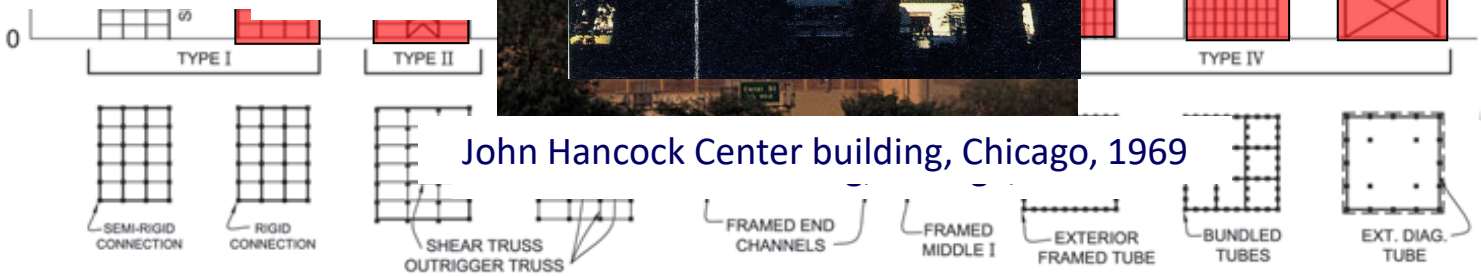
Lateral-load resisting systems

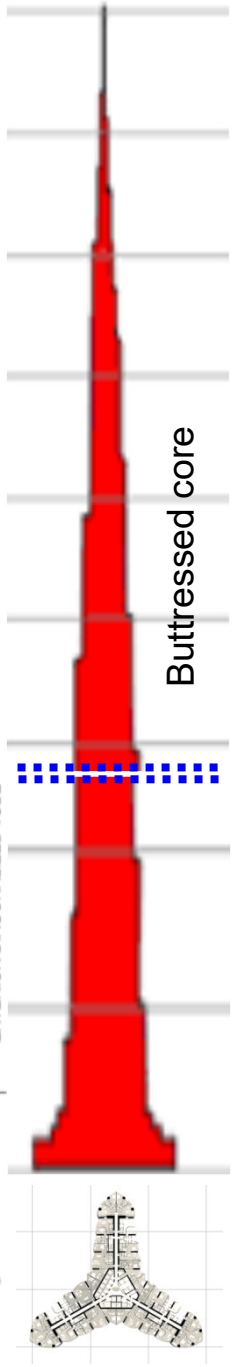
- A rigid unbraced frame may be capable of resisting lateral loads without relying on an additional bracing system in case of a low to medium-height building
- High-rise building systems should use structural systems that are effective in resisting the larger lateral loads
- Types of lateral-load resisting systems (R. Plank, M. McEvoy):
  - Shear frames: beams and columns connected with rigid joints
  - Shear truss: bracing between columns to form vertical shear trusses
  - Shear truss-frames: shear frames + shear trusses
  - Shear truss-frame-outrigger and belt trusses: internal core is connected to perimeter frames by deep girders – outriggers.
  - Framed tubes: close spacing columns on the exterior frames forming a vertical tube. The tube behaves as a cantilever
  - Truss tubes: the same system as framed tubes, tied by a system of diagonals
  - Bundled or modular tubes: framed or trusses tubes grouped together like cells
  - Super-frame: megaframe in the overall form of a Vierendeel frame
  - Composite systems: mixed RC and steel systems (concrete shear walls or concrete framed tubes combined with various structural steel framings
- Recommended limits for typical multistory frames are given in the next table

First Wisconsin Center Building
1974



John Hancock Center building, Chicago, 1969





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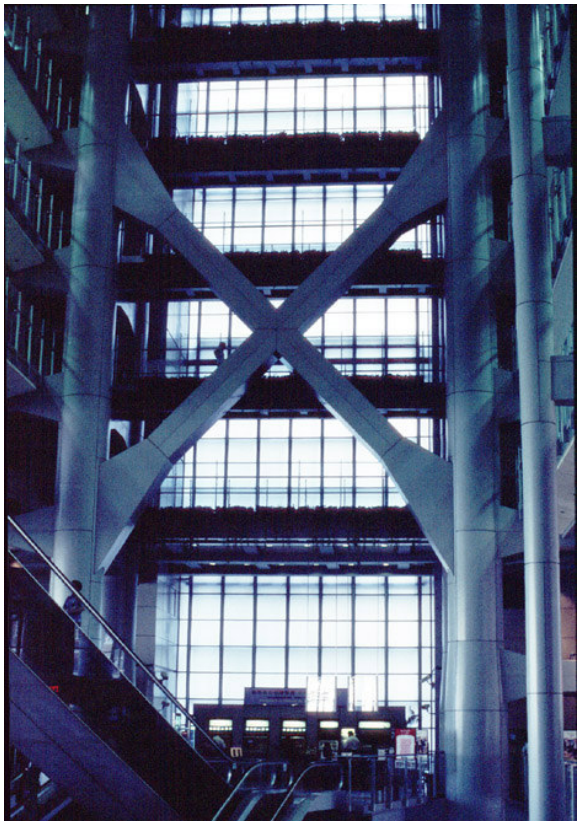
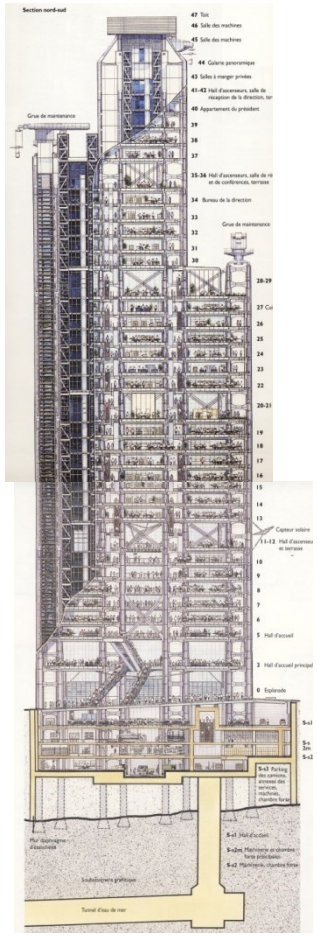
Structural systems for multistory buildings

Hongkong and Shanghai Bank
Completed 1986
180 meters height





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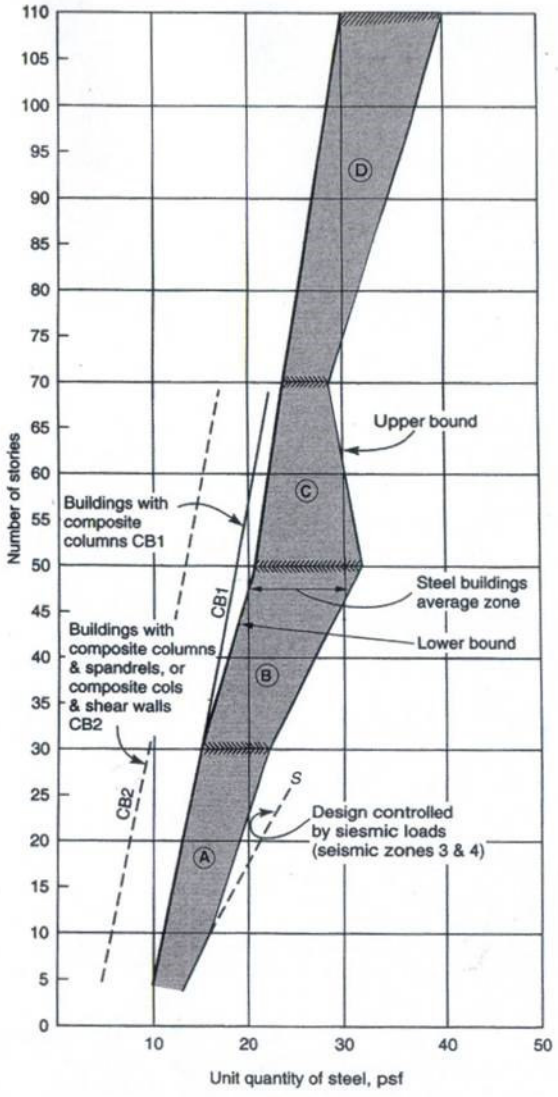
Structural systems

- Even for high seismic areas, for buildings with more then 25-30 stories, the wind load becomes predominant in design
- However, seismic design philosophy should taken into account (structural system, local detailing, ….)



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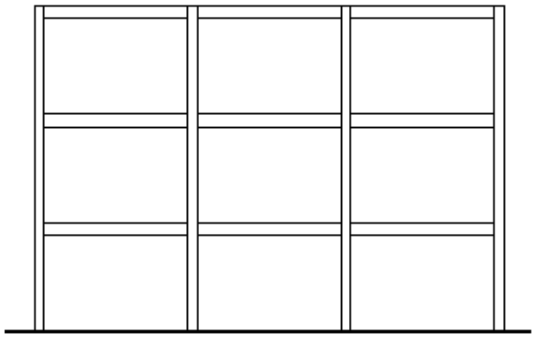
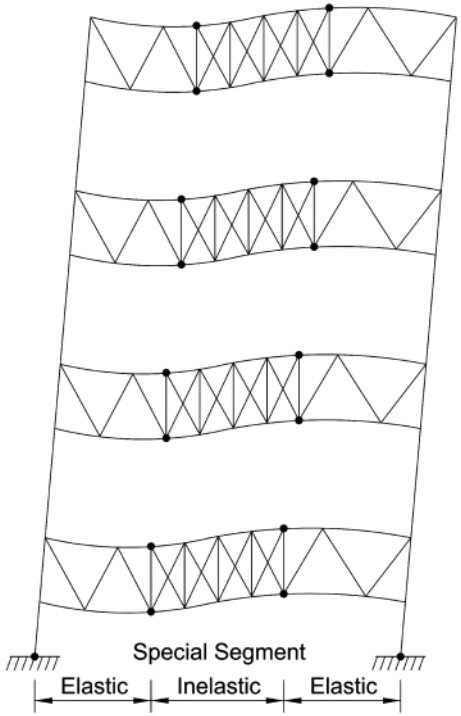
Types of frames

- Homogenous frames
  - Unbraced frames (moment resisting frames)



  - Staggered truss systems



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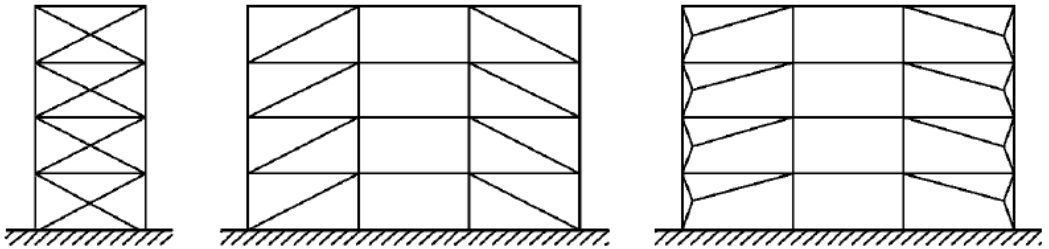
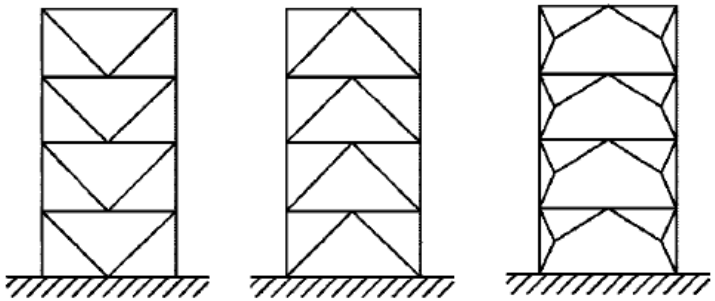
- Homogenous frames
  - Centrically braced frames

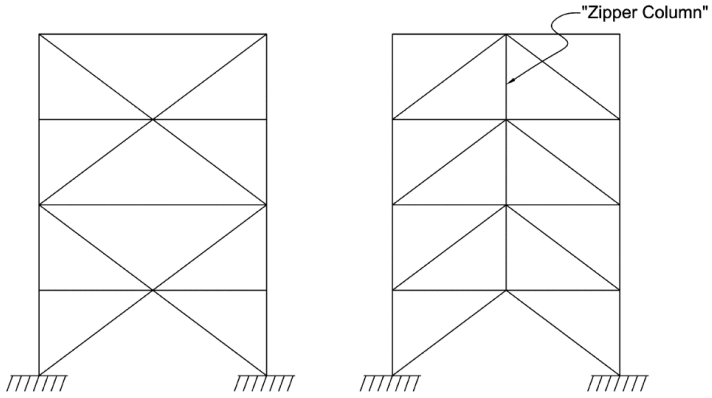


One-story x-bracing

single diagonal



Chevron bracing (V or inverted V)



Two-story x-bracing

"Zipper-column" with inverted-V bracing.

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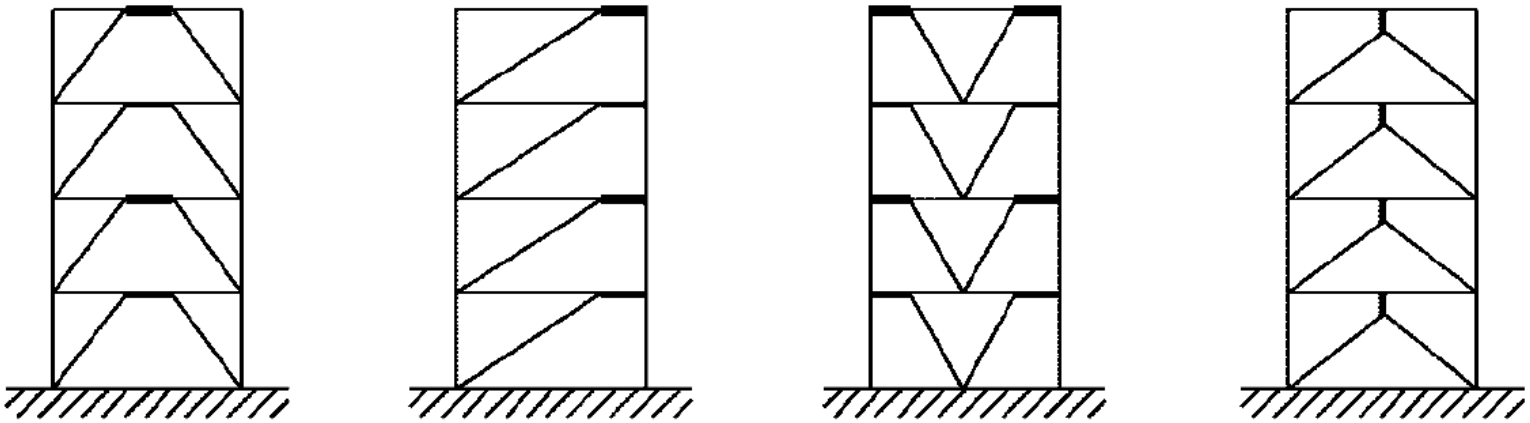
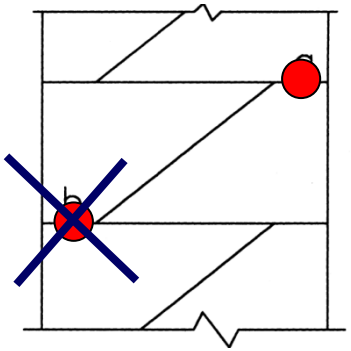
- Homogenous frames
  - Eccentricallv braced frames



In some bracing arrangements, such as that illustrated in figure (right) with links at each end of the brace, links may not be fully effective. If the upper link has a significantly lower design shear strength than that for the link in the story below, the upper link will deform inelastically and limit the force that can be developed in the brace and to the lower link. When this condition occurs the upper link is termed an active link and the lower link is termed an inactive link. The presence of potentially inactive links in an EBF increases the difficulty of analysis.



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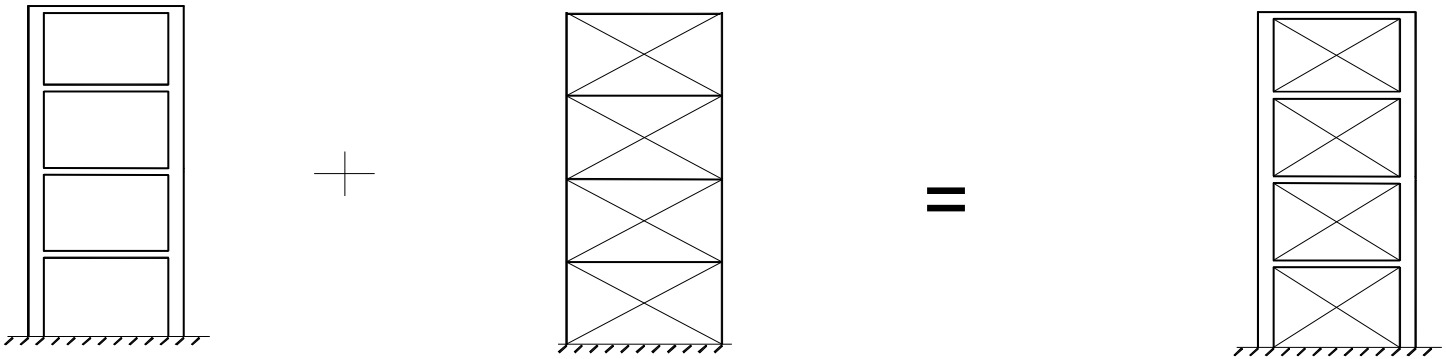
- Dual frames

Flexible system + **Rigid system**

Predominantly acting under gravitational loads

Predominantly acting under lateral loads



Dual system - Structural system with the following features: (1) an essentially complete space frame that provides support for gravity loads; (2) resistance to lateral load provided by moment frames that are capable of resisting at least 25 percent of the base shear, and concrete or steel shear walls, or steel braced frames (EBF, CBF); and (3) each system designed to resist the total lateral load in proportion to its relative rigidity.

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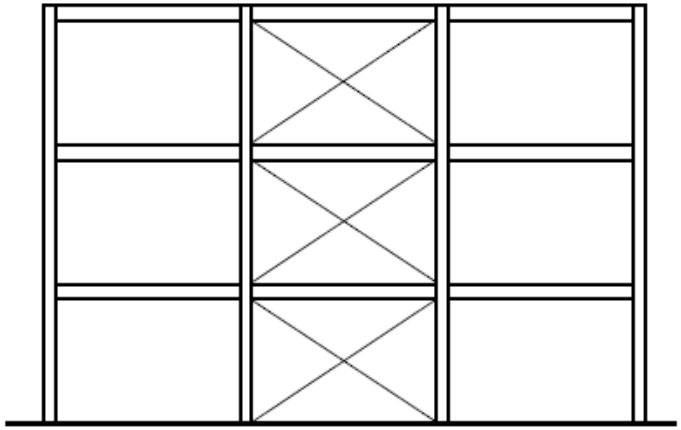
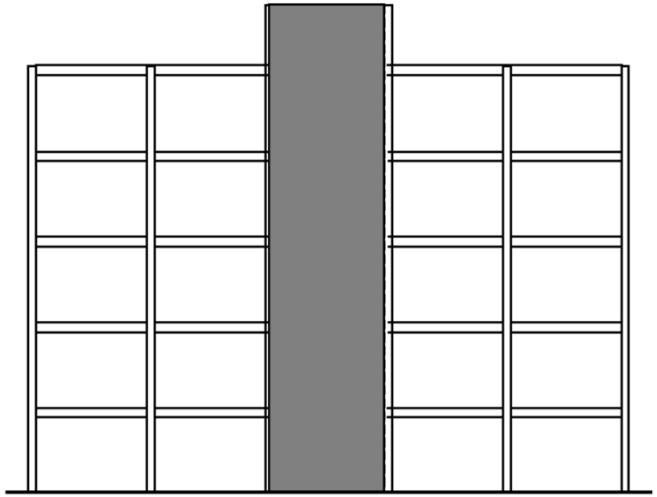
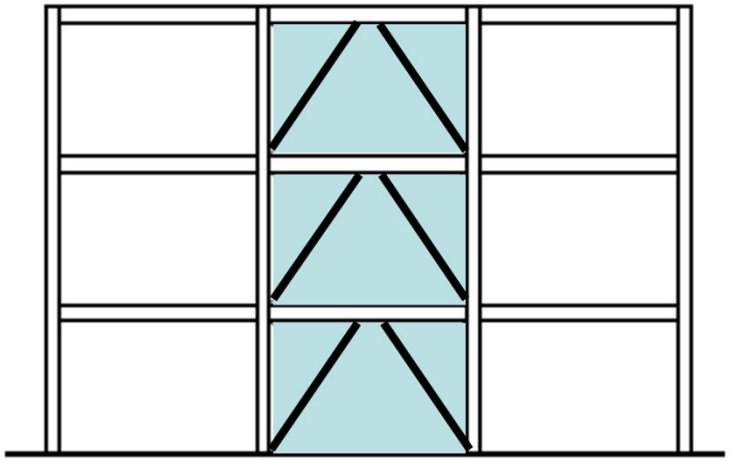
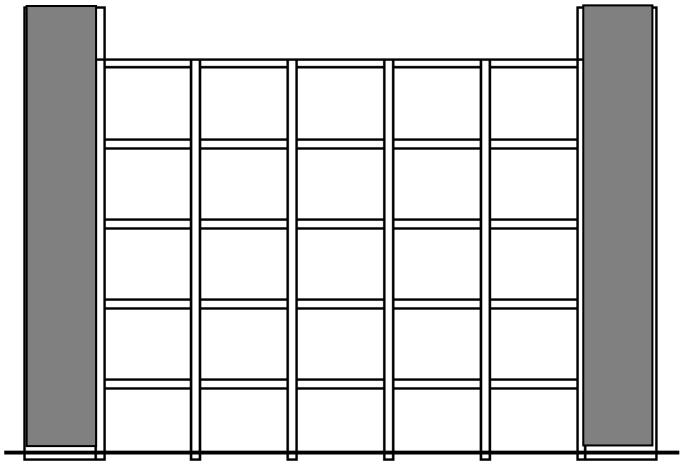
Examples of dual frames









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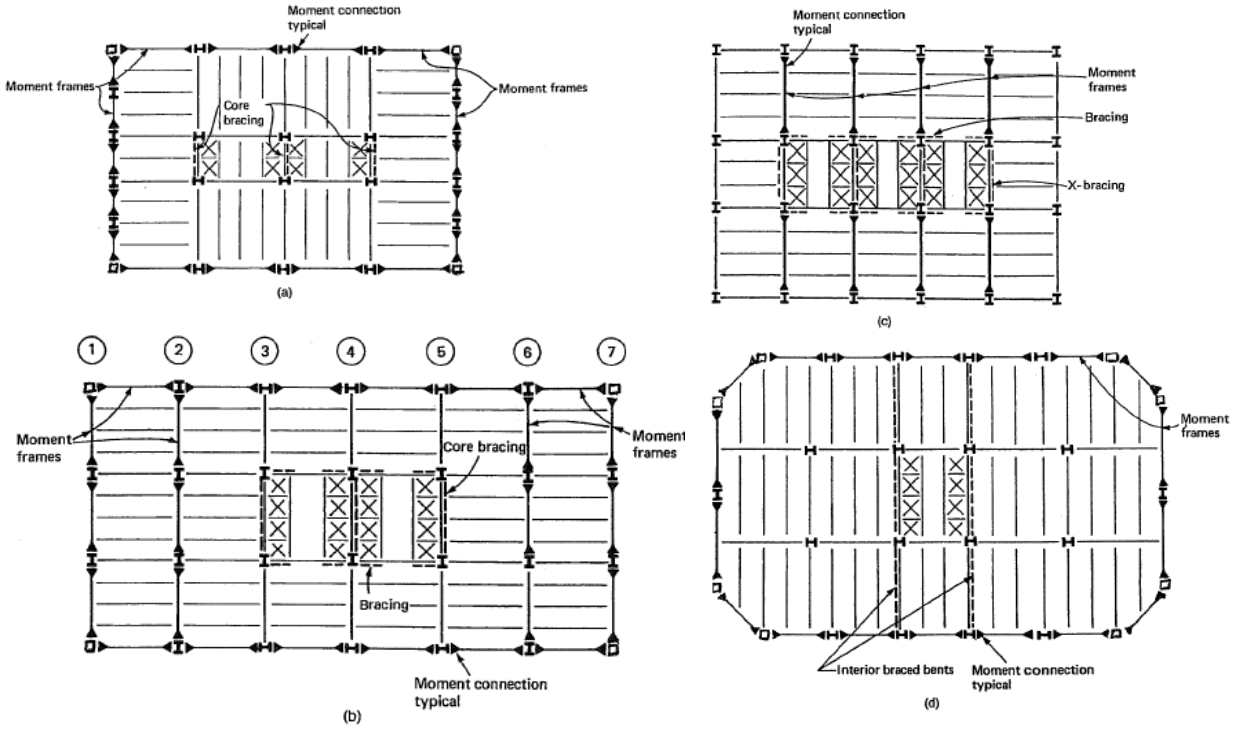
Dual frames

– Interaction between braced and unbraced spans



Schematic plans showing interacting braced and rigid frames: (a) braced core and perimeter frames; (b) braced core and interior and exterior frames; (c) braced core and interior frames; (d) full-depth interior bracing and exterior frames

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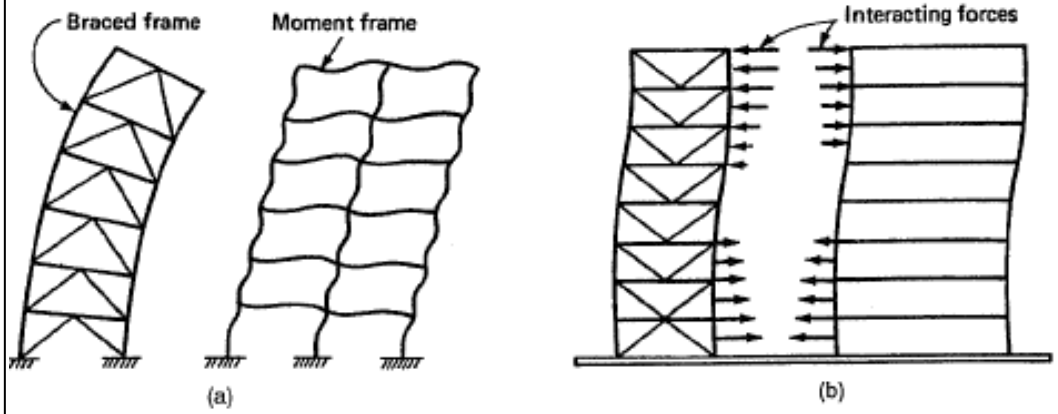
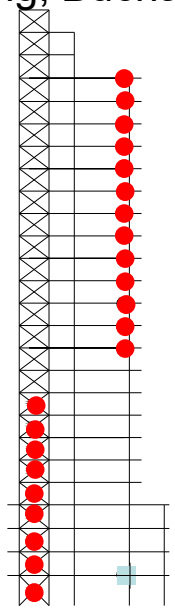
Interaction between braced and unbraced spans



Interaction between braced and moment frames: a) characteristic deformation shape; b) variation of shear forces resulting from interaction

Distribution of plastic hinges, a_g = 0,16g, TCI building, Bucharest



- MRFs act predominantly in shear (no web effect) while BFs act in bending
- Diaphragm effect of the slab assures the interaction between MRF and BF → better behavior

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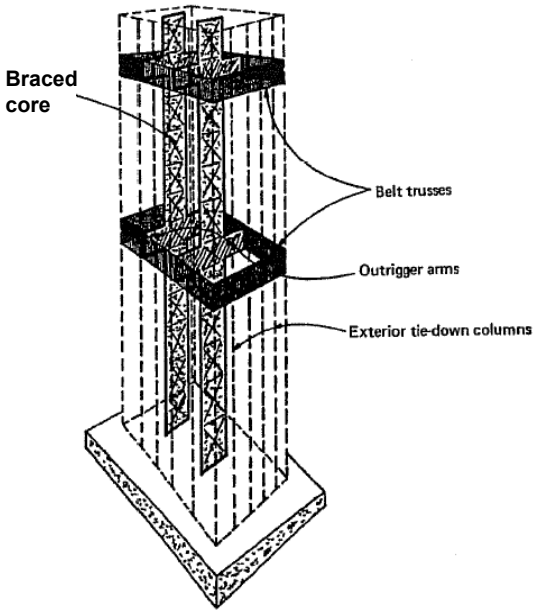
Outrigger and belt truss system

- The outriggers couple the columns and the core
- The lateral deflections are smaller than if the core is freestanding
- Belt trusses around the building



Advantages:

- Reduce building deflections and core bending stresses
- Reduce the rotational reactions
- Minimize the structural cost penalty associated with stability of slender buildings
- Effective for improving 3D behavior of irregular buildings

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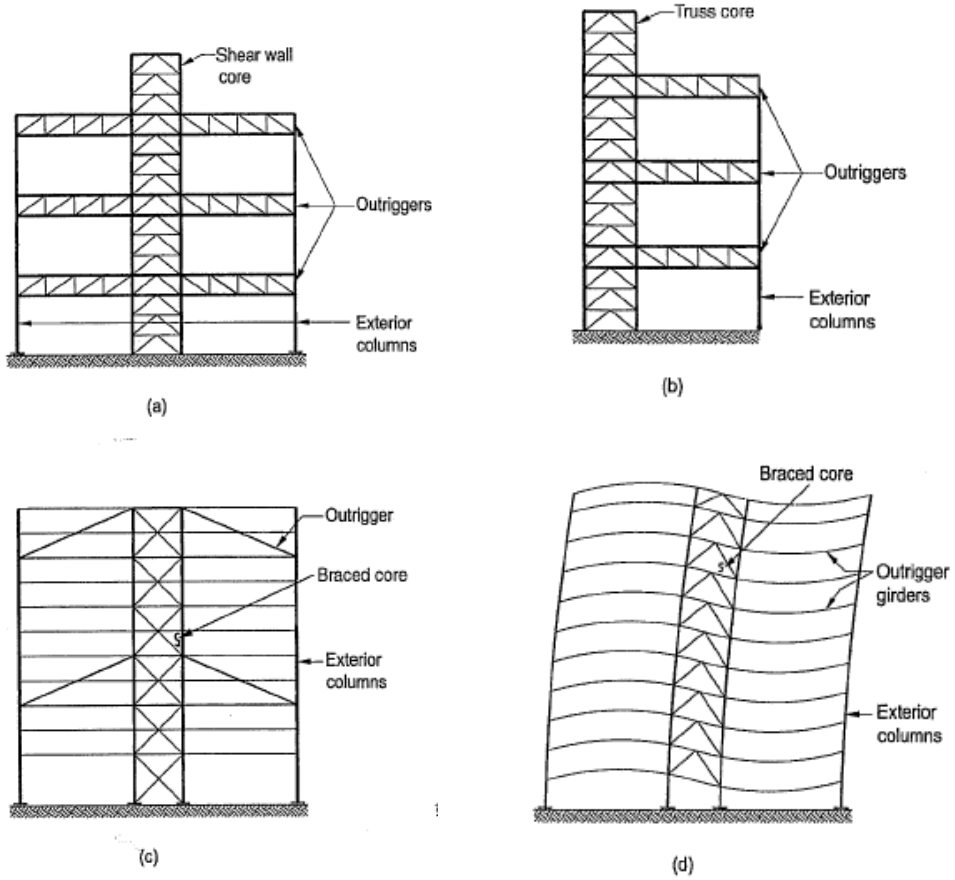
Outrigger and belt truss system - arrangements



(a) outrigger system with a central core: (b) outrigger system with offset core; (c) diagonals acting as outriggers; (d) floor girders acting as outriggers.

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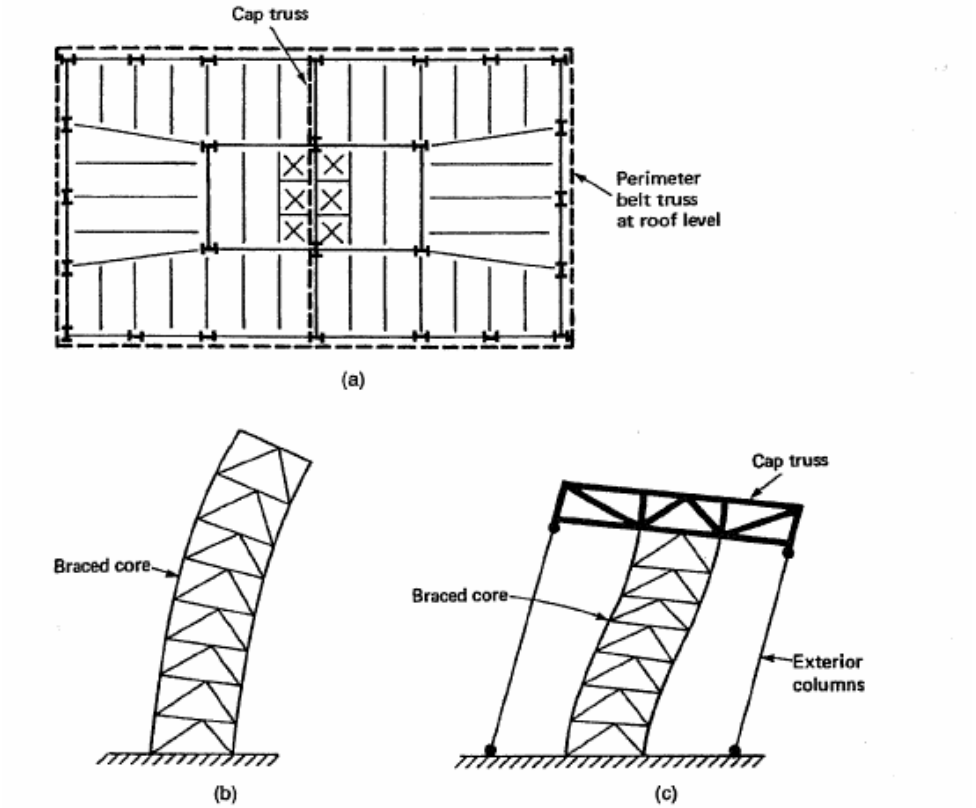
Outrigger and belt trus system - behaviour



a) Building plan with cap truss; b) cantilever bending of core; c) tie-down action of cap truss

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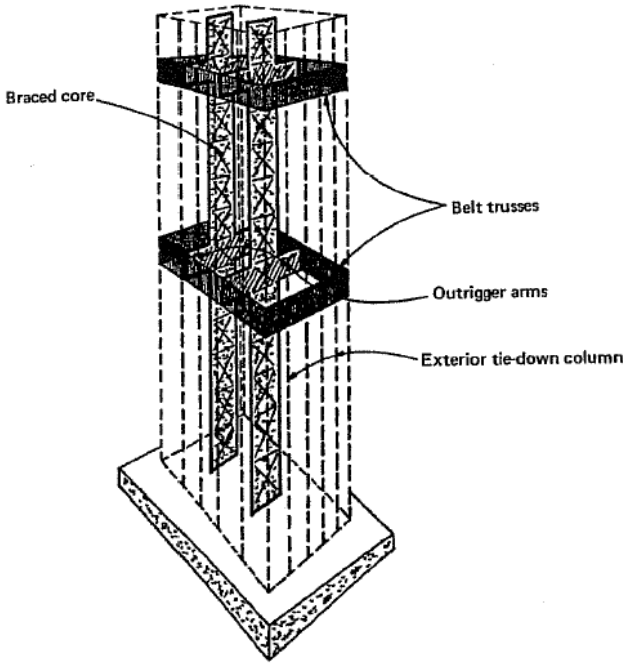
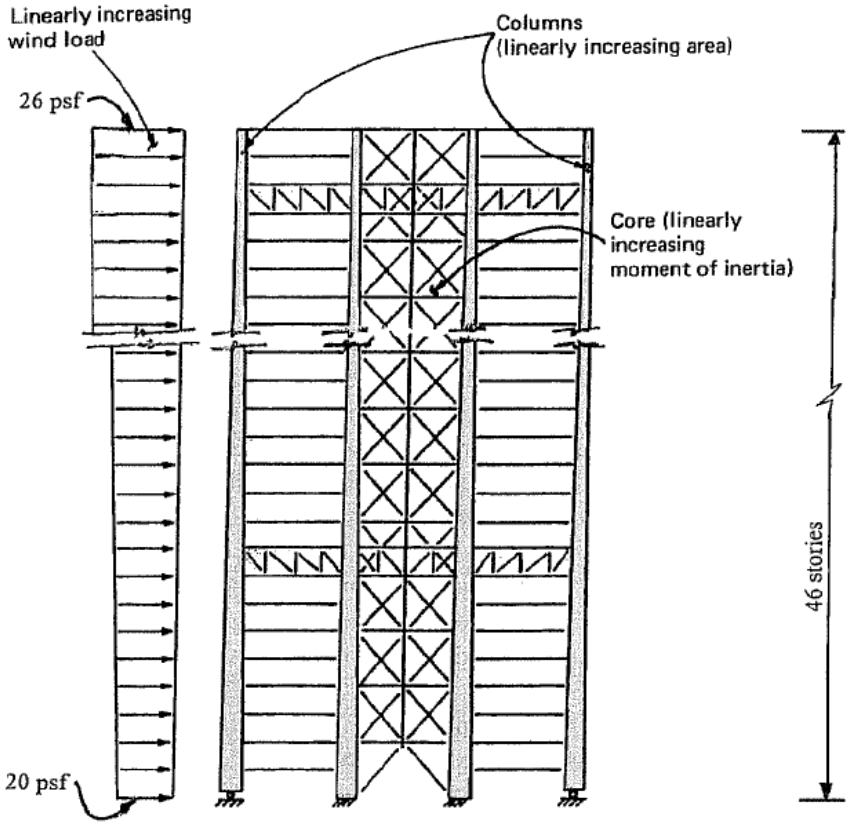
Optimum location of two outriggers



Schematic structural system for a building with two outrigger and belt trusses



Conceptual model and loading diagram for a building with two outriggers and belt trusses

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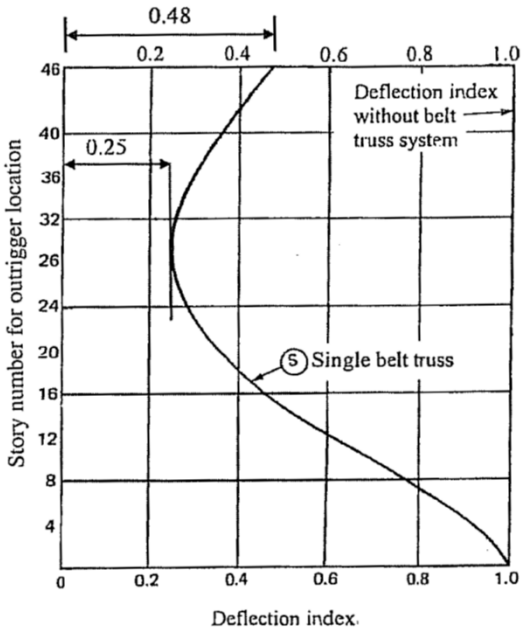
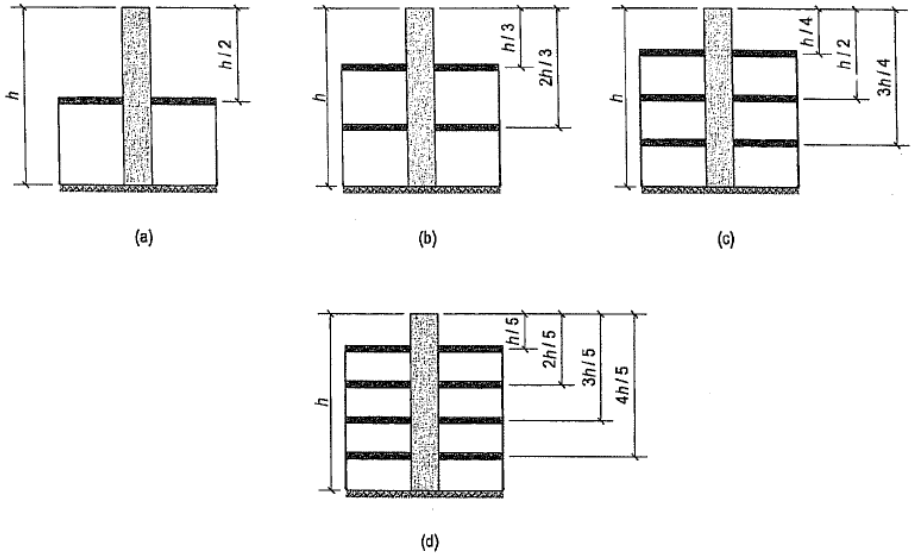
Outriggers and belt trusses at several locations



Deflection index vs. level of the outrigger



Optimum location of belt and outrigger trusses: a) one outrigger; b) two outriggers; c) three outriggers; d) for outriggers

Note:

Deflection index = $\frac{\text{Top displacement with/without outriggers}}{\text{Top displacement with outriggers}}$

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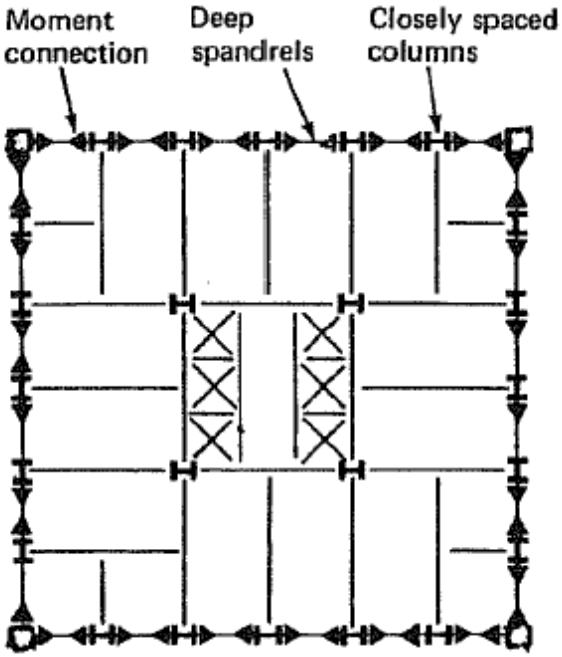
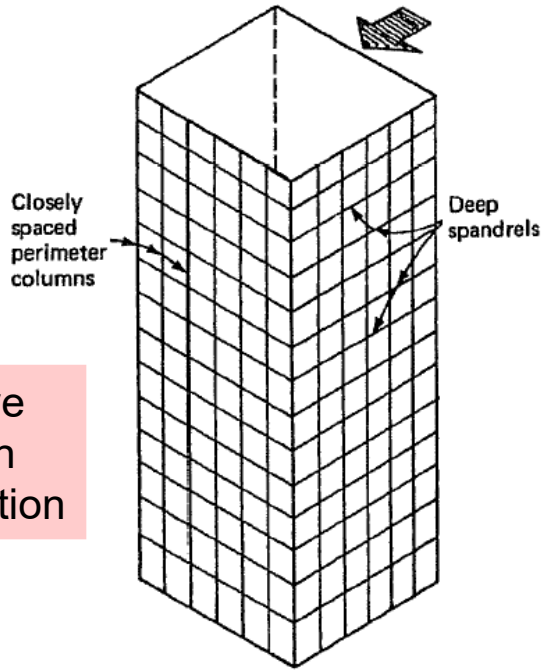
Tube effect

- Structures with closely spaced columns and deep spandrels (tube effect)



Schematic plan of framed tube

Columns have major axis on perimeter direction



Isometric view of framed tube

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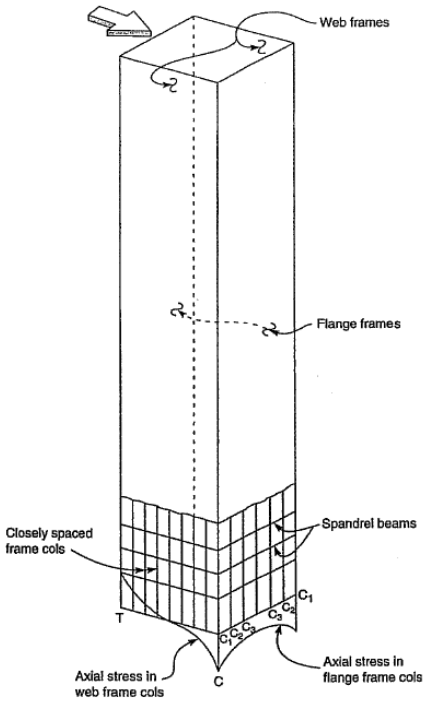
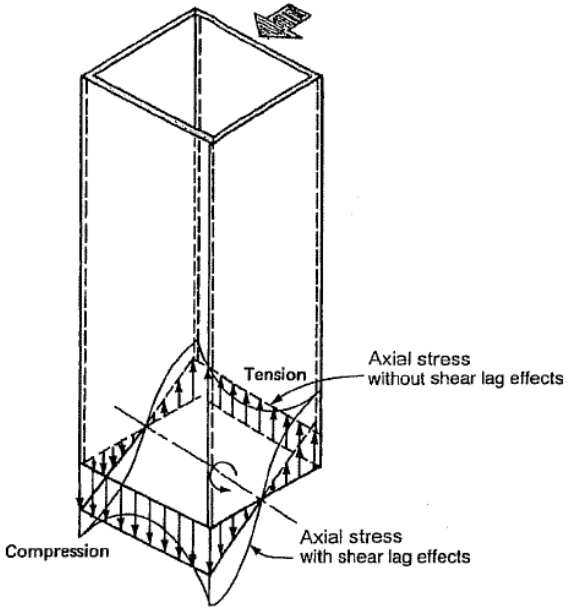
Shear lag effect



Bending effect and “shear lag” in case of a tube with free transversal displacement

Important: distribution of axial stresses in the square tube with/without “shear lag” effect



Bending effect and “shear lag” in case of a tube with closely spaced columns

“Shear lag” effect in the tube. Important: distribution of axial stresses is different comparing to classical bending theory

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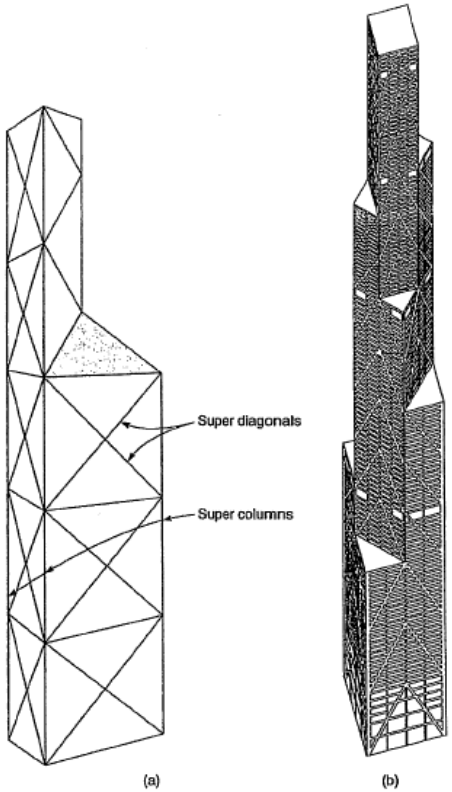
Tube building with diagonals

- Structures with closely columns (tube) and perimetral bracings



a) Tube building with diagonals on multiple stories; b) Building with rotated tubes and super diagonals

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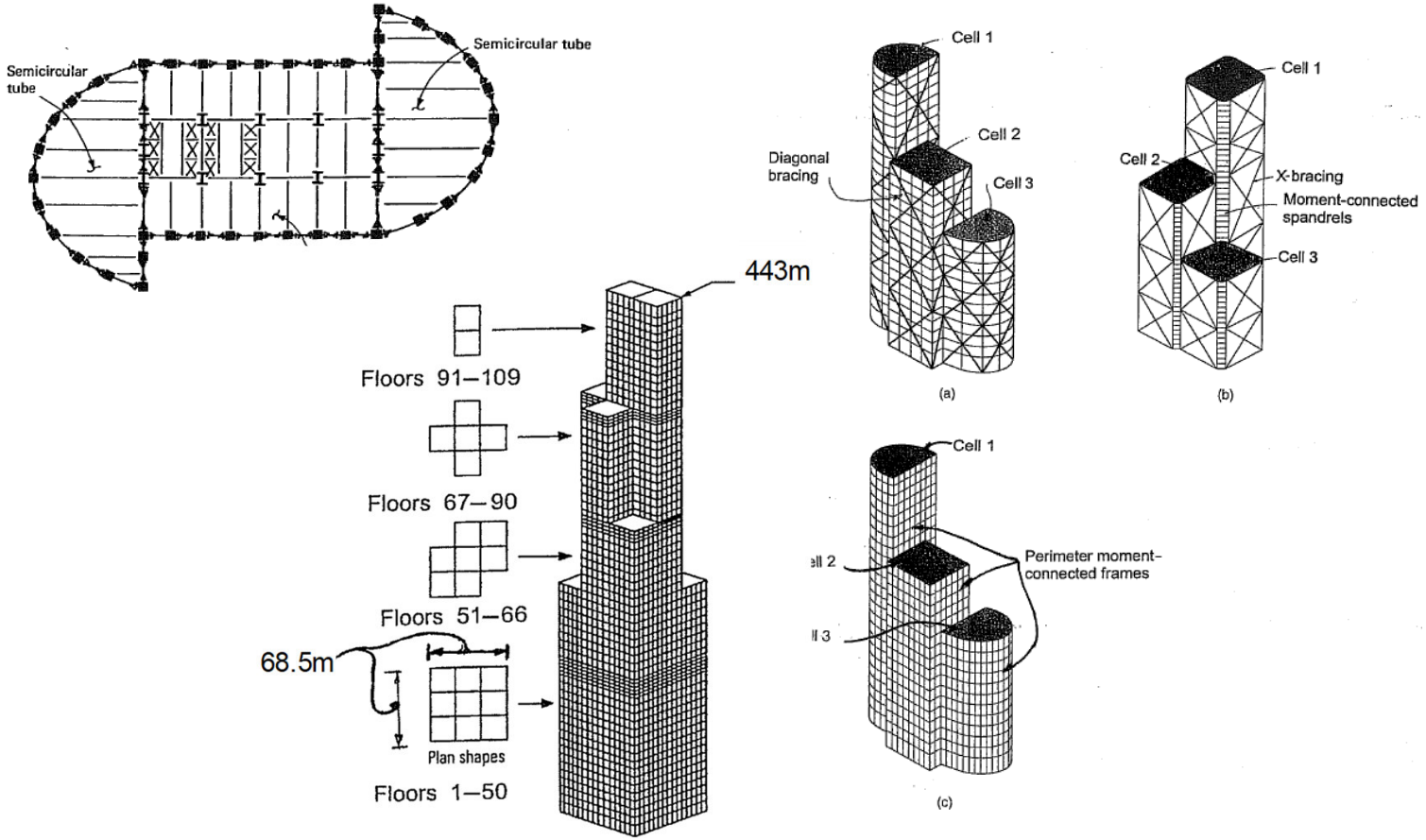
Multiple tubes

- Structures with multiple tubes



Concept of a structure with multiple tubes: a) perimetral diagonal bracings; b) X bracings and moment connected spandrels; c) perimeter moment connected frames

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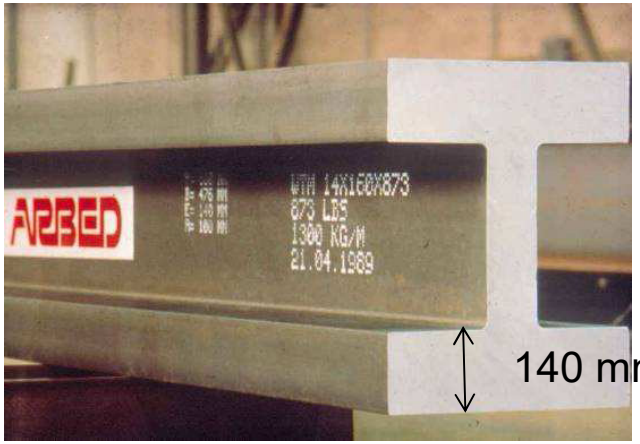
Beams and column elements

- Hot rolled profiles, welded sections, with/without composite action (concrete)

Jumbo sections for High-rise buildings



900 high-rise buildings in the U.S.A.

1500 high-rise buildings worldwide

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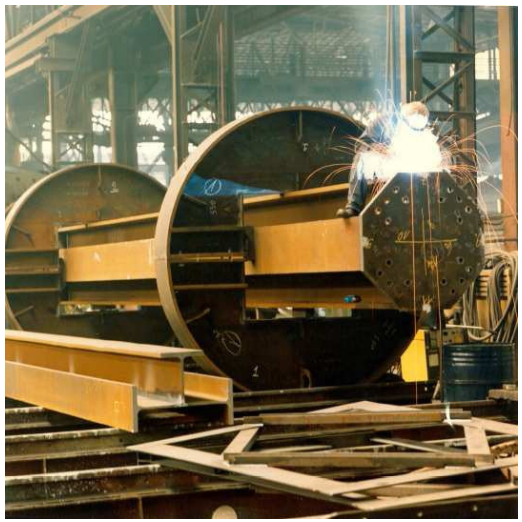
Beams and column elements

Optimized solutions based on rolled profiles



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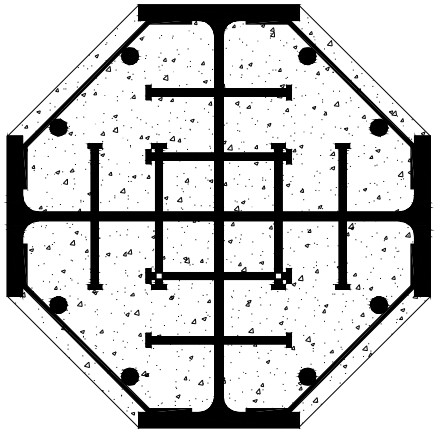
Beams and column elements

Composite action:

- Partially encased columns
- Fully encased columns



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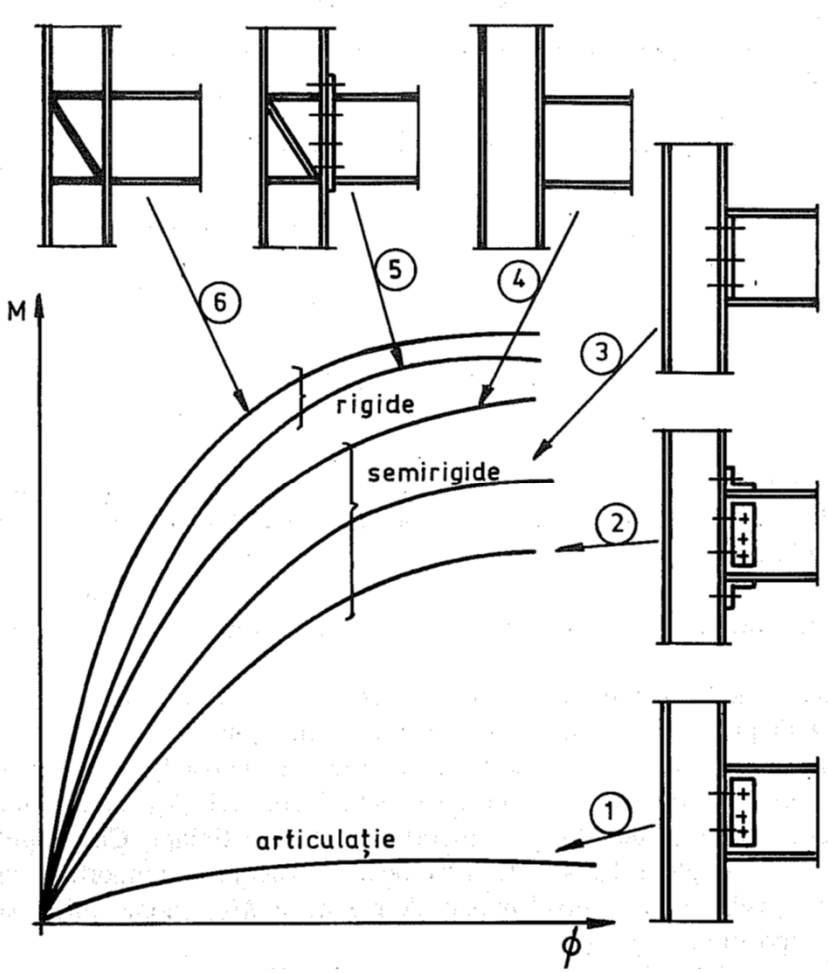
Beam-to-column connections



Moment-rotation curves for different types of joints

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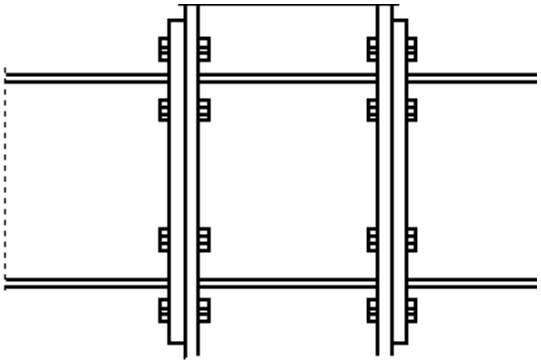
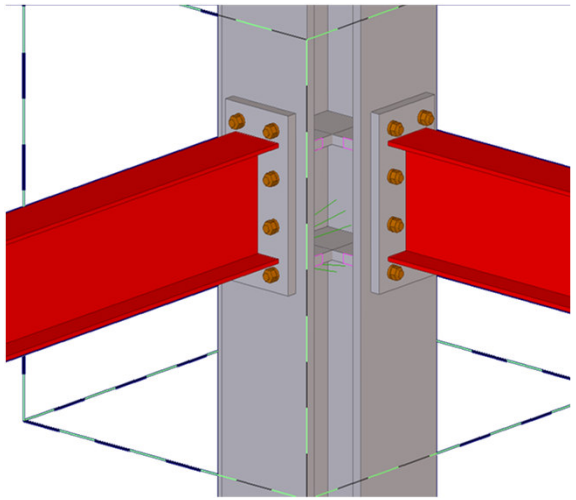
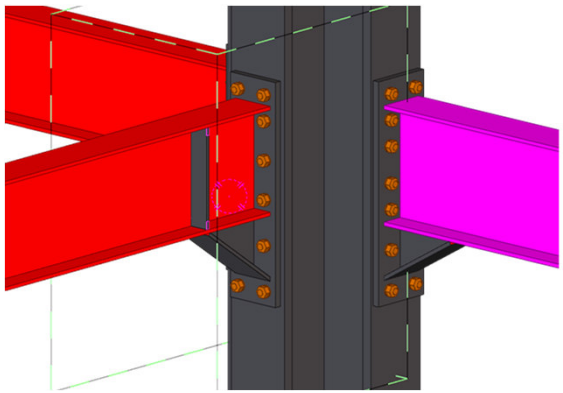
Beam to column joints





Extended end plate bolted joint



Haunched extended end plate bolted joint

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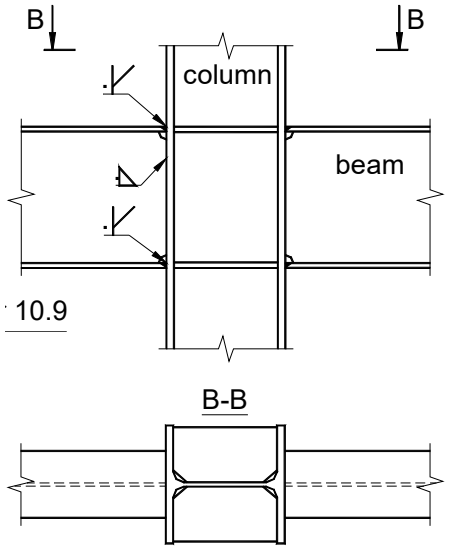
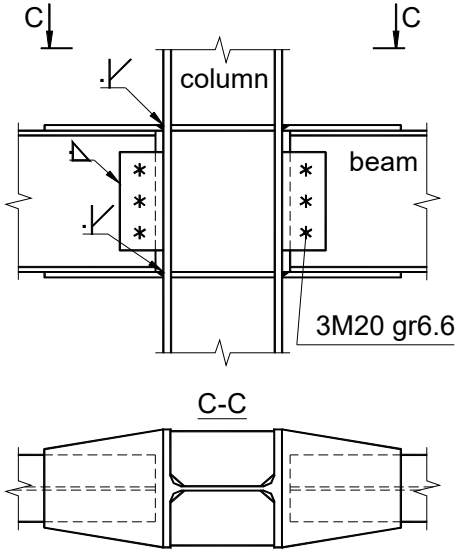
Beam to column joints



Welded joint



Cover plates welded joint

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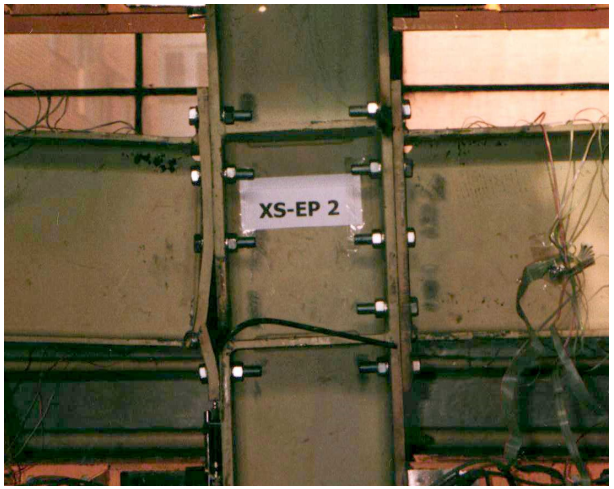
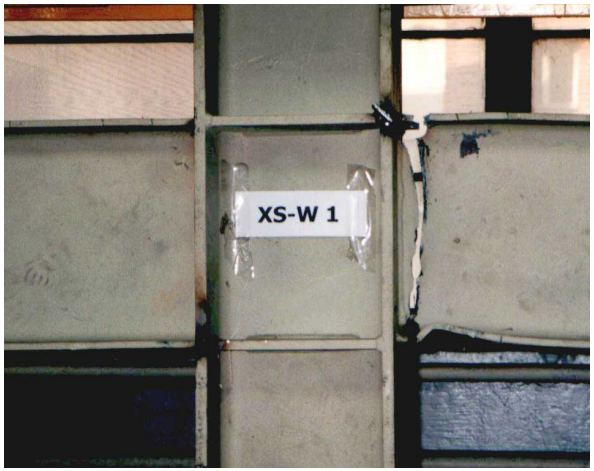
Beam to column joints

Example of beam-to-column joints – failure modes





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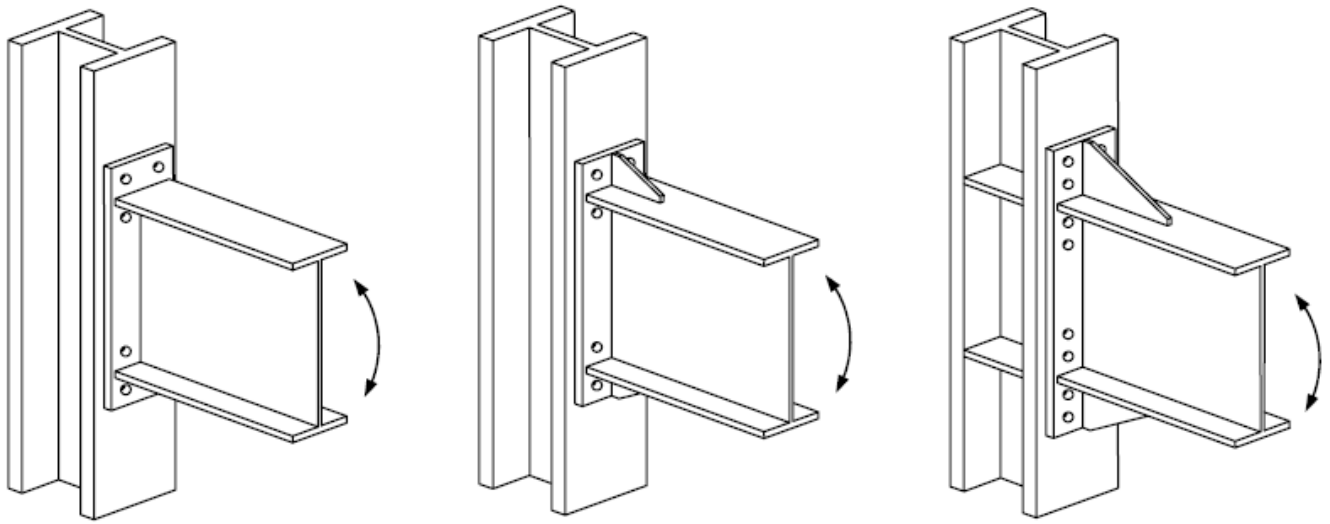
Prequalification of beam-to-column joints

- Strength and stiffness properties of beam to column joints can be evaluated with EN 1993-1-8 recommendations
- Plastic rotation capacity is a key issue in the seismic performance of the structure – no provisions to evaluate this characteristic
  - Provision of qualifying test
  - Use of a prequalified connection



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PARAMETRIC LIMITATIONS ON PREQUALIFICATION

Parameter	Four Bolt Unstiffened (4E)		Four Bolt Stiffened (4ES)		Eight Bolt Stiffened (8ES)	
	Maximum, in. (mm)	Minimum in. (mm)	Maximum in. (mm)	Minimum in. (mm)	Maximum in.(mm)	Minimum in. (mm)
t_p	2-1/4 (57)	1/2 (13)	1-1/2 (38)	1/2 (13)	2-1/2 (64)	3/4 (19)
b_p	10-3/4 (273)	7 (178)	10-3/4 (273)	10-3/4 (273)	15 (381)	9 (229)
g	6 (152)	4 (102)	6 (152)	3-1/4 (83)	6 (152)	5 (127)
p_f	4-1/2 (114)	1-1/2 (38)	5-1/2 (140)	1-3/4 (44)	2 (51)	1-3/4 (44)
p_b	-	-	-	-	3-3/4 (95)	3-1/2 (89)
d	55 (1400)	25 (635)	24 (610)	13-3/4 (349)	36 (914)	18-1/2 (470)
t_{bf}	3/4 (19)	3/8 (10)	3/4 (19)	3/8 (10)	1 (25)	5/8 (16)
b_{bf}	9-1/4(235)	6 (152)	9 (229)	6 (152)	12-1/4 (311)	7-3/4 (197)

where:

t_p = thickness of the end-plate, in. (mm)
b_p = width of the end-plate, in. (mm)
g = horizontal distance between bolts, in. (mm)
p_f = vertical distance between beam flange and the nearest row of bolts, in. (mm)
p_b = distance between the inner and outer row of bolts in an eight bolt connection, in. (mm)
d = depth of the connecting beam, in. (mm)
t_{bf} = thickness of beam flange, in. (mm)
b_{bf} = width of beam flange, in. (mm)

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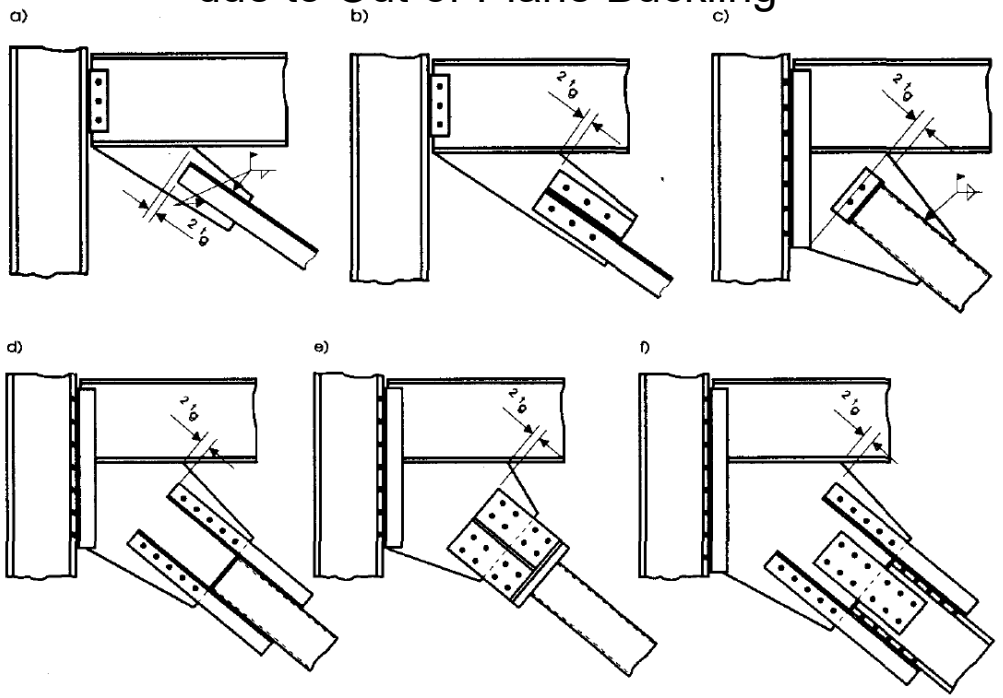
Connections for concentrically braced frames

- Brace connections must be designed either to allow brace end rotation if a pinned end condition is assumed or to permit a plastic hinge to develop at the brace ends if brace fixity is anticipated.

Connection Detail Allowing Brace End rotation due to Out-of-Plane Buckling



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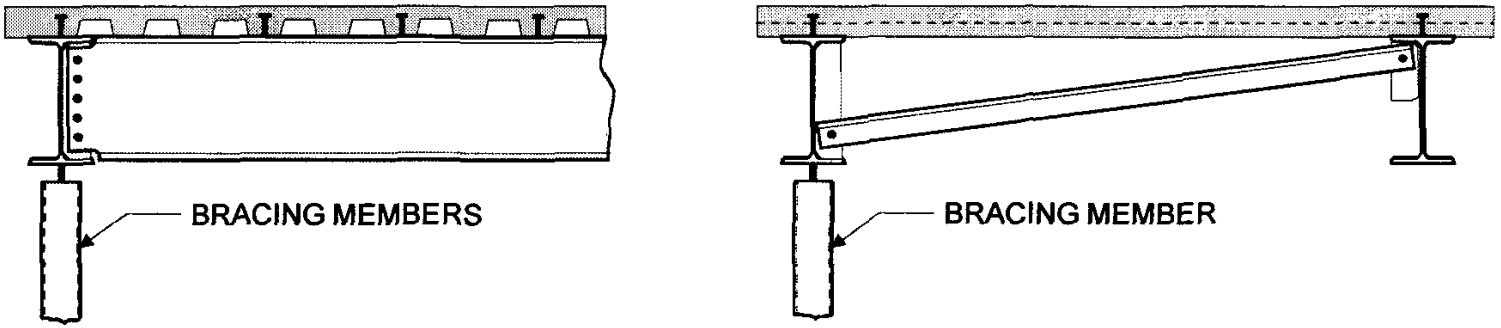
Connections for concentrically braced frames

- When V- and inverted V-Bracing are used, out-of-plane stability of the beams when the bracing members buckle in compression is critical to achieve a stable response
- Code provisions require that lateral bracing be provided at both beam flanges at the point of intersection of the bracing members, as shown in figure



Lateral Bracing of Beams in V- and inverted V-Bracing

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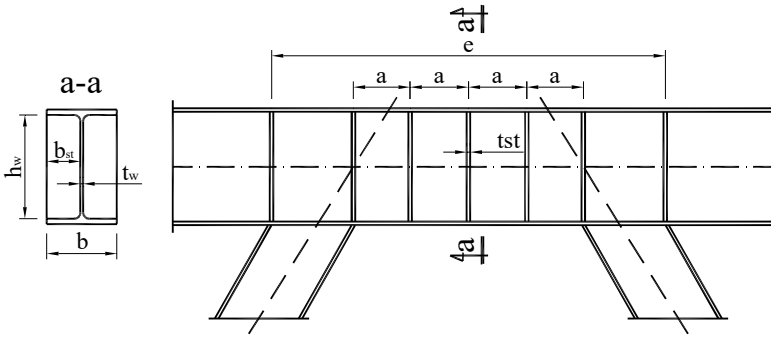
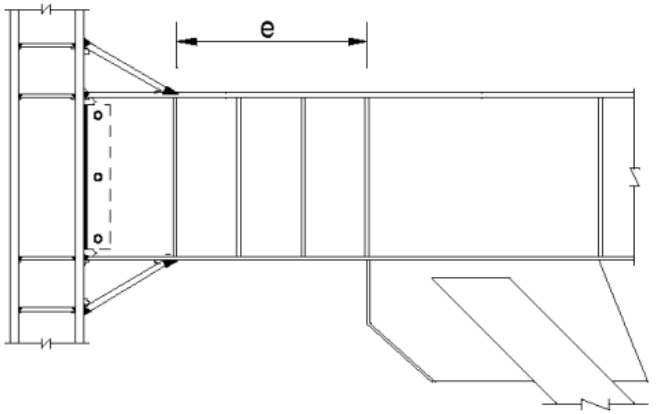
Connections, detailings for eccentrically braced frames

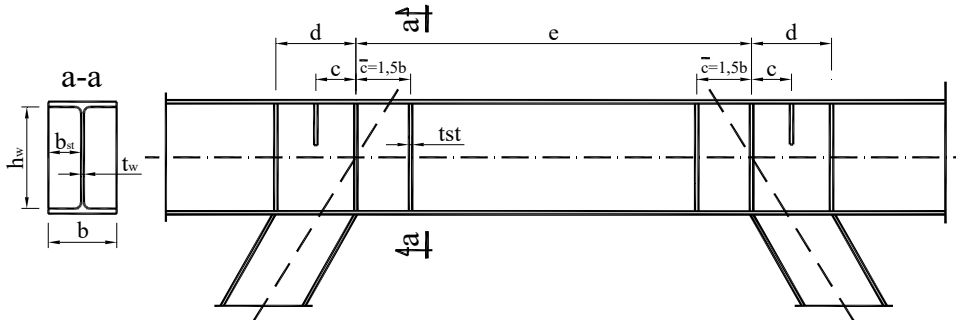
Short link



Example of link-to-column connections.



Long link



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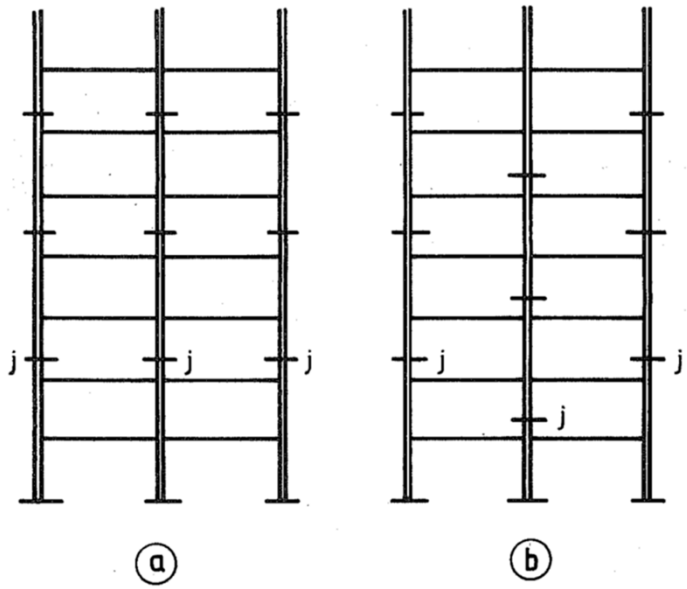
Column splices

- Columns are delivered at the construction site in sections
- Sections are jointed on site by means of welded or bolted connections
- Splice connections can be at same level (a) or staggered (b)



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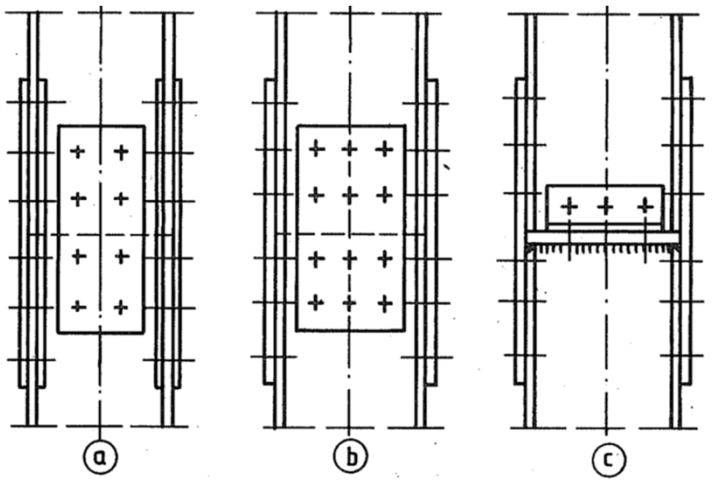
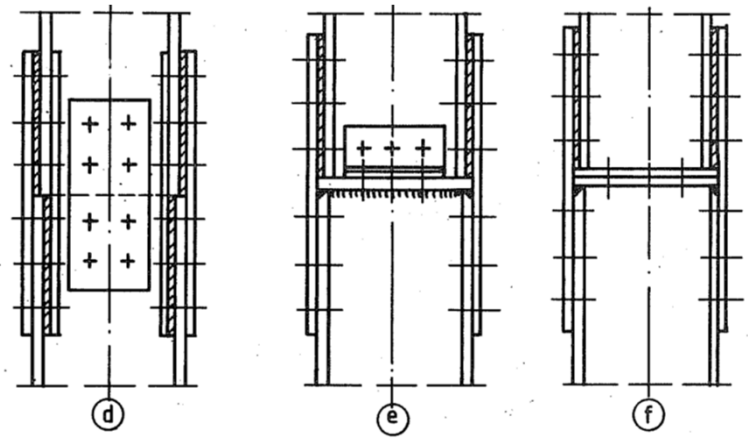
Splice bolted connections



Columns sections are identical

a) double cleat bolted connection
b) single cleat bolted connection
c) single cleat and cap plate bolted connection



Columns sections are different

d) double cleat bolted connection
e), f) single cleat and cap plate bolted connection

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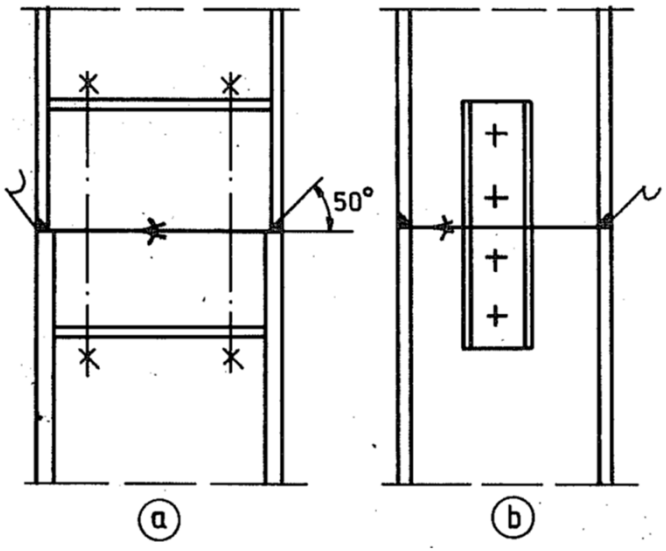
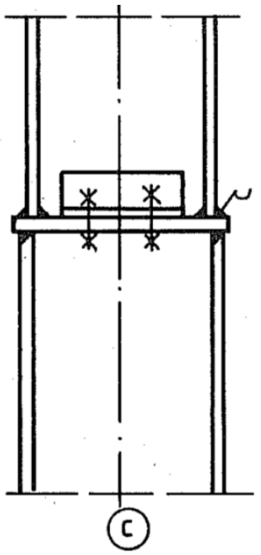
Splice welded connections

Columns sections are identical

a) Butt weld connection

b) Butt weld connection (bolted cleat)





Columns sections are different

c) cap plate welded connection

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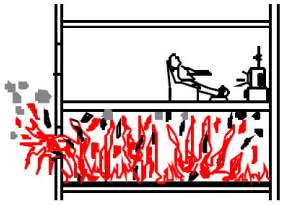
Floor systems

- Functions of the floor slabs
  - Resistance (supporting vertical loads)

  - Protective functions:
    - Heat (temperature difference, thermal insulation)
    - Noise (airborne noise, impact noise reduction and absorption)
    - Fire
    - Humidity (tightness, vapor barrier)
    - Handling with horizontal force (seismic zones)



REI 30, REI 60,…

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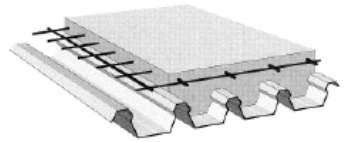
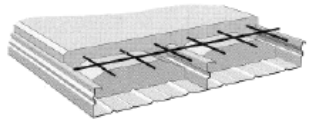
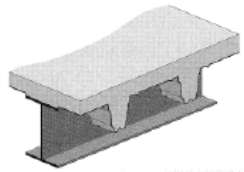
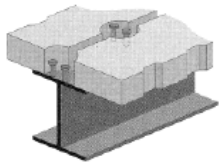
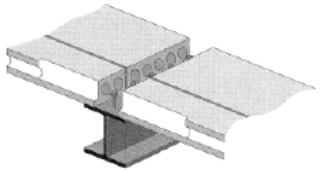
Floor systems - types

Type	Application	Example
Composite slab	10cm≤ d ≤ 30cm 2kN/m^2 ≤ q ≤ 6kN/m^2 for spans **2m ≤ L ≤ 6m**	
Prefabricated composite slabs	Sheeting / Rockwool / Concrete for d = 20cm ⇒ q = 2kN/m^2 for spans **4m ≤ L ≤ 7m**	
Slabs with deep rolled profile sheetings	h = ca. 200mm, d = ca. 25cm for spans **L ≤ 6m**	
Concrete slab	on-site or partially prefabricated d = ca. 20cm ⇒ q = 5kN/m^2 for spans **L ≤ 6m**	
Prestressed hollow core slabs	16cm≤ d ≤ 40cm (1.20m width) 2.5kN/m^2 ≤ q ≤ 6kN/m^2 for spans **8m ≤ L ≤ 14m**	

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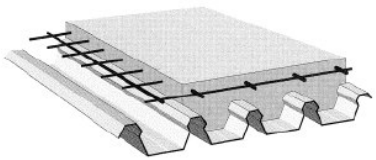
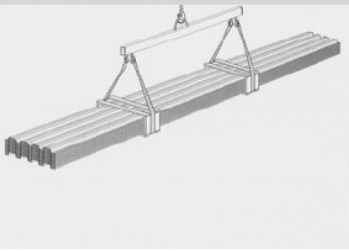
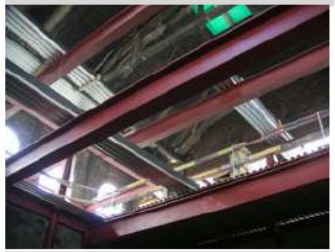
Type	Application	Example
Composite slab	10cm≤ d ≤ 30cm 2kN/m^2 ≤ q ≤ 6kN/m^2 for spans **2m ≤ L ≤ 6m**	

Compact stacking	Optimized transport	Distributed loads	Easy storage of loads
			
400 m2 per 10 m-length package	A lorry can deliver 1000 to 2000 m2 or more	by a simple crane operation	on steel beams

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Composite slab - Concreting

- Several floor levels can be prepared prior to the pouring phase
- Pouring on several storeys can be performed without the need for props
- The intermediate levels below the active level can be freely accessed

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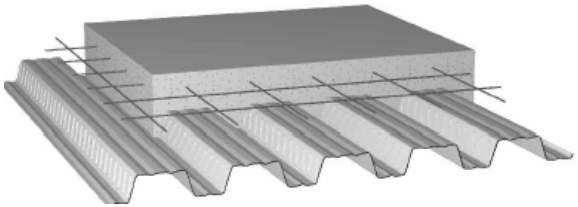
Composite slab - Advantages

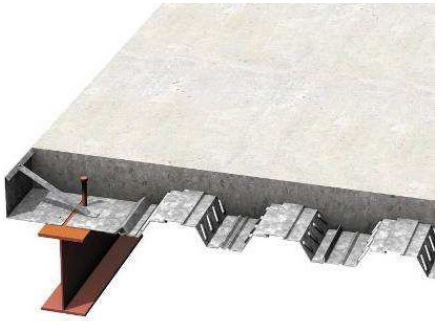


Steel sheet acts as:

- lower rebars as sheet is contributing
- formwork for the concrete
- work platform with immediate effect
- protection of underneath levels
- stabilization element against lateral torsional buckling of the steel beam
- diaphragm directly after connection to the beam

Furthermore:

- leads to lighter structure ⇒ structural savings
- leads to fast installation
- it is placed by hand ⇒ no lifting equipment
- it is easy to implement openings in it
- gives possibility of welding studs
- avoids visible cracks
- avoids surface cracks in case of fire
- is galvanized for durability



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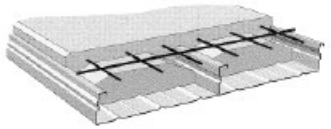
Type	Application	Example
Prefabricated composite slabs	Sheeting / Rockwool / Concrete for d = 20cm ⇒ q = 2kN/m^2 for spans **4m ≤ L ≤ 7m**	

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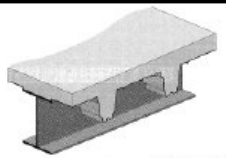
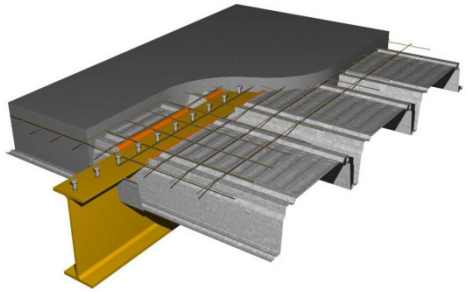
Slabs with deep rolled profile sheetings	h = ca. 200mm, d = ca. 25cm for spans **L ≤ 6m**	



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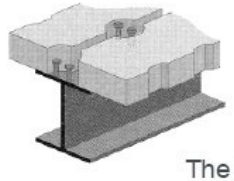
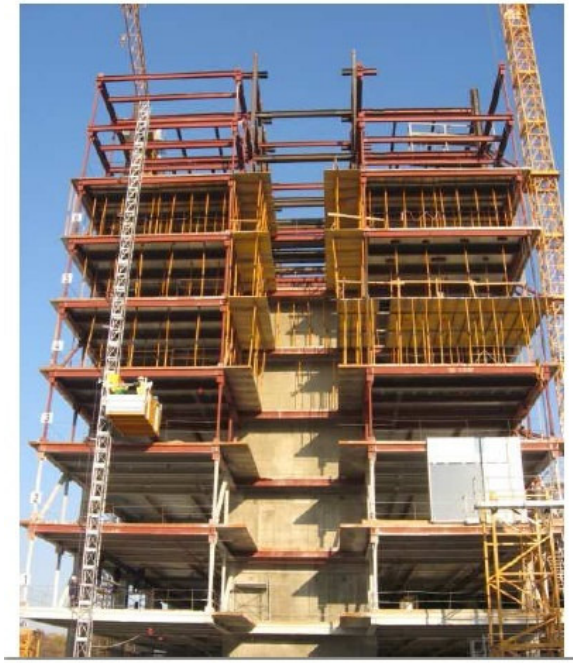
Type	Application	Example
Concrete slab	on-site or partially prefabricated d = ca. 20cm ⇒ q = 5kN/m^2 for spans **L ≤ 6m**	 The Gate – allpic (RO)



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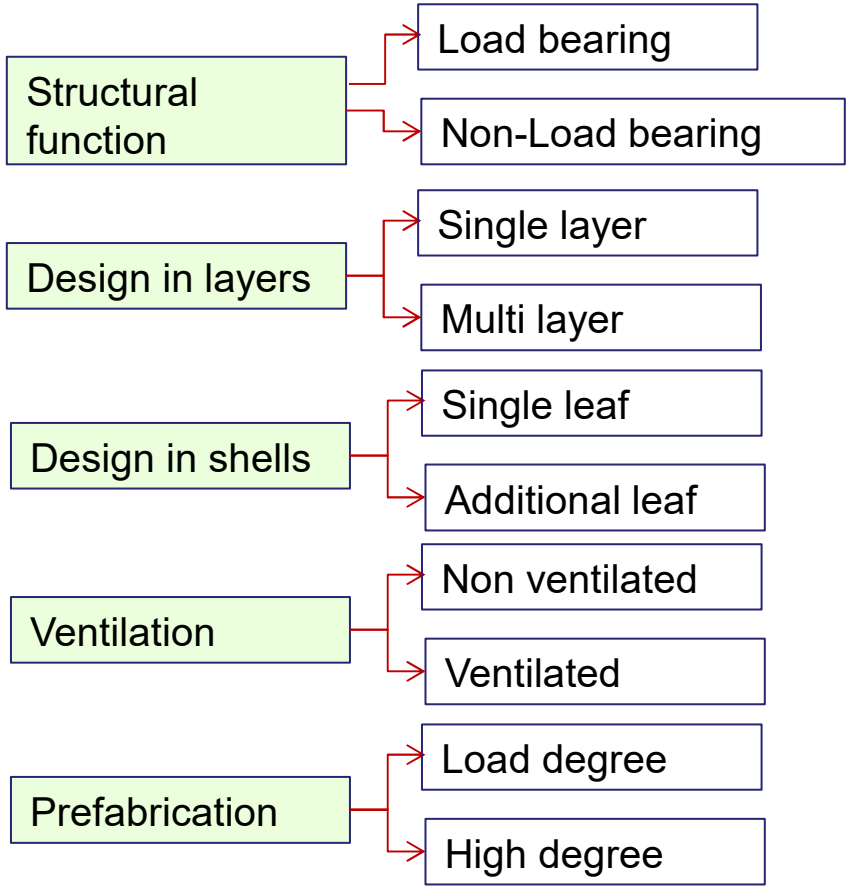
Façade systems

Classification according to structural criteria



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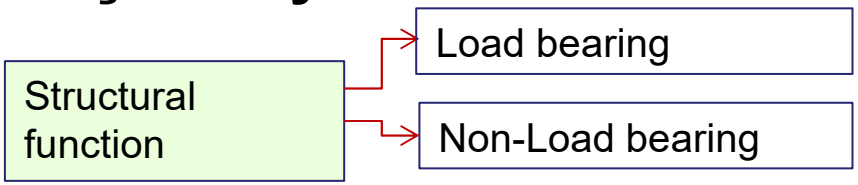
Façade systems



Structural function:

-Non-bearing façades do not contribute to the bearing resistance,

-Loads are:

• vertical (self-weight, life-loads, snow, plants, racks…)

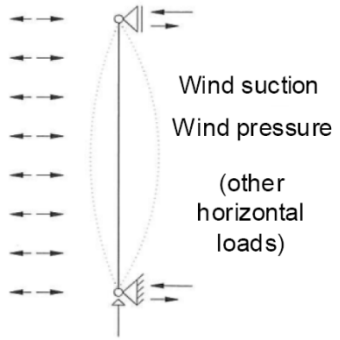
• horizontal (wind, life-loads as impacts, …)

• reactive forces (thermal or hybrid)

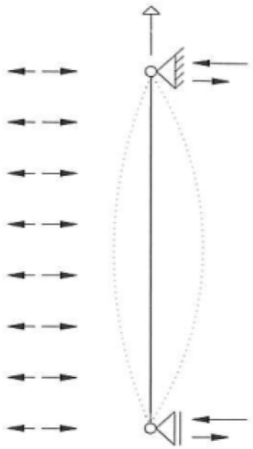
⇒ one fixed and one sliding support,

-Standing or hanging (no stability problem, fixed after erection) are options in design.

Standing facade



Hanging facade



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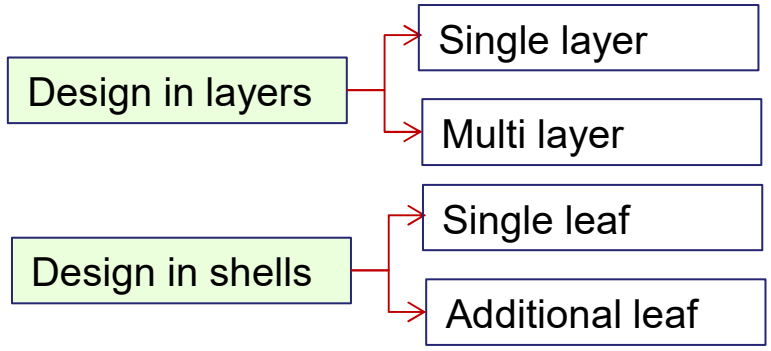

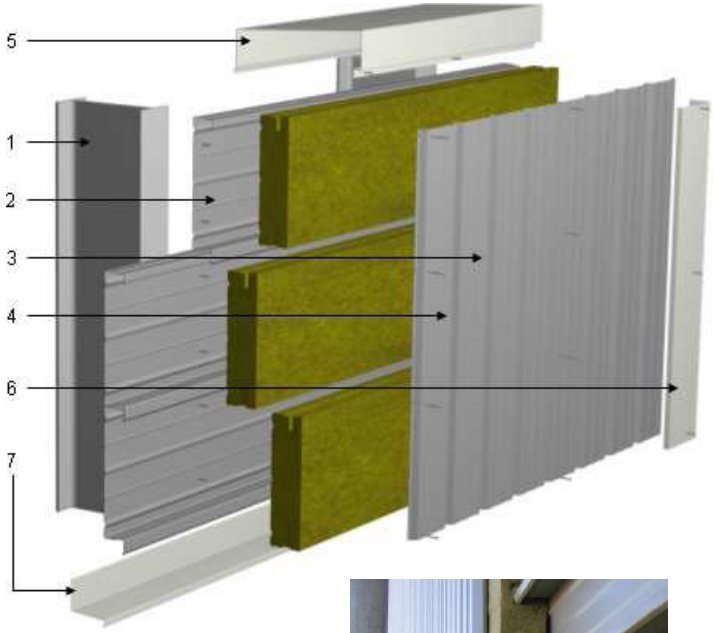

Layers / Shells:

-Non-bearing are defined as layers,
-Bearing are defined as shells,
-Layers/Shells with varying material, thickness and structure may be combined to comply with functional subtasks.

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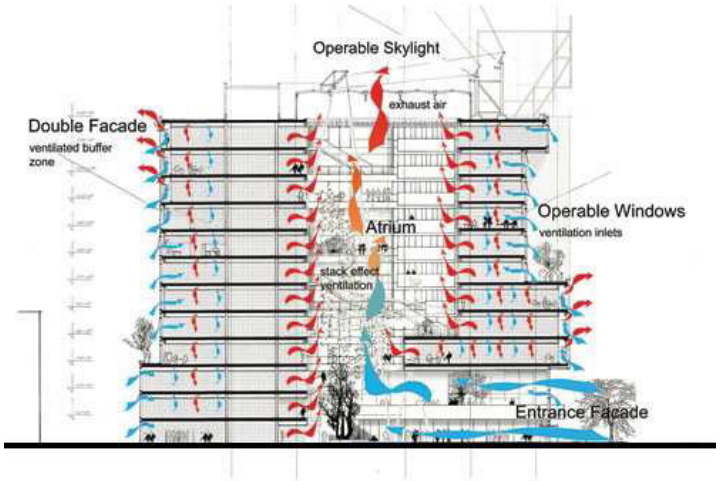
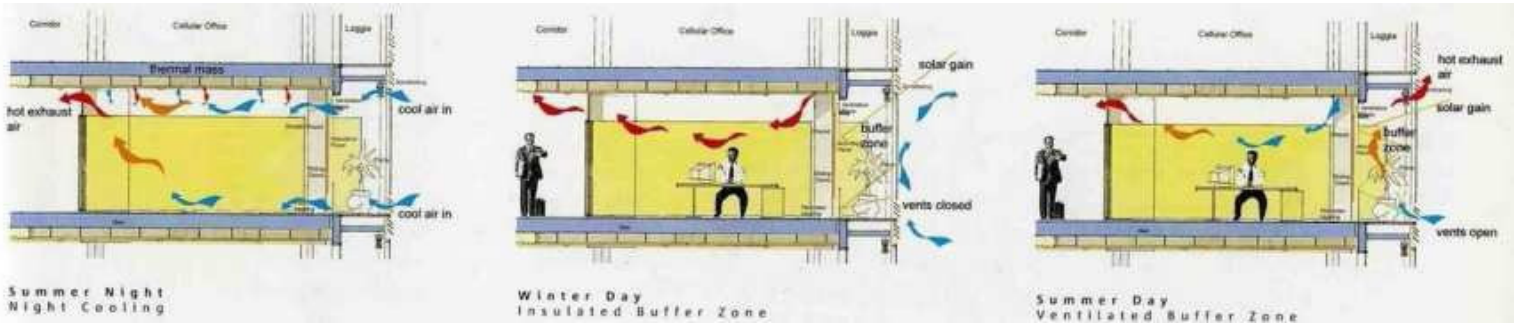
Façade systems

Ventilation → Non ventilated
Ventilation → Ventilated

Ventilation:

-Ventilated façades incorporate one or multiple air layers to evacuate heat and condensate via thermal uplift.





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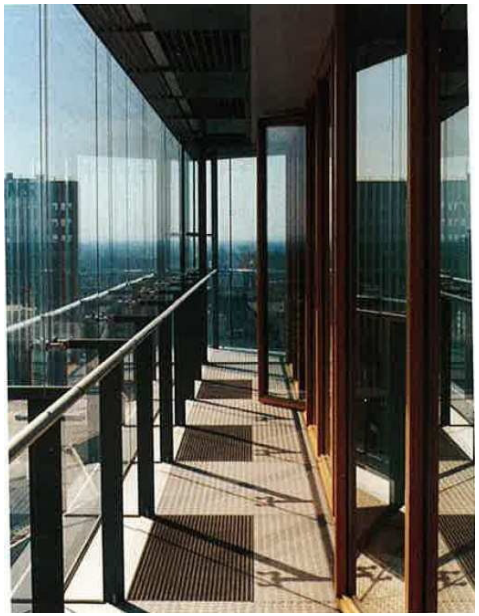
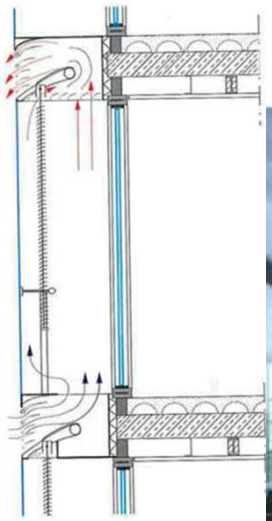
Façade systems

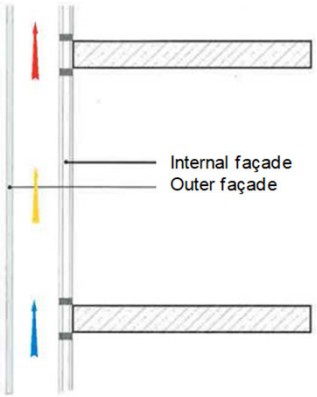
Conception – Ventilation – Double-skin façades

Corridor Façades
most effective in ventilation but disadvantages in room-to-room noise insulation





Multi-storey Façades
for glass façades without openings or if mechnical ventilation is used anyway for cooling



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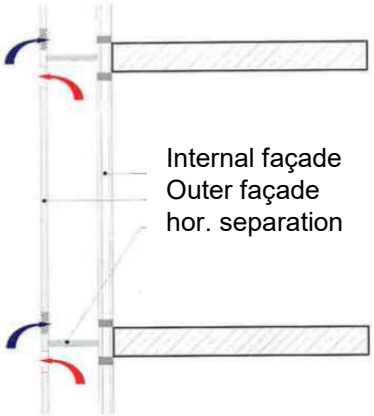
Façade systems

Conception – Ventilation – Double-skin façades

Box-type Window
for pier façades and high requirements on privacy/confidentially between rooms

Shaft-box Façade
suitable for high noise insulation requirements due to small openings



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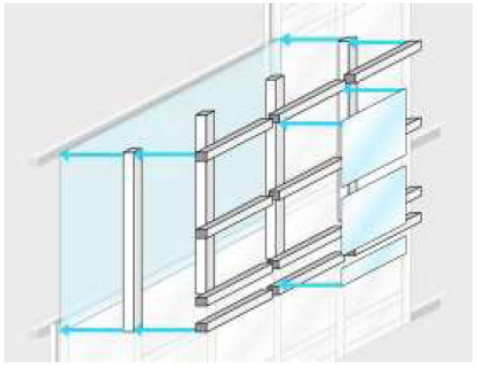
Façade systems

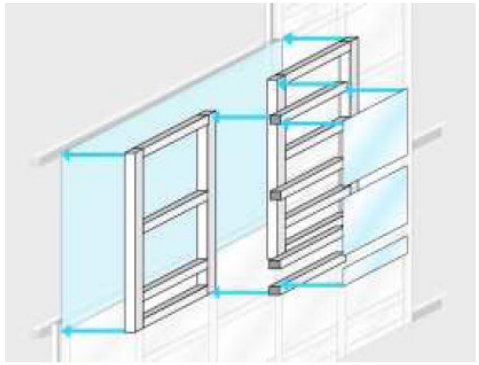
Prefabrication
- Load degree
- High degree

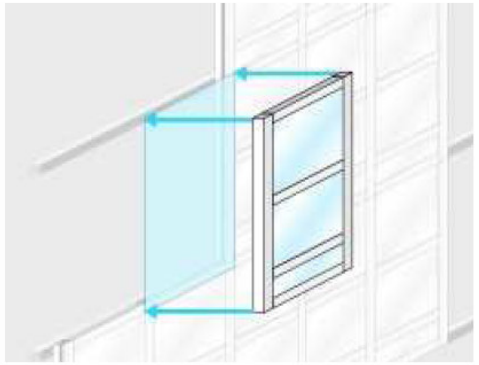
Conception – Prefabrication and Installation principles

Post-Beam-Façade

Element-Façade







Single-Beam-Mounting

Ladder-Mounting

Element-Mounting

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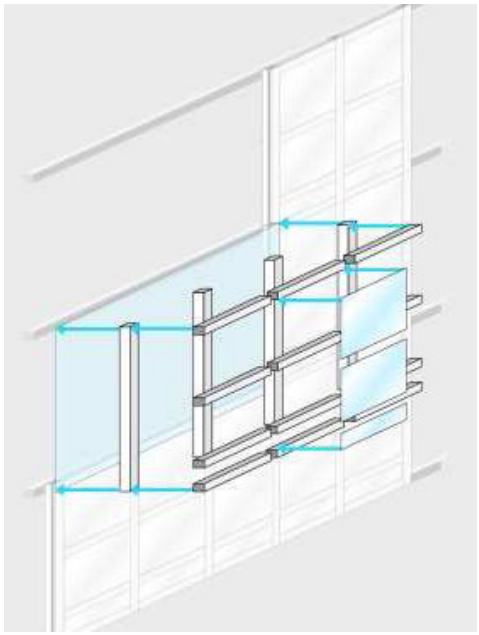
Principle – Single-Beam-Mounting:

-Delivery and mounting of single components on-site

-Fabrication in conventional way with vertical posts and transversal beams

-Glazing and further additional infills are installed in a separate working step from the external side

⇒ **Low degree of prefabrication**



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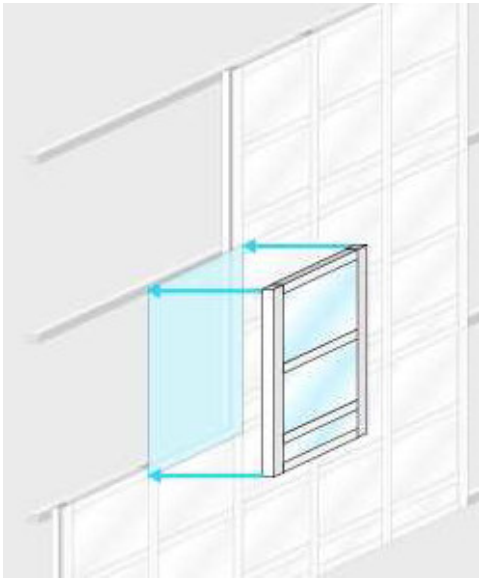
Principle – Element - Mounting:

-Delivery of fully prefabricated elements to site

-Elements are connected and sealed together and to the building

⇒ **High degree of prefabrication**



Element Façade – Prefabrication in shop and transport to site

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Element Façade – Mounting of elements on-site

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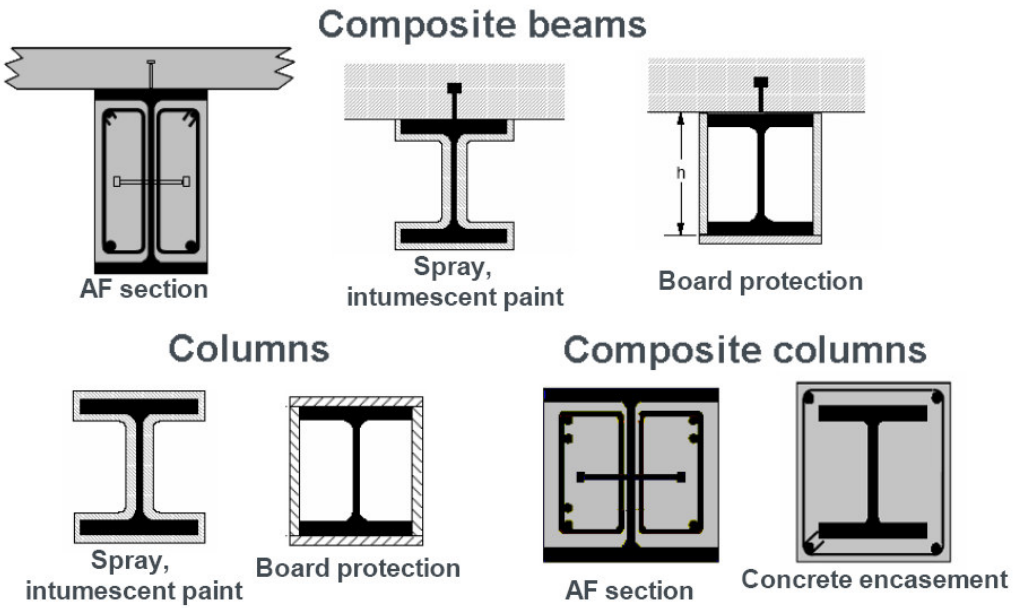
Fire protection systems

- There are numerous fire protection technologies currently available for protecting structural elements during a fire and providing a fire resistance
- Application of insulating materials is one of the most common means of protection: Insulating Board Systems, Mineral Fiber Systems, Spray Applied Fireproofing, Intumescent Paints, Concrete Encasement



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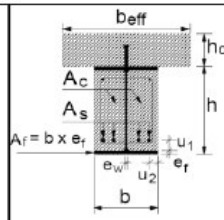
Fire resistance

Composite beams – EN 1994-1-2

Table 4.1:



Condition for application:
slab: $h_c \geq 120$ mm
$b_{eff} \leq 5$ m
steel section: $b / e_w \geq 15$
$e_f / e_w \leq 2$
additional reinforcement area, related to total area between the flanges: $A_s/(A_c + A_s) \leq 5\%$

		Standard Fire Resistance				
		R30	R60	R90	R120	R180
1	Minimum cross-sectional dimensions for load level $\eta_{fi,t} \leq 0,3$					
	min b [mm] and additional reinforcement A_s in relation to the area of flange A_s / A_f					
1.1	$h \geq 0,9 \times$ min b	70/0,0	100/0,0	170/0,0	200/0,0	260/0,0
1.2	$h \geq 1,5 \times$ min b	60/0,0	100/0,0	150/0,0	180/0,0	240/0,0
1.3	$h \geq 2,0 \times$ min b	60/0,0	100/0,0	150/0,0	180/0,0	240/0,0
2	Minimum cross-sectional dimensions for load level $\eta_{fi,t} \leq 0,5$					
	min b [mm] and additional reinforcement A_s in relation to the area of flange A_s / A_f					
2.1	$h \geq 0,9 \times$ min b	80/0,0	170/0,0	250/0,4	270/0,5	-
2.2	$h \geq 1,5 \times$ min b	80/0,0	150/0,0	200/0,2	240/0,3	300/0,5
2.3	$h \geq 2,0 \times$ min b	70/0,0	120/0,0	180/0,2	220/0,3	280/0,3
2.4	$h \geq 3,0 \times$ min b	60/0,0	100/0,0	170/0,2	200/0,3	250/0,3
3	Minimum cross-sectional dimensions for load level $\eta_{fi,t} \leq 0,7$					
	min b [mm] and additional reinforcement A_s in relation to the area of flange A_s / A_f					
3.1	$h \geq 0,9 \times$ min b	80/0,0	270/0,4	300/0,6	-	-
3.2	$h \geq 1,5 \times$ min b	80/0,0	240/0,3	270/0,4	300/0,6	-
3.3	$h \geq 2,0 \times$ min b	70/0,0	190/0,3	210/0,4	270/0,5	320/1,0
3.4	$h \geq 3,0 \times$ min b	70/0,0	170/0,2	190/0,4	270/0,5	300/0,8

Tabular design data EN 1994 -1-2

Minimum cross-sectional dimensions and minimum additional reinforcement for composite beams with partial concrete encasement

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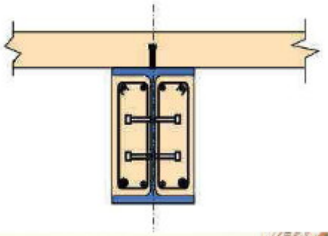
Composite construction – Fire safety

Partially encased Beams



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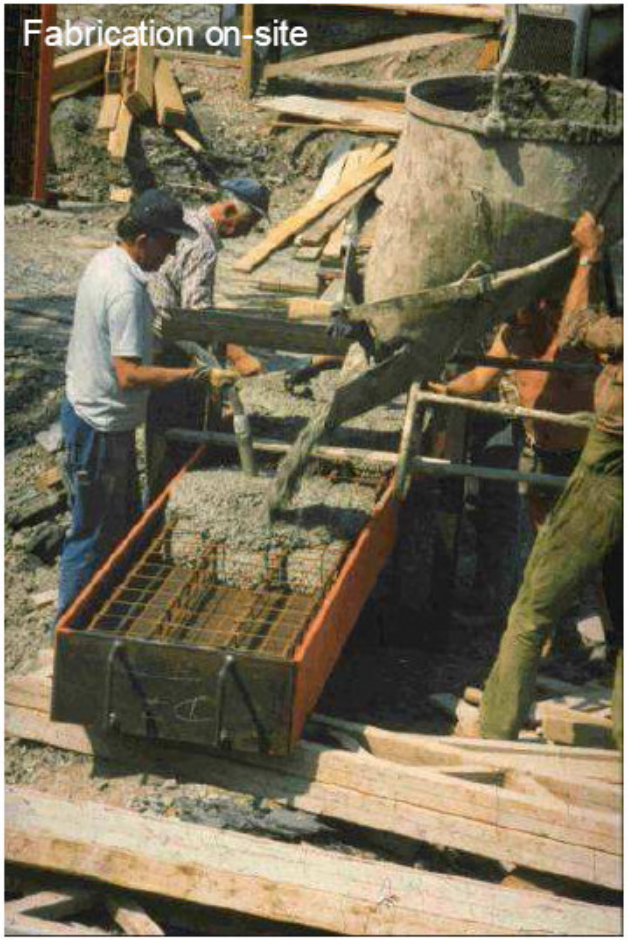
Composite construction

Partially encased Beams



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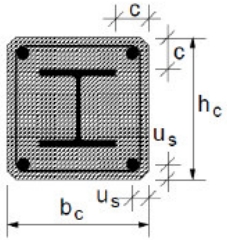
Composite columns – EN 1994-1-2

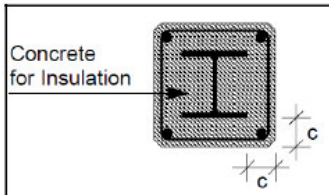
Table 4.4: Minimum cross-sectional dimensions, minimum concrete cover of the steel section and minimum axis distance of the reinforcing bars, of composite columns made of totally encased steel sections.



		Standard Fire Resistance					
		R30	R60	R90	R120	R180	R240
1.1	Minimum dimensions h_c and b_c [mm]	150	180	220	300	350	400
1.2	minimum concrete cover of steel section c [mm]	40	50	50	75	75	75
1.3	minimum axis distance of reinforcing bars u_s [mm]	20*	30	30	40	50	50
	or						
2.1	Minimum dimensio						
2.2	minimum concrete cove						
2.3	minimum axis distance o						

NOTE: *) These values ha

Table 4.5: Minimum concrete cover for a steel section with concrete acting as fire protection



	Standard Fire Resistance				
	R30	R60	R90	R120	R180
Concrete cover c [mm]	0	25	30	40	50

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Composite construction – Fire safety

Partially encased composite columns

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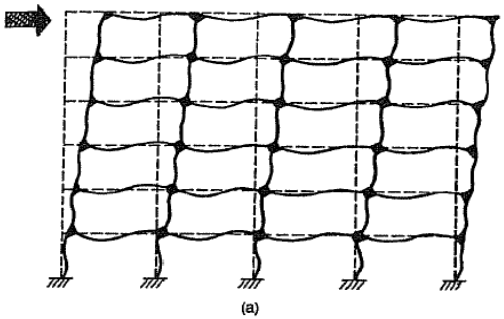
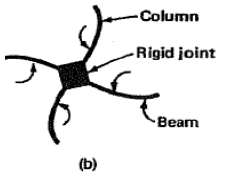
Response of frames to lateral loads

- **Moment frames**
  - Lateral loads are resisted primarily by the rigid frame action
  - Rigid joints
  - SCWB configuration
  - Limiting the lateral drifts





seismic

$\Delta_{adm} = H_e/50 = 0.02H_e$ SLU

$\Delta_{adm} = H_e/125 = 0.008H_e$ SLS

wind

$\Delta_{adm} = H/500 = 0.002H$ SLS

•A frame is considered rigid when its beam-to-column connections have sufficient rigidity to hold virtually unchanged the original angles between intersecting members.
•In this system, shown in figure, lateral loads are resisted primarily by the rigid frame action; that is, by the development of shear forces and bending moments in the frame members and joints. The continuity at both ends of beams also assists in resisting gravity loads more efficiently by reducing positive moments in beam spans. Moment frames have certain advantages in building applications due to their flexibility in architectural planning. They may be placed at the building exterior without restrictions on their depths. They may also be located throughout the interior of the building with certain limitations on beam depths to allow for passage of mechanical and air conditioning ducts. Because there are no bracing elements, they are considered architecturally more versatile than other systems such as braced frames or shear walls.

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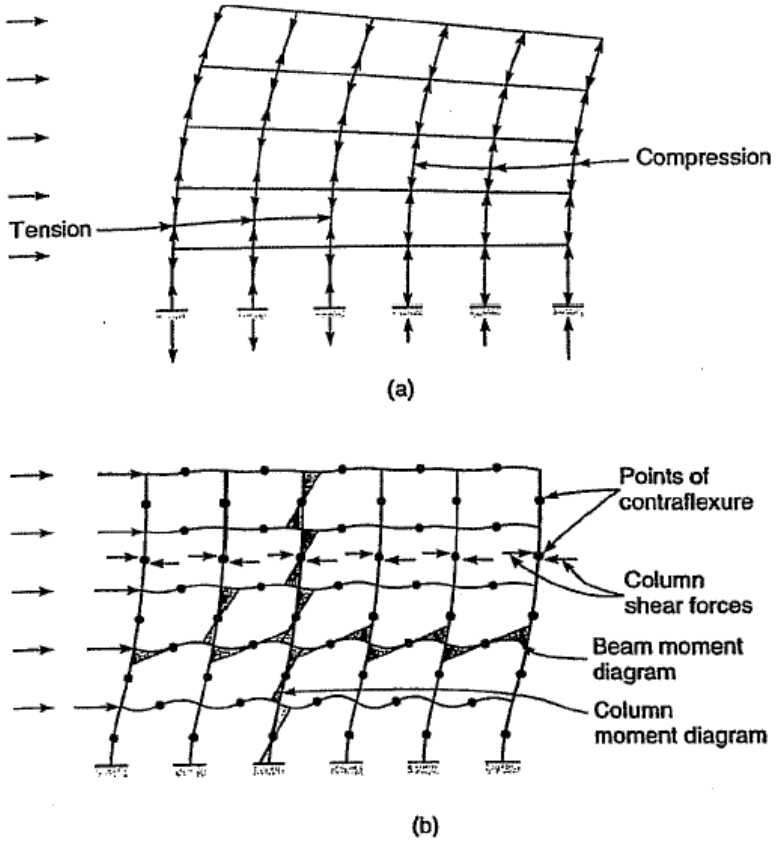
Deflection characteristics



Rigid frame deflections: (a) forces and deformations caused by external overturning moment; (b) forces and deformations caused by external shear.

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Concentrically braced frames

- Rigid frame systems are not efficient for buildings taller than about 20 stories because the shear racking component of deflection due to the bending of columns and beams causes the story drift to be too great.
- Addition of diagonal or V-braces within the frame transforms the system into a vertical truss, virtually eliminating the bending of columns and beams.
- High stiffness is achieved because the horizontal shear is now primarily absorbed by the web members and not by the columns.
- The webs resist lateral forces by developing internal axial actions and relatively small flexural actions. In simple terms, braced frames may be considered cantilevered vertical trusses resisting lateral loads primarily through the axial stiffness of columns and braces. The columns act as the chords in resisting the overturning moment, with tension in the windward column and compression in the leeward column. The diagonals work as the web members resisting the horizontal shear in axial compression or tension, depending on the direction of inclination.

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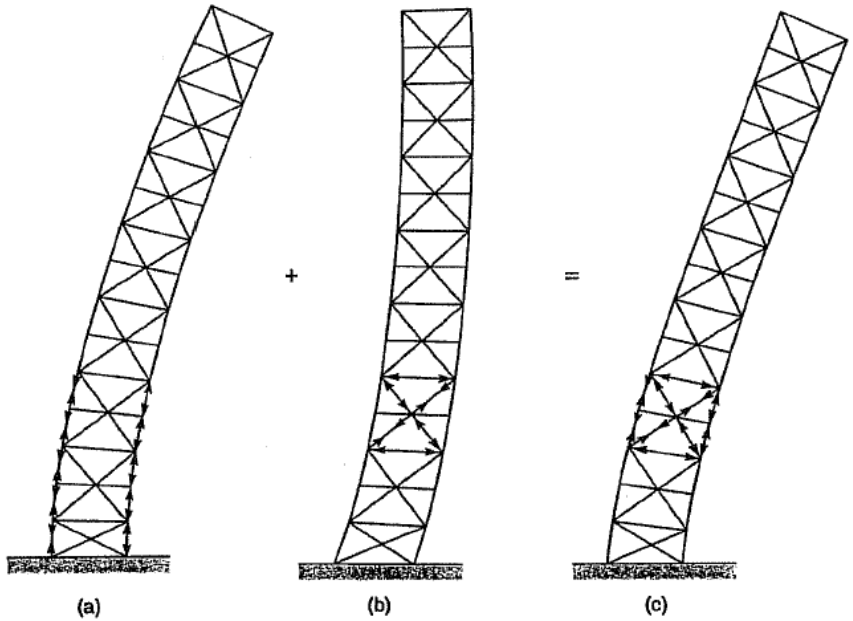
Concentrically braced frames

- Axial deformation of columns – bending effect – a)
- Axial deformation of braces (web effect) – shear effect – b)



Braced frame deformation: (a) **flexural** deformation; (b) **shear** deformation; (c) **combined** configuration.

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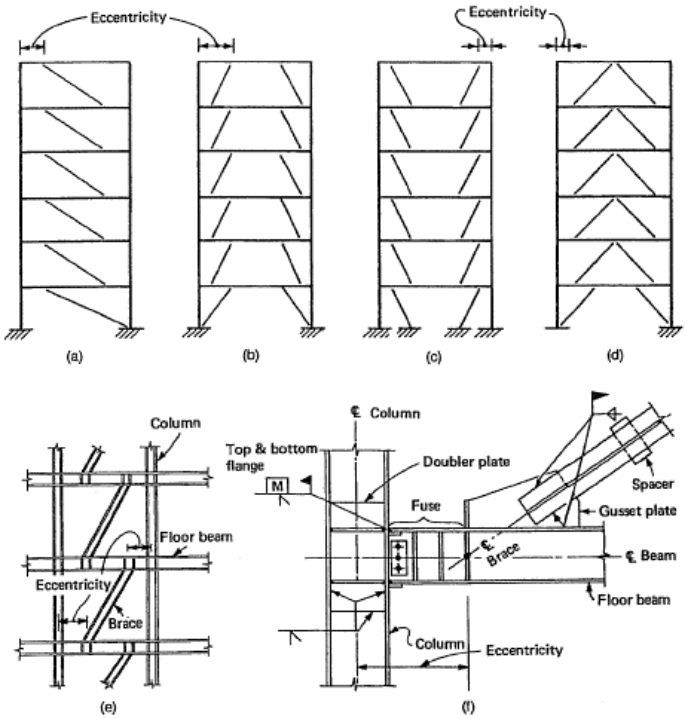
Ecentrically braced frames

- An eccentric bracing system attempts to combine the strength and stiffness of a braced frame with the inelastic behavior and energy dissipation characteristics of a moment frame.
- The system is called eccentric because deliberate eccentricities are inserted between beam- to-column or beam-to-brace connections.
- The eccentric beam element acts as a fuse by limiting large forces from entering into and causing buckling of braces.
- The eccentric segment of the beam, called the link, undergoes flexural or shear yielding prior to formation of plastic hinges in other bending members and well before buckling of any compression members.



Thus, the system maintains stability even under large inelastic deformations. The required stiffness during wind or minor earthquakes is maintained because no plastic hinges are formed under these loads and all behaviour is entirely elastic.

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Wind load vs. seismic load

Wind load

- Strong winds may cause a variety of problems, particularly in tall buildings
- Modern tall buildings are even more prone to wind action, due to their lightweight walls and partitions, which reduce the mass and the damping
- Even for high seismic areas, for buildings with more then 25-30 stories, the wind load governs the design
- Attention should be paid to the following criteria:
  - Strength and stability
  - Fatigue of members and connections
  - Excessive lateral deformations (may cause cracking of claddings or permanent deformations to nonstructural elements)
  - Excessive vibrations that cause discomfort to the occupants

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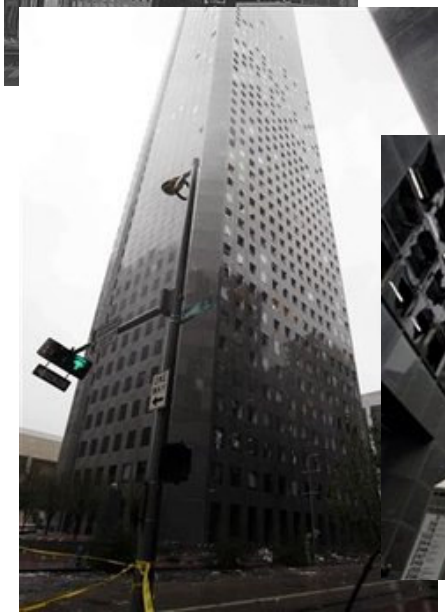
Wind load



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Seismic load

- Many of European areas are under seismic risk
- Southern Europe experienced very damaging earthquakes during the last decades.
- Many existing structures have inadequate protection against strong earthquakes. The vulnerability is very much increasing, due to the rapid grow of the construction industry.
- Seismic loading requires an understanding of the structural behavior under inelastic cyclic deformations
- Behavior under such loading is fundamentally different from wind loading (and gravity loading). It is necessary to pay more attention to type of analysis and detailing requirements, in order to assure acceptable seismic performance beyond the elastic range.
- Some structural damage in members and connections can be expected under design ground motion, as the majority of modern seismic codes allow inelastic energy dissipation in the structural system

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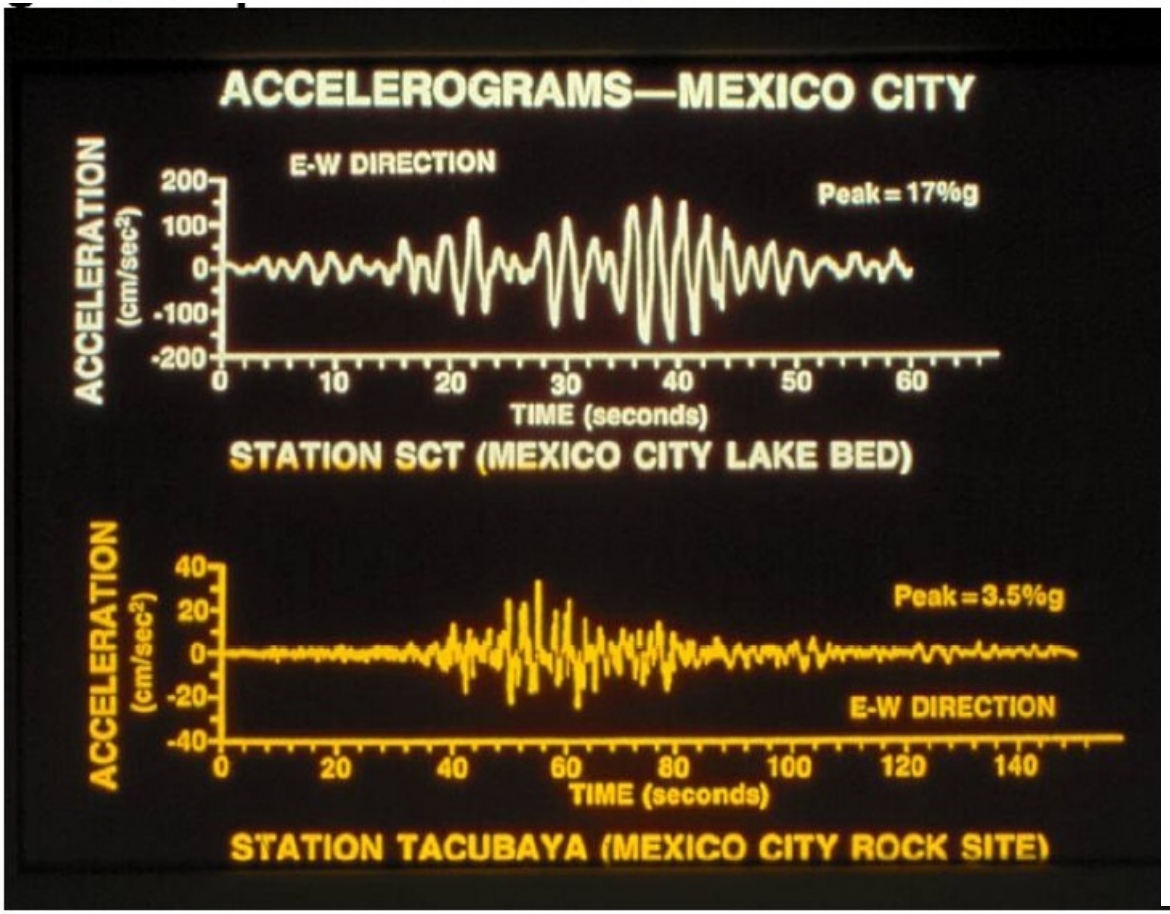
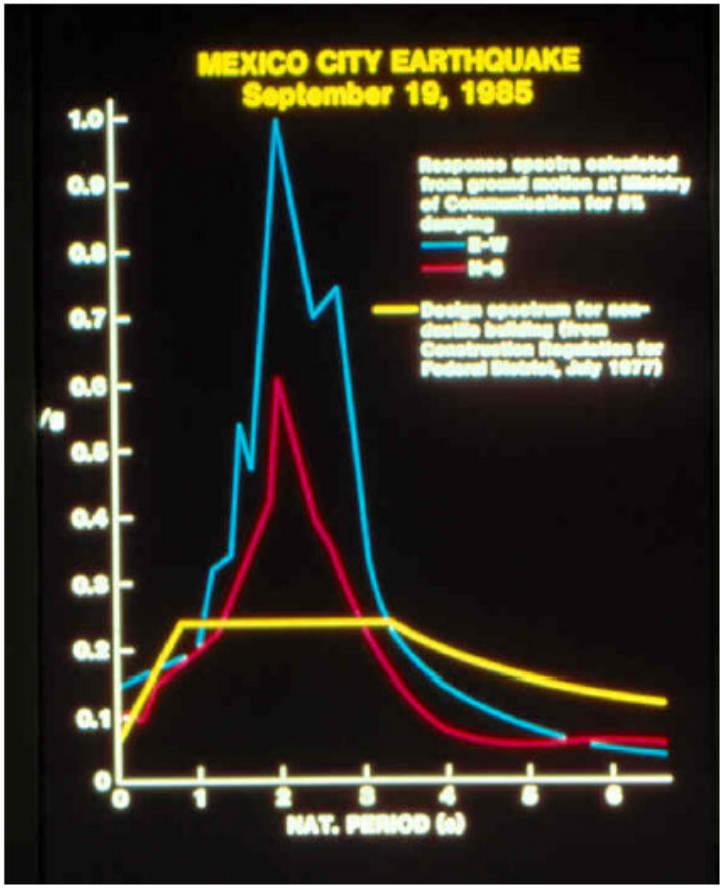
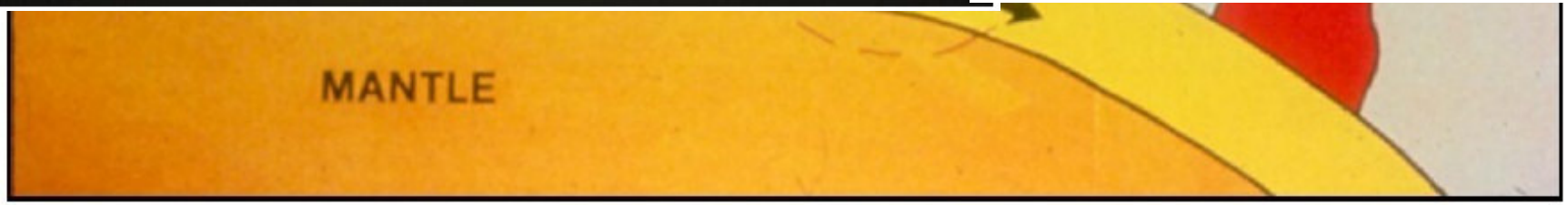
Local effects on site

- Soil may act as a filter
- It can modify frequency content of the ground motion
- Amplification of the ground motion (or reduction) may be recorded on site
- Duration of the ground motion is increased

1985 Mexico City







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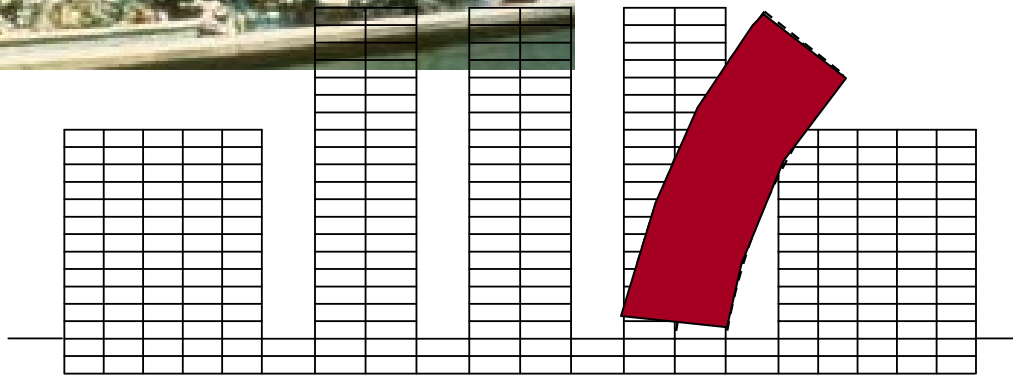
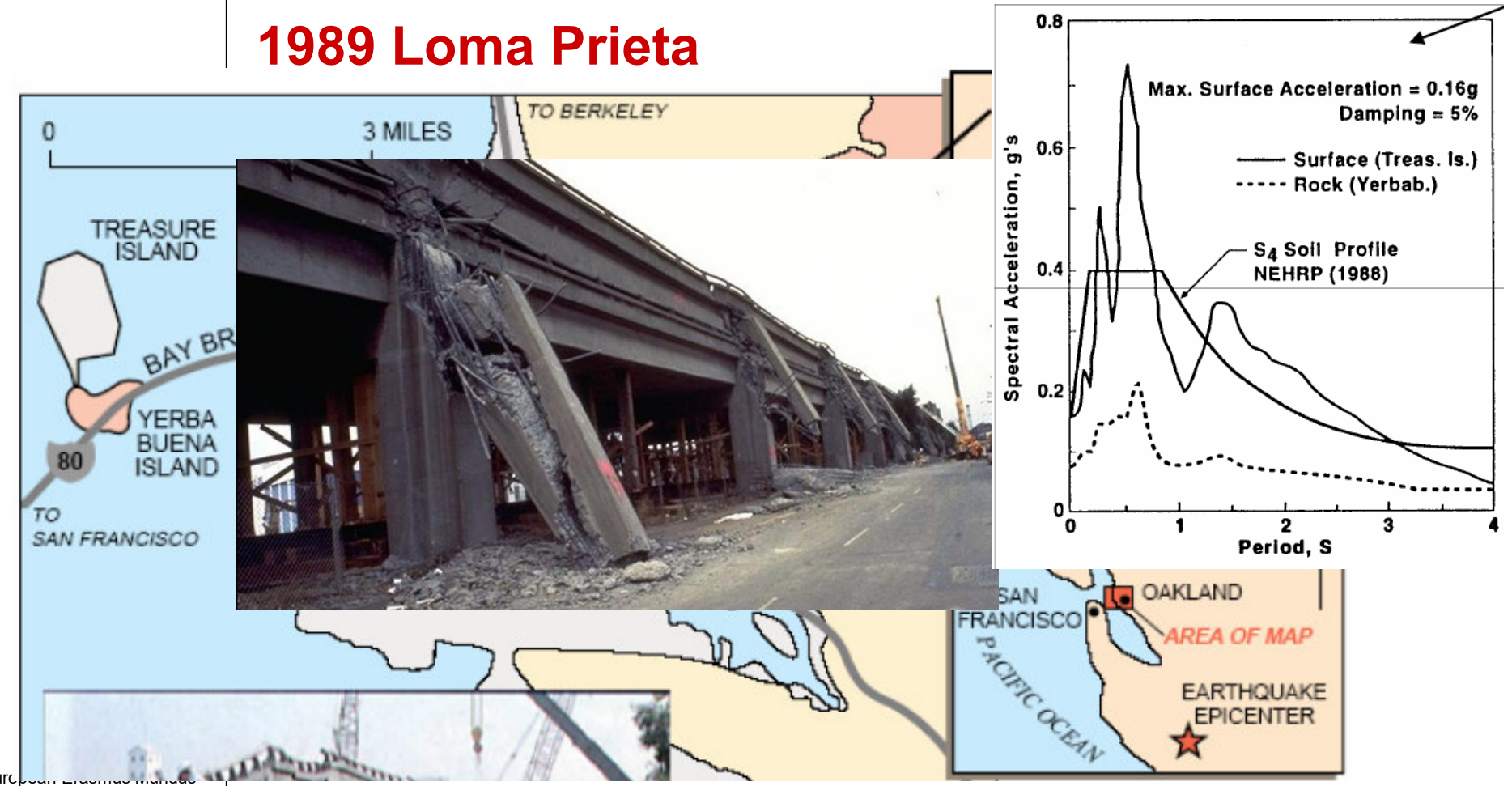
1985 Mexico City - Pino Suarez



1989 Loma Prieta



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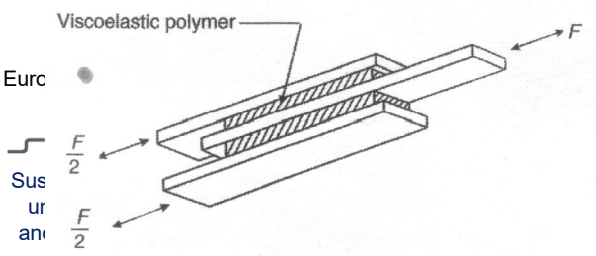
Effect of damping

- Structural response may be reduced by an increase of the damping properties
- Damping of the structural elements is limited
- One option for increasing the damping is the introduction of external damping devices
  - Viscoelastic passive dampers
  - Passive control (tuned mass dampers)
  - Active control (tuned active dampers)
- These systems are effective both against winds and earthquakes

Viscoelastic passive dampers

World Trade Center, 417 m

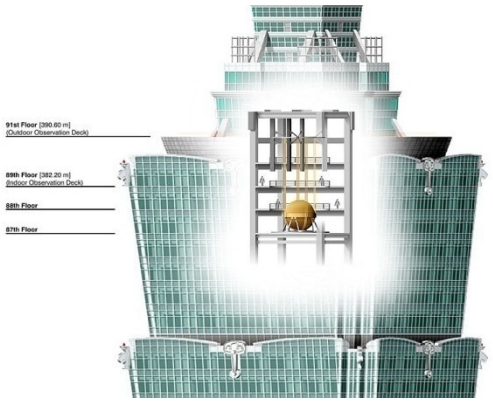
10 000 dampers in the structure, about 100 dampers at the ends of the floor trusses at each floor from the 7th to the 107th



Tuned mass dampers

Burj al Arab, Dubai, 321 m

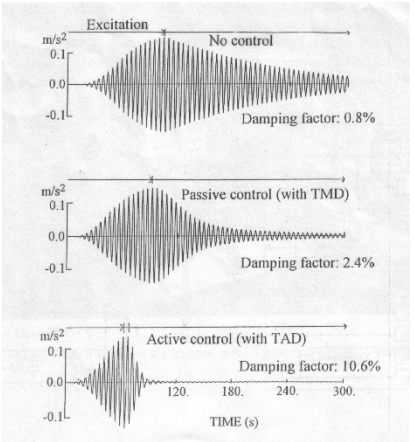
Taipei 101 , 509.2 m



Tuned active dampers

Yokohama Landmark Tower, 295.8 m

Reduction of the acceleration:
29% ÷ 39%



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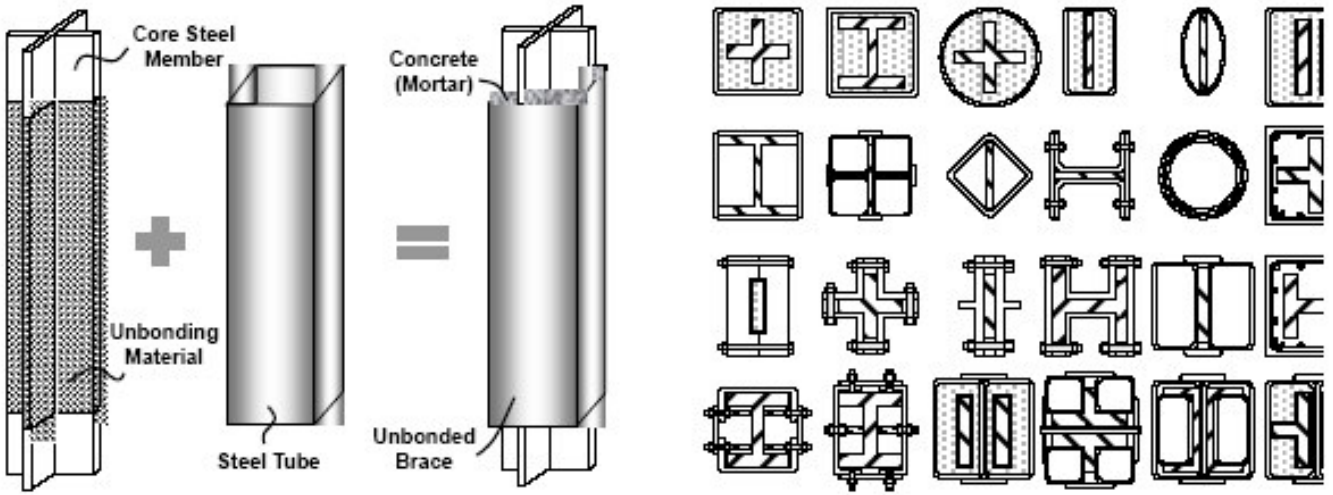
New structural systems for seismic applications

- Buckling-restrained braced frames (BRB)
- Steel plate shear walls (SPSW)
- Systems with removable dissipative members (RDM)



Schematic and typical types of buckling restrained braces

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Application of BRB - Tzu-Chi Culture Building, Taiwan

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Steel plate shear walls

- The Steel Plate Shear Walls (SPSWs) application has increased in recent years. Design requirements for SPSWs are already implemented in the AISC 2005.
- One of the most important application of steel plate shear walls in a very highly seismic area is the 35-story high-rise in Kobe, Japan.
- The structure was constructed in 1988 and was subjected to the 1995 Kobe earthquake.

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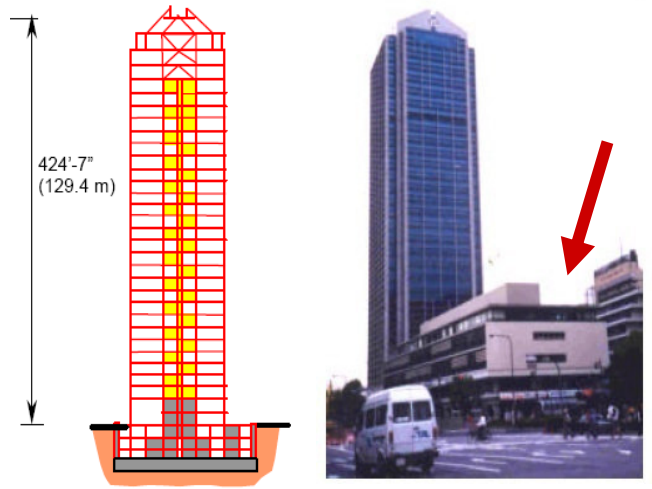
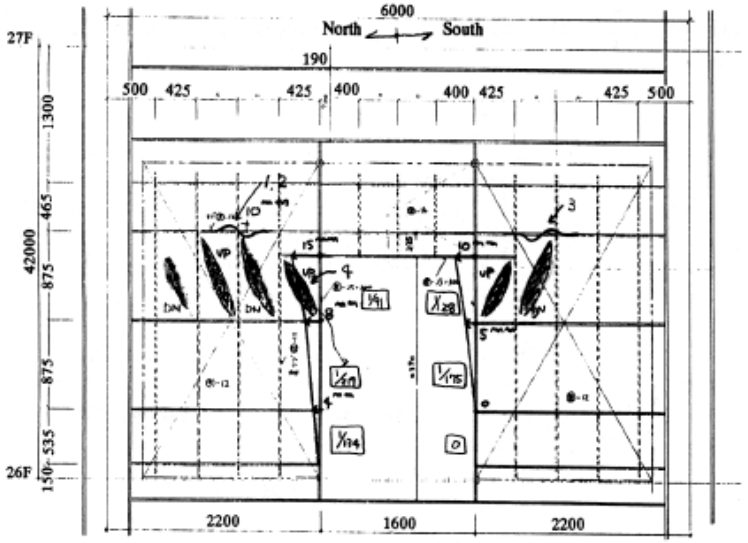
- Studies of this structure (Fujitani et al., 1996) (AIJ, 1995) have indicated that the damage was minor and consisted of local buckling of stiffened steel plate shear walls on the 26th story (Fujitani et al., 1996)
- Interesting to note the adjacent building was heavily damaged during the same earthquake, suffering a partial collapse due to a soft story mechanism





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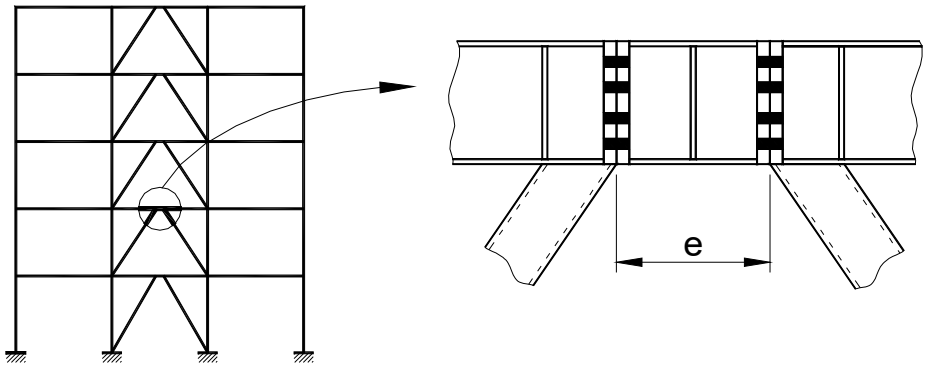
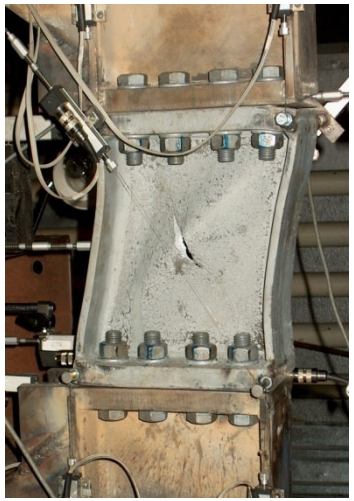
Systems with removable dissipative members

- Systems with removable dissipative members (RDM) allow replacement of the dissipative elements damaged as a result of a moderate to strong earthquake, reducing the repair costs. Application of this principle to eccentrically braced frames, where links act as dissipative zones, is presented in figure.



Removable bolted link concept



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Principals of design and actions

- Safe and economical design of buildings is a complex issue
- Computation of design loads requires a complex analysis involving considerations such as building use, location and geometry
- All building codes specify minimum design loads, which include dead load, imposed load, wind load, and, when applicable, snow load, crane load and earthquake load
- Excepting earthquake load, all the other loads do not require any greater understanding of structural behavior beyond elastic point
- Structural behavior is fundamentally different for seismic loading, and there are a number of detailed requirements and provisions necessary to assure acceptable seismic performance
- Structures should also be able to resist abnormal loads due to fire, impact or explosion, so as to limit the effects of local collapse and to prevent or minimize progressive collapse

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Structural design

- Design situations shall be classified as follows :
  - persistent design situations, which refer to the conditions of normal use
  - transient design situations, which refer to temporary conditions applicable to the structure, e.g. during execution or repair
  - accidental design situations, which refer to exceptional conditions applicable to the structure or to its exposure, e.g. to fire, explosion, impact or the consequences of localised failure
  - seismic design situations, which refer to conditions applicable to the structure when subjected to seismic events
- Design for limit states shall be based on the use of structural and load models for relevant limit states:
  - Ultimate limit states (ULS)
  - Serviceability limit states (SLS)
- The verifications shall be carried out for all relevant design situations and load cases. The requirements should be achieved by the partial factor method

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Limit states

- Ultimate limit state

– The limit states that concern:

- the safety of people, and/or
- the safety of the structure

shall be classified as ultimate limit states.

– The following ultimate limit states shall be verified where they are relevant :

- loss of equilibrium of the structure or any part of it, considered as a rigid body
- failure by excessive deformation, transformation of the structure or any part of it into
- a mechanism, rupture, loss of stability of the structure or any part of it, including supports and foundations
- failure caused by fatigue or other time-dependent effects

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Limit states

- Serviceability limit state
  - The limit states that concern :
    - functioning of the structure or structural members under normal use
    - the comfort of people
    - the appearance of the construction works

  shall be classified as serviceability limit states
  - The verification of serviceability limit states should be based on criteria concerning the following:
    - the deformations that affect the appearance, the comfort of users, or the functioning of the structure (including equipments), or that cause damage to finishes or non-structural members
    - vibrations
    - damage that is likely to adversely affects the appearance, the durability, or the operation of the structure.

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Serviceability (SLS) - informative

- ridge vertical deflection:

$$f_{max} \leq L/250$$

- eaves horizontal deflection

$$\delta \leq H/100$$

- crane girder horizontal deflection :

$$\delta^{C} = H_{cr}/700$$

- crane beam mid span vertical deflection:

$$f_{max}^{cr} \leq T/600$$

Where:
- L = Main frame span [m];
- T = Bay between main frames [m];
- H = eave height of main frame [m];
- Hcr = level of the crane beam [m];
- Q = lifting capacity of crane [t]

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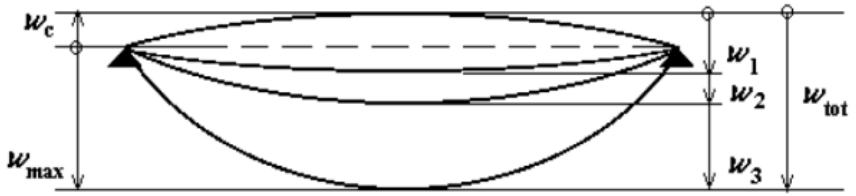
Deformations and horizontal displacements

- Vertical deflections are represented schematically below:



Definitions of vertical deflections

where:

- w_c is the precamber in the unloaded structural member
- w_1 is the initial part of the deflection under permanent loads of the relevant combination of actions
- w_2 is the long-term part of the deflection under permanent loads
- w_3 is the additional part of the deflection due to the variable actions of the relevant combination of actions
- w_{tot} is the total deflection as sum of w_1 , w_2 , w_3
- w_{max} is the remaining total deflection taking into account the precamber

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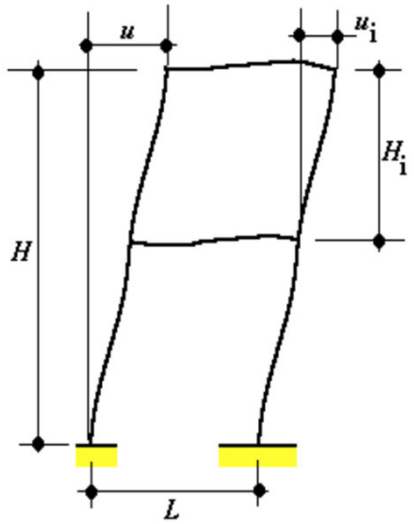
Deformations and horizontal displacements

- Horizontal displacements are represented schematically below:



Definition of horizontal displacements

where:

- u is the overall horizontal displacement over the building height H
- u_i is the horizontal displacement over a storey height H_i

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Classification of actions

- Actions shall be classified by their variation in time as follows:
  - permanent actions (G), e.g. self-weight of structures, fixed equipment and road surfacing, and indirect actions caused by shrinkage and uneven settlements;
  - variable actions (Q), e.g. imposed loads on building floors, beams and roofs, wind actions or snow loads ;
  - accidental actions (A), e.g. explosions, or impact from vehicles.

Certain actions, such as seismic actions and snow loads, may be considered as either accidental and/or variable actions, depending on the site location, see EN 1991 and EN 1998.

In northern areas, roof loads are determined by the expected maximum snow loads. However, in southern areas, where snow accumulation is not a problem, minimum roof live loads should be specified to accommodate the weight of workers, equipment, and materials during maintenance and repair.

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Combination of actions - ULS

- Combinations of actions for persistent or transient design situations (fundamental combinations):

$$\sum_{j\geq 1} \gamma_{G,j} G_{k,j} "+" \gamma_P P "+" \gamma_{Q,1} Q_{k,1} "+" \sum_{i>1} \gamma_{Q,i} \psi_{0,i} Q_{k,i}$$

- Combinations of actions for accidental design situations:

$$\sum_{j\geq 1} G_{k,j} "+" P "+" A_d "+" (\psi_{1,1} \text{ or } \psi_{2,1}) Q_{k,1} "+" \sum_{i>1} \psi_{2,i} Q_{k,i}$$

- Combinations of actions for seismic design situations:

$$\sum_{j\geq 1} G_{k,j} "+" P "+" A_{Ed} "+" \sum_{i\geq 1} \psi_{2,i} Q_{k,i}$$

where:

-G_k, Q_k is the characteristic value of the actions
- A_{Ed} is the design value of seismic action
- A_d is the design value of an accidental action
- γ is the partial factor for actions
- ψ_0 is the factor for combination value of a variable action
- ψ_1 is the factor for frequent value of a variable action
- ψ_2 is the factor for quasi-permanent value of a variable action

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Recommended values of ψ factors for buildings

Action	ψ_0	ψ_1	ψ_2
Imposed loads in buildings, category (see EN 1991-1-1)			
Category A : domestic, residential areas	0,7	0,5	0,3
Category B : office areas	0,7	0,5	0,3
Category C : congregation areas	0,7	0,7	0,6
Category D : shopping areas	0,7	0,7	0,6
Category E : storage areas	1,0	0,9	0,8
Category F : traffic area, vehicle weight ≤ 30kN	0,7	0,7	0,6
Category G : traffic area, 30kN < vehicle weight ≤ 160kN	0,7	0,5	0,3
Category H : roofs	0	0	0
Snow loads on buildings (see EN 1991-1-3)*			
Finland, Iceland, Norway, Sweden	0,70	0,50	0,20
Remainder of CEN Member States, for sites located at altitude H > 1000 m a.s.l.	0,70	0,50	0,20
Remainder of CEN Member States, for sites located at altitude H ≤ 1000 m a.s.l.	0,50	0,20	0
Wind loads on buildings (see EN 1991-1-4)	0,6	0,2	0
Temperature (non-fire) in buildings (see EN 1991-1-5)	0,6	0,5	0

NOTE The ψ values may be set by the National annex.
* For countries not mentioned below, see relevant local conditions.

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Combination of actions - SLS

- Characteristic combination (normally used for irreversible limit states):

$$\sum_{j\geq 1} G_{k,j} \text{ "+" } P \text{ "+" } Q_{k,1} \text{ "+" } \sum_{i>1} \psi_{0,i} Q_{k,i}$$

- Frequent combination (normally used for reversible limit states):

$$\sum_{j\geq 1} G_{k,j} \text{ "+" } P \text{ "+" } \psi_{1,1} Q_{k,1} \text{ "+" } \sum_{i>1} \psi_{2,i} Q_{k,i}$$

- Quasi-permanent combination (normally used for long-term effects and the appearance of the structure):

$$\sum_{j\geq 1} G_{k,j} \text{ "+" } P \text{ "+" } \sum_{i\geq 1} \psi_{2,i} Q_{k,i}$$

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Seismic design

- When applicable, the seismic hazard shall be taken into account in the early stages of the conceptual design of a building, thus enabling the achievement of a structural system which, within acceptable costs, satisfies the fundamental requirements:
  - structural simplicity
  - uniformity, symmetry and redundancy
  - bi-directional resistance and stiffness
  - torsional resistance and stiffness
  - diaphragmatic behaviour at roof level
  - adequate foundation

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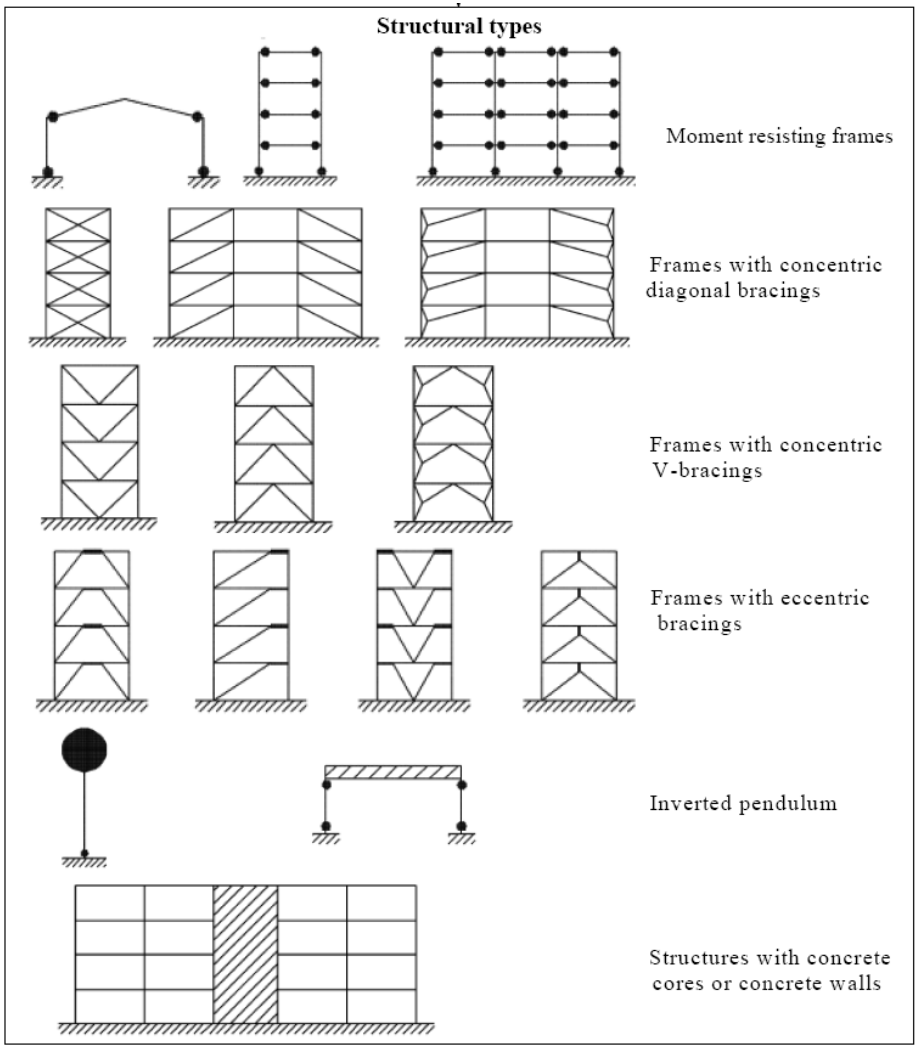
Flow chart and design criteria for Steel Frames in seismic areas



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Design concepts, ductility classes

- Earthquake resistant steel buildings shall be designed in accordance with one of the following concepts:
  - Concept a) Low-dissipative structural behaviour
  - Concept b) Dissipative structural behaviour

Design concept	Structural ductility class	Range of the reference values of the behaviour factor q
Concept a) Low dissipative structural behaviour	DCL (Low)	≤ 1,5 - 2
Concept b) Dissipative structural behaviour	DCM (Medium)	≤ 4 also limited by the values of Table 2
	DCH (High)	only limited by the values of Table 2

Table 1: Design concepts, structural ductility classes and upper limit reference values of the behaviour factors

Ductility class	Reference value of behaviour factor q	Required cross-sectional class
DCM	$1,5 < q \leq 2$	class 1, 2 or 3
	$2 < q \leq 4$	class 1 or 2
DCH	$q > 4$	class 1

Table 2: Requirements on cross-sectional class of dissipative elements depending on ductility class and reference behaviour factor

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Types of structures, behaviour factors

- Steel buildings shall be assigned to one of the following structural types, according to the behaviour of their primary resisting structure under seismic actions:
  a) Moment resisting frames, are those in which the horizontal forces are mainly resisted by members acting in an essentially flexural manner.
  b) Frames with concentric bracings, are those in which the horizontal forces are mainly resisted by members subjected to axial forces.
  c) Frames with eccentric bracings, are those in which the horizontal forces are mainly
  resisted by axially loaded members, but where the eccentricity of the layout is such that energy can be dissipated in seismic links by means of either cyclic bending or cyclic shear.
  d) Inverted pendulum structures, are structures in which dissipative zones are located at the bases of columns.
  e) Structures with concrete cores or concrete walls, are those in which horizontal forces are mainly resisted by these cores or walls.
  f) Moment resisting frames combined with concentric bracings.
  g) Moment resisting frames combined with infills.

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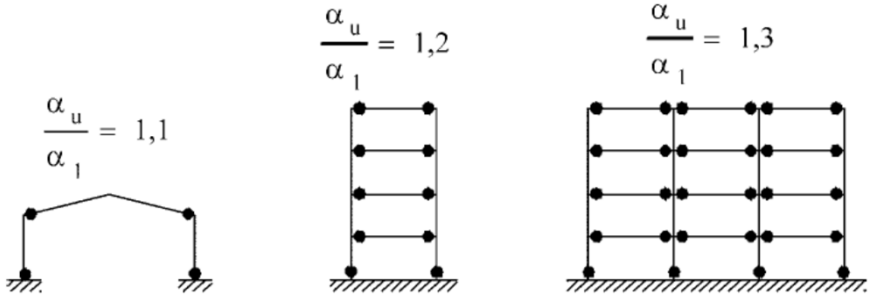
Upper limit of reference values of behaviour factors for systems regular in elevation

STRUCTURAL TYPE	Ductility Class	
	DCM	DCH
a) Moment resisting frames	4	$5\alpha_u/\alpha_1$
b) Frame with concentric bracings Diagonal bracings V-bracings	 4 2	 4 2,5
c) Frame with eccentric bracings	4	$5\alpha_u/\alpha_1$
d) Inverted pendulum	2	$2\alpha_u/\alpha_1$
e) Structures with concrete cores or concrete walls	See section **5**	
f) Moment resisting frame with concentric bracing	4	$4\alpha_u/\alpha_1$
g) Moment resisting frames with infills Unconnected concrete or masonry infills, in contact with the frame Connected reinforced concrete infills Infills isolated from moment frame (see moment frames)	2	2
	See section **7**	
	4	$5\alpha_u/\alpha_1$



Moment resisting frames (dissipative zones in beams and at bottom of columns)

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Dissipative vs. non-dissipative structures

- When dissipative concept is applied, the design seismic action is considerably smaller than the one required for an elastic response of the structure (*q* of the order of 2 ÷ 6):
  – Advantageous for high-seismicity regions and/or middle-rise and high-rise structures
  – special design requirements should be fulfilled, in order to assure the necessary ductility of the structure (like structural layout, ductility of members and connections, steel quality, detailing)

- When low dissipative concept is applied, the only requirement is that seismic action (computed based on a behaviour factor $q = 1$) is considered when establishing action effects
  – Values of q factor $1.0 < q < 2.0$ can be adopted only when they can be justified theoretically and/or experimentally.
  – Economical for low-seismicity areas and/or light structures, like single-storey steel frame buildings

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Flow chart and design criteria for Steel Frames in seismic areas

Definition of Limit State:
DLS (Damage Limitation State)
ULS (Ultimate Limit State)

↓

Definition of earthquake level:
Level 1 for DLS
Level 2 for ULS

↓

Representations of the seismic action:
Elastic response spectrum:
- Horizontal elastic response spectrum
- Vertical elastic response spectrum
- Design spectrum for elastic analysis

Alternative representations of the seismic action - Time - history representation:
- Artificial accelerograms
- Recorded or simulated accelerograms

↓

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Flow chart and design criteria for Steel Frames in seismic areas

Seismic design of building:

Principles of conceptual design:
– structural simplicity;
– uniformity, symmetry and redundancy;
– bi-directional resistance and stiffness;
– torsional resistance and stiffness;
– diaphragmatic behaviour at storey level;
– adequate foundation

Type of analysis:
1. Simplified analysis - lateral force method of analysis"
2. Simplified dynamic analysis - modal response spectrum analysis
3. Advanced analysis
   - non-linear static (pushover) analysis
   - non-linear time history (dynamic) analysis,

Design of steel building
- For the design of steel building EN 1993-1-1 applies
- Additional rules regarding seismic design are those given in EN 1998-1

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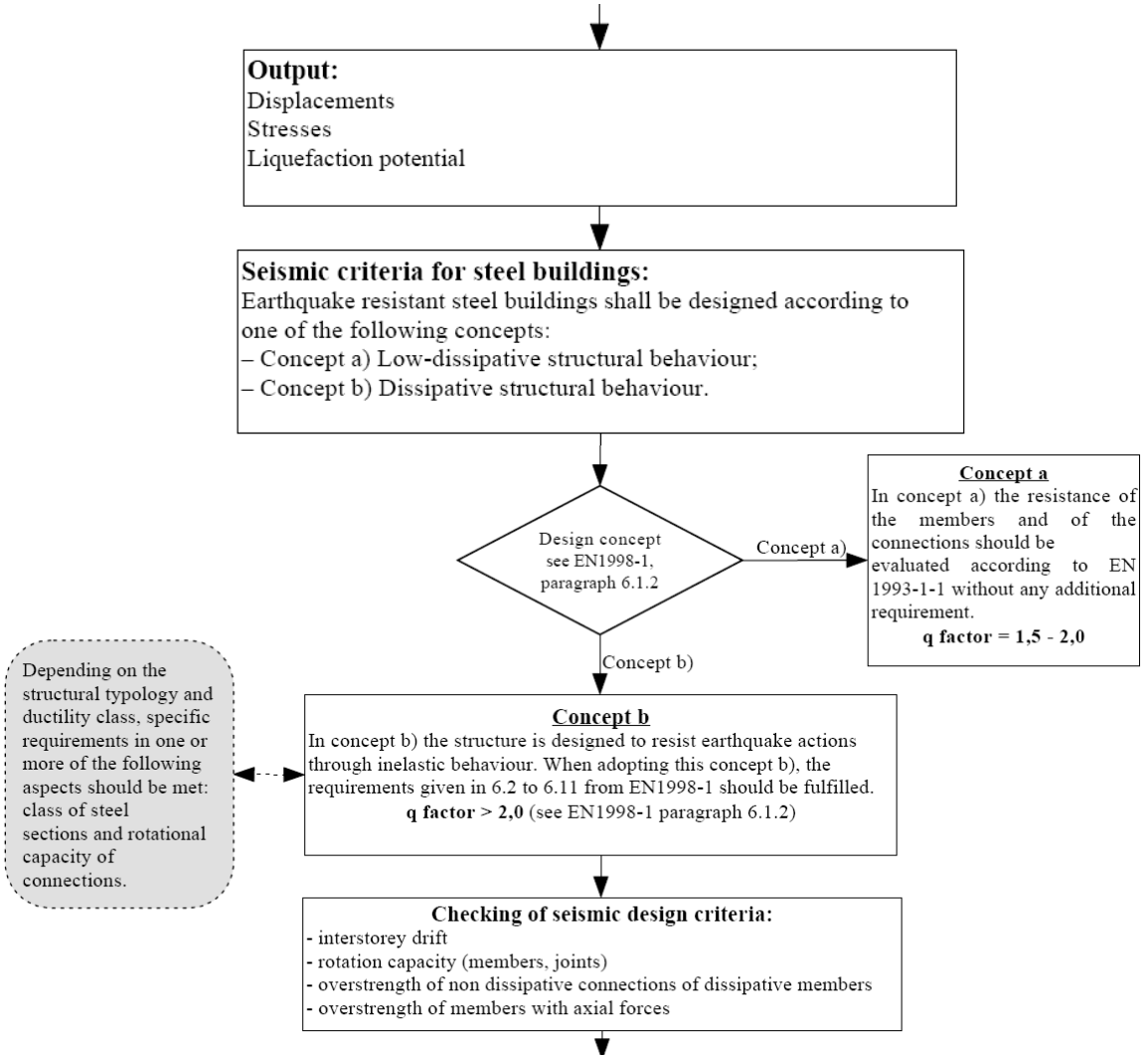

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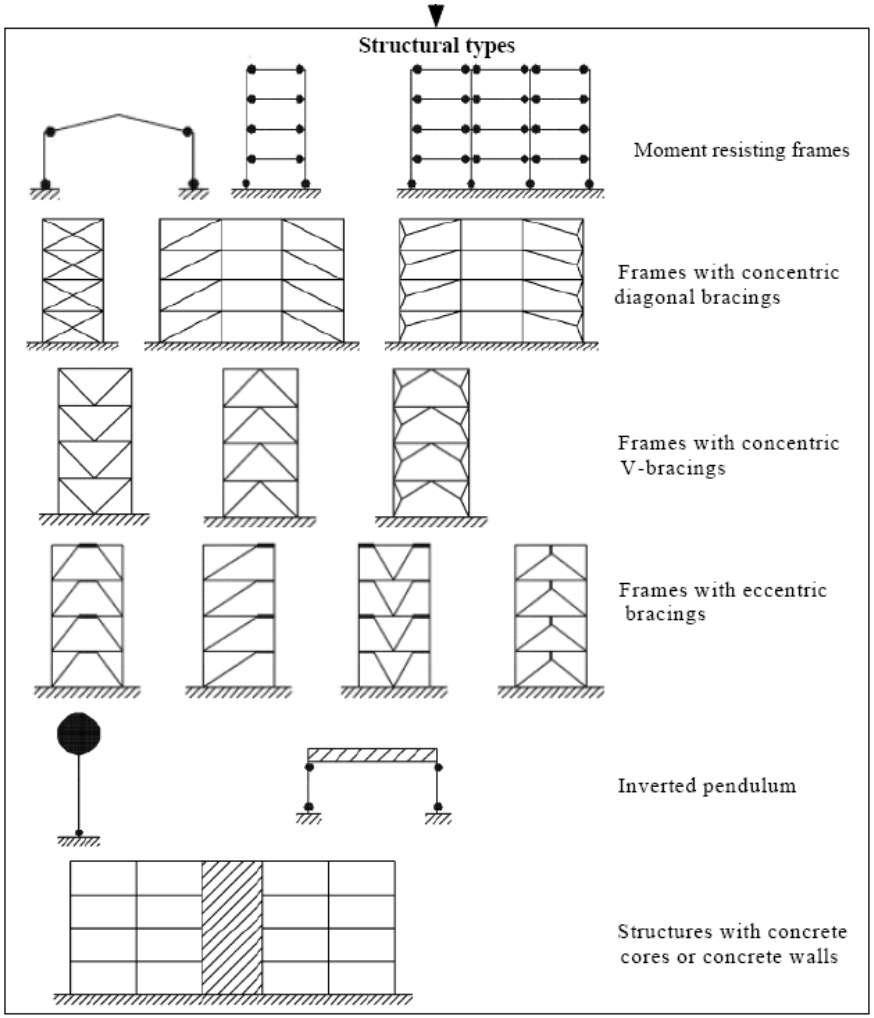

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Accidental loadings

- Accidental design situations are design situations involving exceptional conditions of the structure or its exposure, including fire, explosion, impact or local failure

In essence, the objective is to ensure that buildings do not suffer disproportionate collapse under accidental loading. Largely, this is assured in steel framed buildings by designing connections appropriately.

A structure shall be designed and executed in such a way that it will not be damaged by events such as: – explosion, – impact, and – the consequences of human errors, to an extent disproportionate to the original cause.

Note 1 - The events to be taken into account are those agreed for an individual project with the client and the relevant authority.

Note 2 - Further information is given in EN 1991-1-7.

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Accidental loadings

Potential damage shall be avoided or limited by appropriate choice of one or more of the following:

- *avoiding, eliminating or reducing the hazards to which the structure can be subjected;*
- *selecting a structural form which has low sensitivity to the hazards considered;*
- *selecting a structural form and design that can survive adequately the accidental removal of an individual member or a limited part of the structure, or the occurrence of acceptable localised damage;*
- *avoiding as far as possible structural systems that can collapse without warning;*
- *tying the structural members together.*

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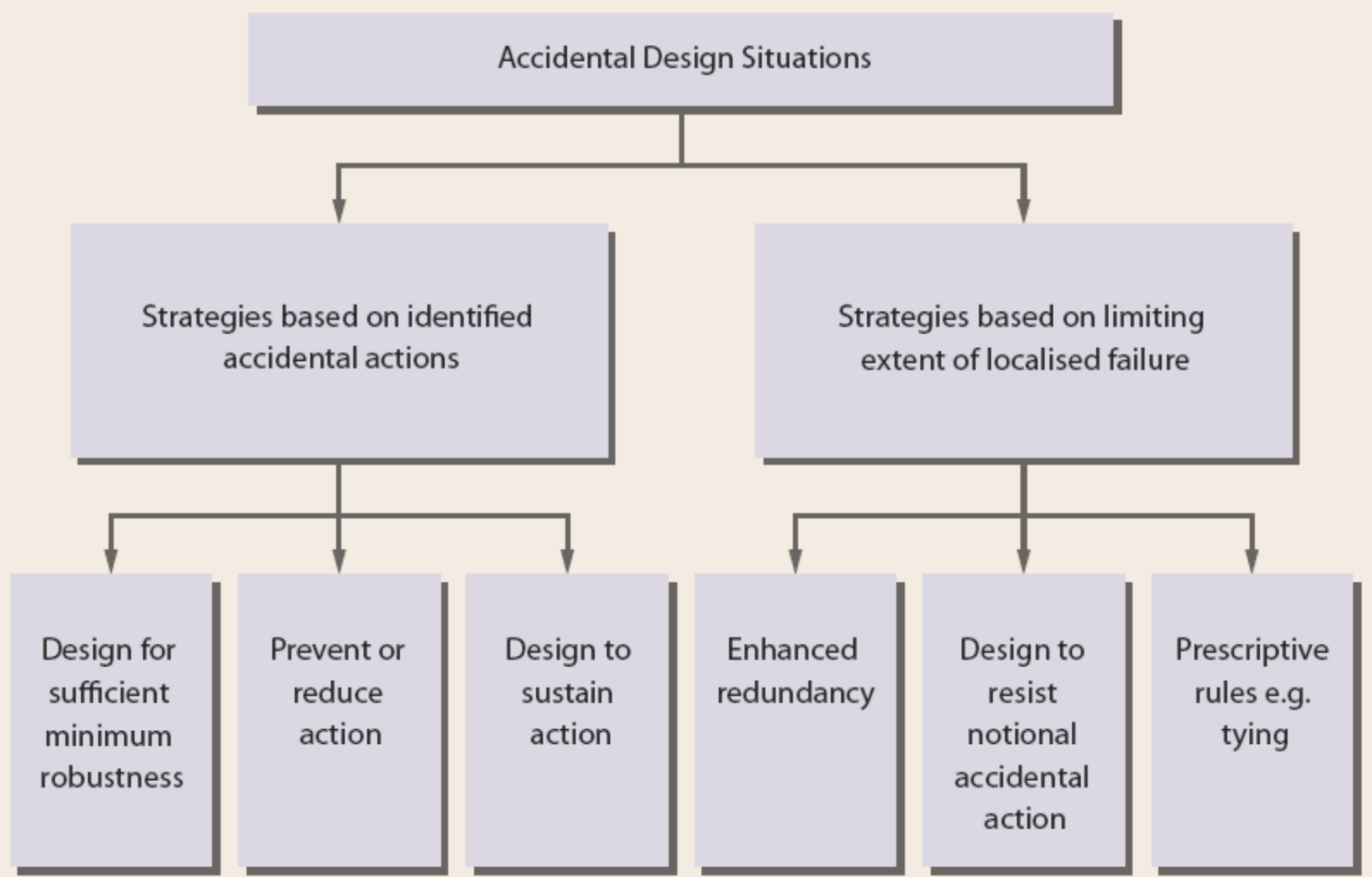
Strategies for accidental design situations



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Case study: Design of a low rise building

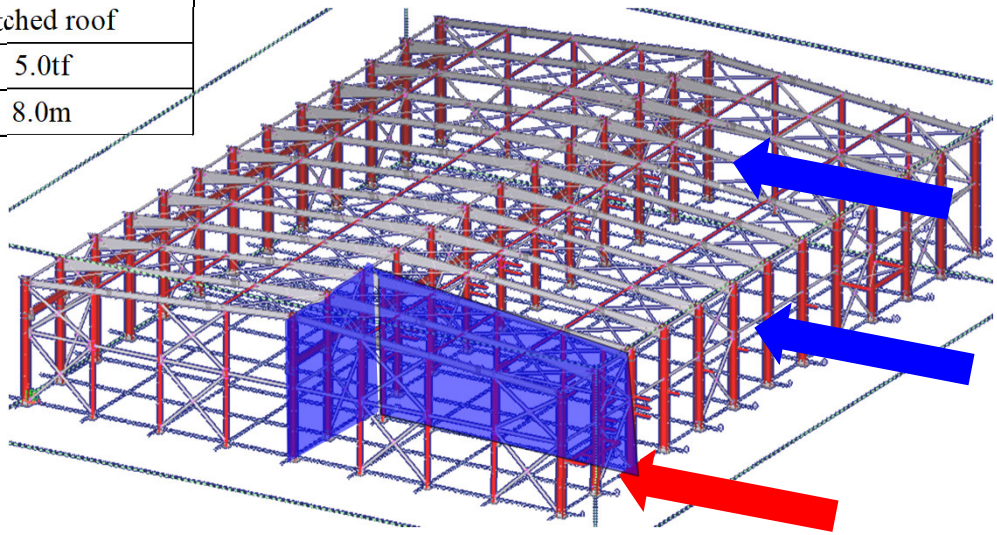
General data

Location	Timisoara
Span:	26,79m
No. of spans	2
Bay	6,0m
No. of bays	12
Length	53,59 m
Clear height at the eave	10,10m
Height at the eave	+ 10,60
Roof slope	4º
Roof system	Pitched roof
Overhead crane	5.0tf
Level of the crane rail	8.0m

Destination:
- sandwich panels
- products made of expanded polystyrene

Demands:
- 4.5 h fire resistance - wall
-2 h fire resistance – columns
- 1 h fire resistance - beams

3D view



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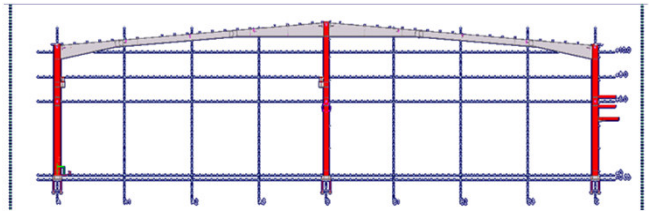
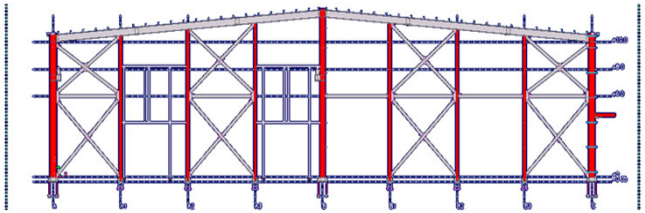
Structural system

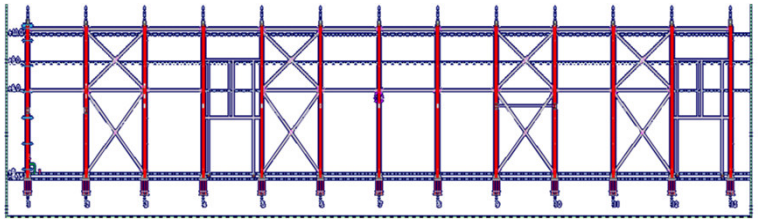
Intermediate transversal frame



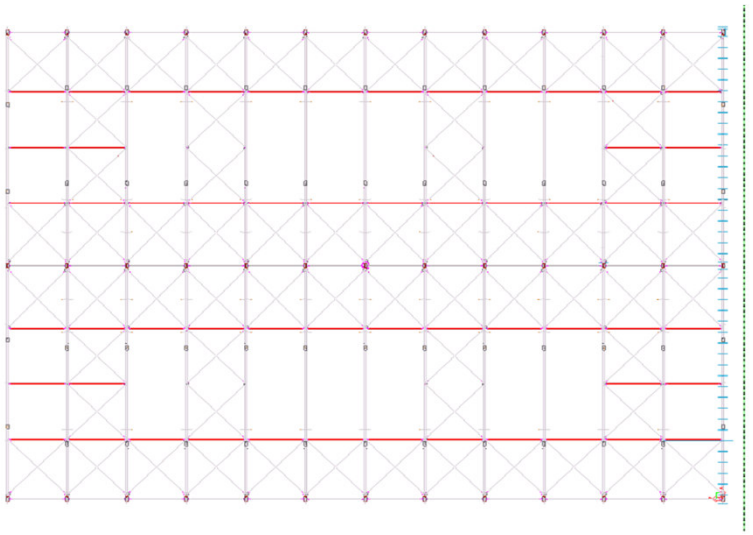
End-wall transversal frame



End-wall longitudinal frame



Top view with the horizontal bracings



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Evaluation of actions

- Dead load (SREN 1991-1-1)
  - Self weight of the steel skeleton (automatically calculated in the analysis)
  - Dead load from the roof
    - Sheeting 0.15
    - Insulation 0.15
    - Purlins 0.05

    Total g^n=0,35 kN/m^2
  - Dead load from the walls
    - Sheeting 0.10
    - Insulation 0.15
    - Purlins 0.05

    Total g^n=0,30 kN/m^2
  - Dead load from equipments (electrical equipment, ventilation, etc.)

    g^n=0,30 kN/m^2

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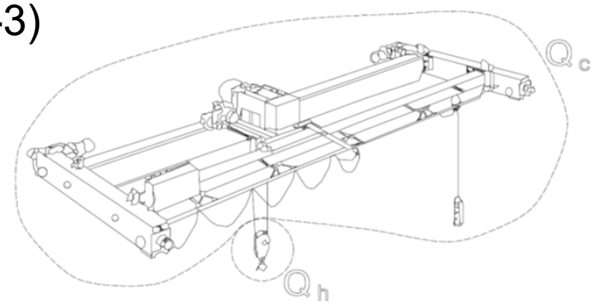
Evaluation of actions



- Overhead traveling crane loads (SREN 1991-3)
  - Capacity – 5 000 kg
  - Wheel loads
- Max. R 2999 kg
- Min. R 459 kg
- Driving and braking forces 0.56 kN
- Inertial forces crane traveling min. HM 0.28 kN
- Inertial forces crane traveling max. HM 1.71 kN
- Skewing forces (including factor 1.1) S = 9.77 kN
  max. Hs = 8.38 kN
  min. Hs = 1.38 kN
- Max buffer forces max Pu = 7.6 kN
- Weight of the crane 1851 kg
- Weight of the trolley 373 kg
- Weight of the crane girder 1025 kg
- Weight of the crane travel units 290 kg
- Weight of the electrical equipment 163 kg
- Lifting 6/1m/min
- Cross traveling 5-30m/min
- Crane traveling 10/40m/min

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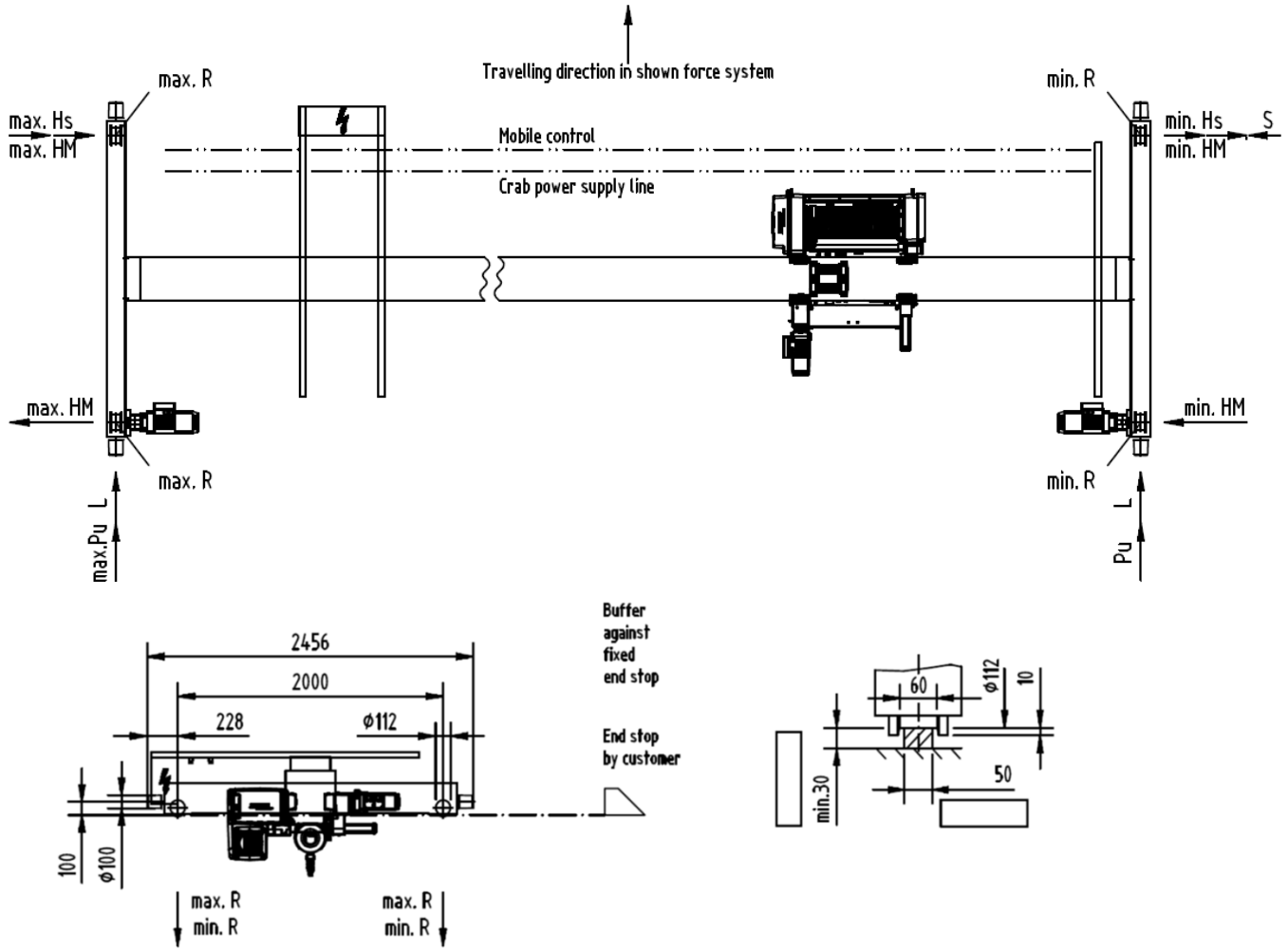
Overhead traveling crane



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Overhead traveling crane

Dynamic factors φ_i

Dynamic factors	Effects to be considered	To be applied to
φ_1	– excitation of the crane structure due to lifting the hoist load off the ground	selfweight of the crane
φ_2 or φ_3	–dynamic effects of transferring the hoist load from the ground to the crane –dynamic effect of sudden release of the payload if for example grabs or magnets are used	hoist load
φ_4	–dynamic effects induced when crane is travelling on rail tracks or runways	selfweight of the crane and hoist load
φ_5	–dynamic effects caused by drive forces	drive forces
φ_6	–dynamic effects of a test load moved by the drives in the way the crane is used	test load
φ_7	–dynamic elastic effects of impact on buffers	buffer loads

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Overhead traveling crane

Groups of loads and dynamic factors to be considered as one characteristic crane action

		Symbol	Section	Groups of loads									
				ULS							Test load	Acci-dental	
				1	2	3	4	5	6	7	8	9	10
1	Selfweight of crane	Q_c	2.6	φ_1	φ_1	1	φ_4	φ_4	φ_4	1	φ_1	1	1
2	Hoist load	Q_h	2.6	φ_2	φ_3	-	φ_4	φ_4	φ_4	$\eta^{1)}$	-	1	1
3	Acceleration of crane bridge	H_L, H_T	2.7	φ_5	φ_5	φ_5	φ_5	-	-	-	φ_5	-	-
4	Skewing of crane bridge	H_S	2.7	-	-	-	-	1	-	-	-	-	-
5	Acceleration or braking of crab or hoist block	H_{T3}	2.7	-	-	-	-	-	1	-	-	-	-
6	In service wind	F_W^*	Annex A	1	1	1	1	1	-	-	1	-	-
7	Test load	Q_T	2.10	-	-	-	-	-	-	-	φ_6	-	-
8	Buffer force	H_B	2.11	-	-	-	-	-	-	-	-	φ_7	-
9	Tilting force	H_{TA}	2.11	-	-	-	-	-	-	-	-	-	1

[1)] η is the part of the hoist load that remains when the payload is removed, but is not included in the selfweight of the crane.

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Snow load (SR EN 1991-3)

– Snow loads on roofs shall be determined as follows:

$$s = \mu_i \times C_e \times C_t \times s_k$$

where:

μ_i is the snow load shape coefficient
s_k is the characteristic value of snow load on the ground
C_e is the exposure coefficient
C_t is the thermal coefficient

μ_i = 0.8 (pitched roofs, slope < 30°)
s_k = 1.5kN (Timisoara, see snow map)
C_e = 1.0 (normal topography)
C_t = 1.0 (no snow melting caused by heat loss)

$$s_k = 0.8 \times 1.0 \times 1.0 \times 1.5 = 1.2 \text{ kN/m}^2$$

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Snow load map

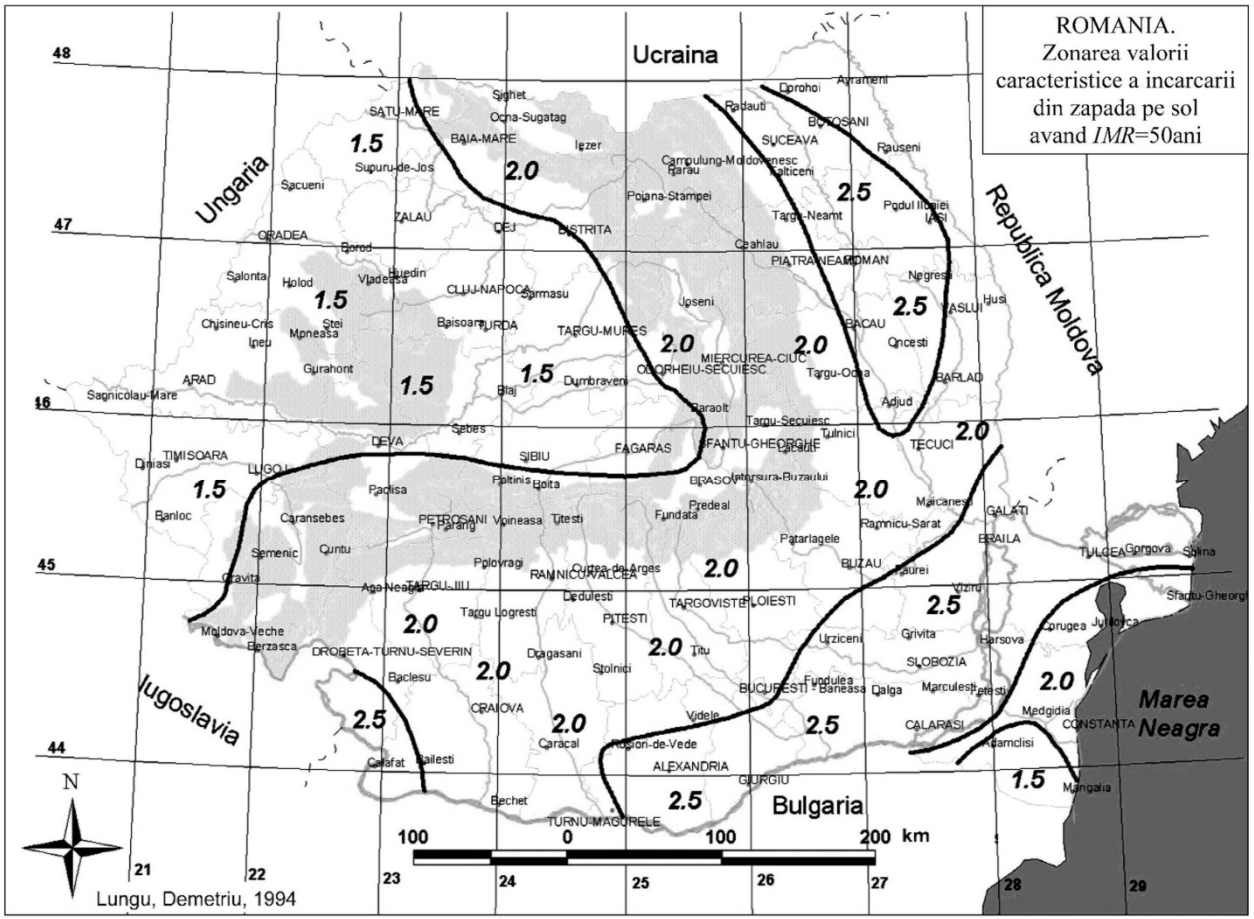

Figura 2.1 Romania - zonarea valorii caracteristice a incarcarii din zapada pe sol $s_{0,k}$, kN/m^2

3. Evaluation of actions

Snow load shape coefficients

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Snow load

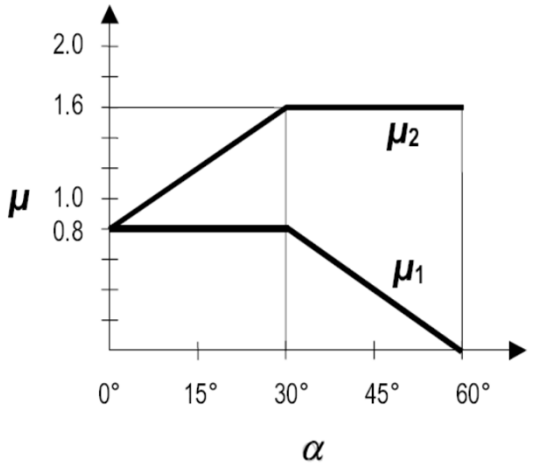
Recommended values of C_e for different topographies

Topography	C_e
Windswept [a]	0,8
Normal [b]	1,0
Sheltered [c]	1,2

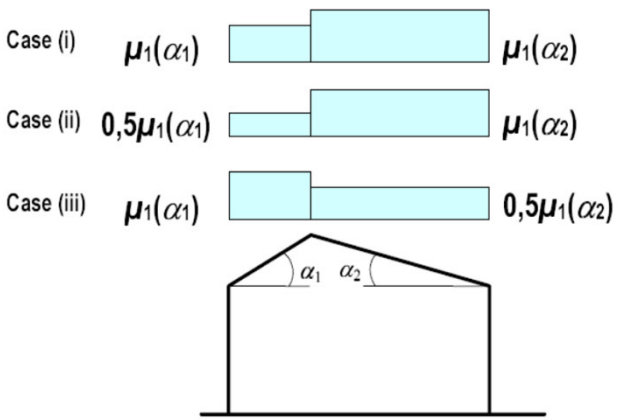
[a] *Windswept topography*: flat unobstructed areas exposed on all sides without, or little shelter afforded by terrain, higher construction works or trees.

[b] *Normal topography*: areas where there is no significant removal of snow by wind on construction work, because of terrain, other construction works or trees.

[c] *Sheltered topography*: areas in which the construction work being considered is considerably lower than the surrounding terrain or surrounded by high trees and/or surrounded by higher construction works.



Angle of pitch of roof α	$0° \le \alpha \le 30°$	$30° < \alpha < 60°$	$\alpha \ge 60°$
μ_1	0,8	$0,8(60 - \alpha)/30$	0,0
μ_2	$0,8 + 0,8\ \alpha/30$	1,6	--



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Wind load (SR EN 1991-3/NA)

– Wind pressure at height z above the ground, on the exterior surfaces is determined with the relation:

$$w(z) = q_{ref}\, C_e(z)\, c_p$$

where:

q_{ref} – the reference wind pressure (for Timisoara the value is 0.4kN/mp, in accordance with Anexa A/ NP-082-04)

$C_e(z)$ – exposure factor at height z

$Ce(z) = C_g(z)\, C_r(z)$ $\qquad z_0 = 0.3$

$C_g(z) = 1+1.35[2I(z)]$ $\qquad k_r(z_0) = 0.22$

$I(z)=\beta^{0.5}/2.5 ln(z/z_0)$ $\qquad \beta^{0.5} = 2.35$

$C_r(z) = k_r^2(z_0)\, (ln\ z/z_0)^2$

C_p - pressure coefficient

q_{ref} = 0.4kN/mp, in accordance with Annex A/ NP-082-04)

$C_e(z)$ – exposure factor at height z

$Ce(z) = 1.71 \times 0.615 = 1.05$

$C_g(z) = 1+1.35 \times 2 \times 0.26 = 1.71$

$I(z)=2.35/(2.5 \times ln(10.6/0.3))=0.26$

$C_r(z) = 0.22^2 \times (ln(10.6/0.3))^2 = 0.615$

C_p = +0.8 pression on windward wall → $w(z) =0.4 \times 1.05 \times 0.8=0.336$ kN/m2

= -0.3 suction on leeward wall → $w(z) =0.4 \times 1.05 \times 0.3=0.126$ kN/m2

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Wind load map

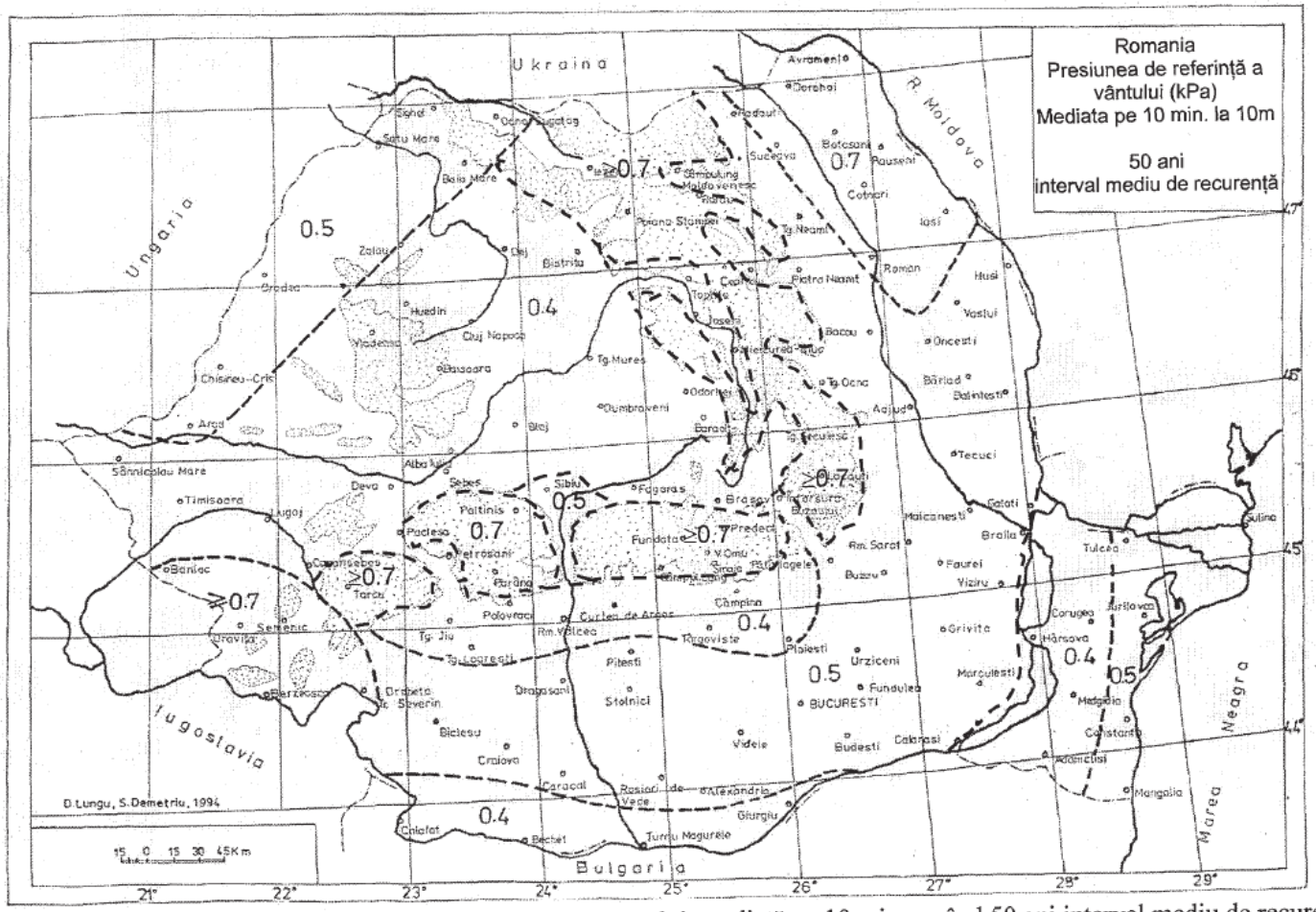

Figura A.2 Valori caracteristice ale presiunii de referință a vântului, mediată pe 10 min., având 50 ani interval mediu de recurență (2% probabilitate anuală de depășire)

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External pressure coefficients for flat roofs

Roof type		Zone F $c_{pe,10}$	Zone F $c_{pe,1}$	Zone G $c_{pe,10}$	Zone G $c_{pe,1}$	Zone H $c_{pe,10}$	Zone H $c_{pe,1}$	Zone I $c_{pe,10}$	Zone I $c_{pe,1}$
Sharp eaves		-1,8	-2,5	-1,2	-2,0	-0,7	-1,2	+0,2	
								-0,2	
With Parapets	h_p/h=0,025	-1,6	-2,2	-1,1	-1,8	-0,7	-1,2	+0,2	
								-0,2	
	h_p/h=0,05	-1,4	-2,0	-0,9	-1,6	-0,7	-1,2	+0,2	
								-0,2	
	h_p/h=0,10	-1,2	-1,8	-0,8	-1,4	-0,7	-1,2	+0,2	
								-0,2	
Curved Eaves	r/h = 0,05	-1,0	-1,5	-1,2	-1,8	-0,4		+0,2	
								-0,2	
	r/h = 0,10	-0,7	-1,2	-0,8	-1,4	-0,3		+0,2	
								-0,2	
	r/h = 0,20	-0,5	-0,8	-0,5	-0,8	-0,3		+0,2	
								-0,2	
Mansard Eaves	α = 30°	-1,0	-1,5	-1,0	-1,5	-0,3		+0,2	
								-0,2	
	α = 45°	-1,2	-1,8	-1,3	-1,9	-0,4		+0,2	
								-0,2	
	α = 60°	-1,3	-1,9	-1,3	-1,9	-0,5		+0,2	
								-0,2	

NOTE 1 For roofs with parapets or curved eaves, linear interpolation may be used for intermediate values of h_p/h and r/h.

NOTE 2 For roofs with mansard eaves, linear interpolation between $\alpha = 30°$, 45° and $\alpha = 60°$ may be used. For $\alpha > 60°$ linear interpolation between the values for $\alpha = 60°$ and the values for flat roofs with sharp eaves may be used.

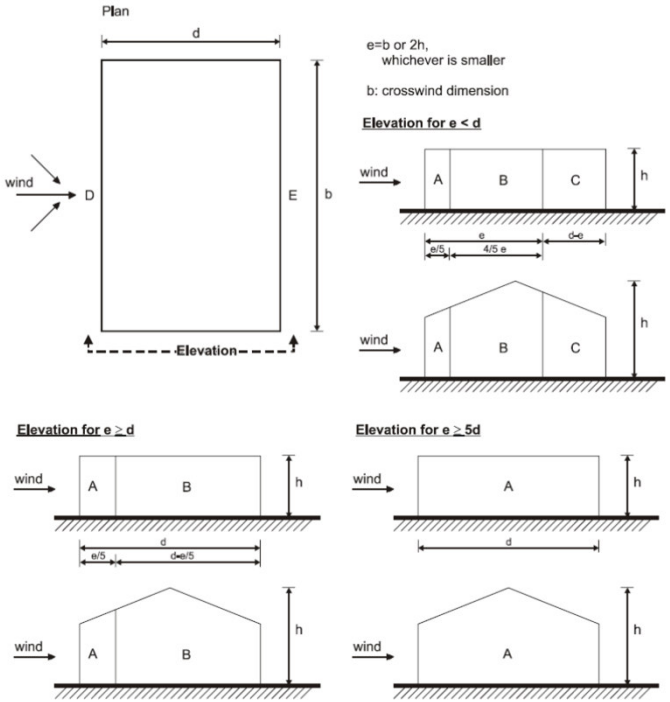
NOTE 3 In Zone I, where positive and negative values are given, both values shall be considered.

NOTE 4 For the mansard eave itself, the external pressure coefficients are given in Table 7.4a "External pressure coefficients for duopitch roofs: wind direction 0° ", Zone F and G, depending on the pitch angle of the mansard eave.

NOTE 5 For the curved eave itself, the external pressure coefficients are given by linear interpolation along the curve, between values on the wall and on the roof.

Recommended values of external pressure coefficients for vertical walls of rectangular plan buildings

Zone	A		B		C		D		E	
h/d	$c_{pe,10}$	$c_{pe,1}$	$c_{pe,10}$	$c_{pe,1}$	$c_{pe,10}$	$c_{pe,1}$	$c_{pe,10}$	$c_{pe,1}$	$c_{pe,10}$	$c_{pe,1}$
5	-1,2	-1,4	-0,8	-1,1	-0,5		+0,8	+1,0	-0,7	
1	-1,2	-1,4	-0,8	-1,1	-0,5		+0,8	+1,0	-0,5	
≤ 0,25	-1,2	-1,4	-0,8	-1,1	-0,5		+0,7	+1,0	-0,3	



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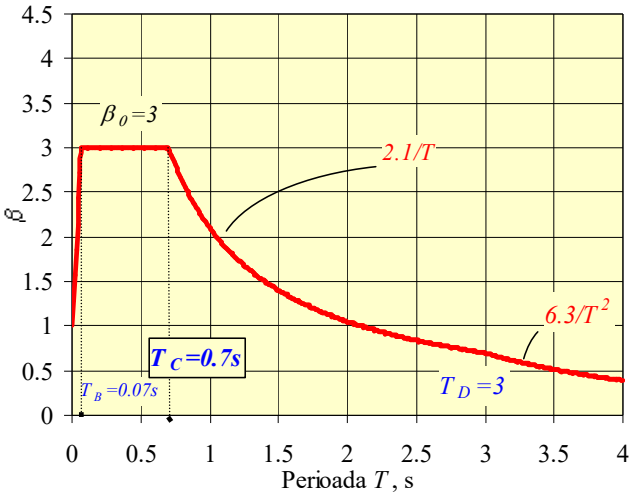
Seismic action (P100-1/2006)

– Is determined in accordance with the seismic code P100-1/2006

– For the structure, the following parameter of the seismic motion are as follows:

- a_g=0.16g (Timisoara, see seismic intensity map)
- T_c=0.7 sec. (Timisoara, see corner period map)
- Horizontal elastic response spectrum β (see figure)
- q factor = 1 (non dissipative structure)
- γ_I=1.0 (class of importance of the building is III)



– Elastic response spectrum is as follows:

$$S_e(T) = a_g \beta(T)$$

– Design response spectrum is as follows:

$0 < T \leq T_B$ $\quad = a_g \dfrac{\beta(T)}{q}$

$T > T_B$ $\quad S_d(T) = a_g \left[1 + \dfrac{\dfrac{\beta_0}{q} - 1}{T_B} T \right]$

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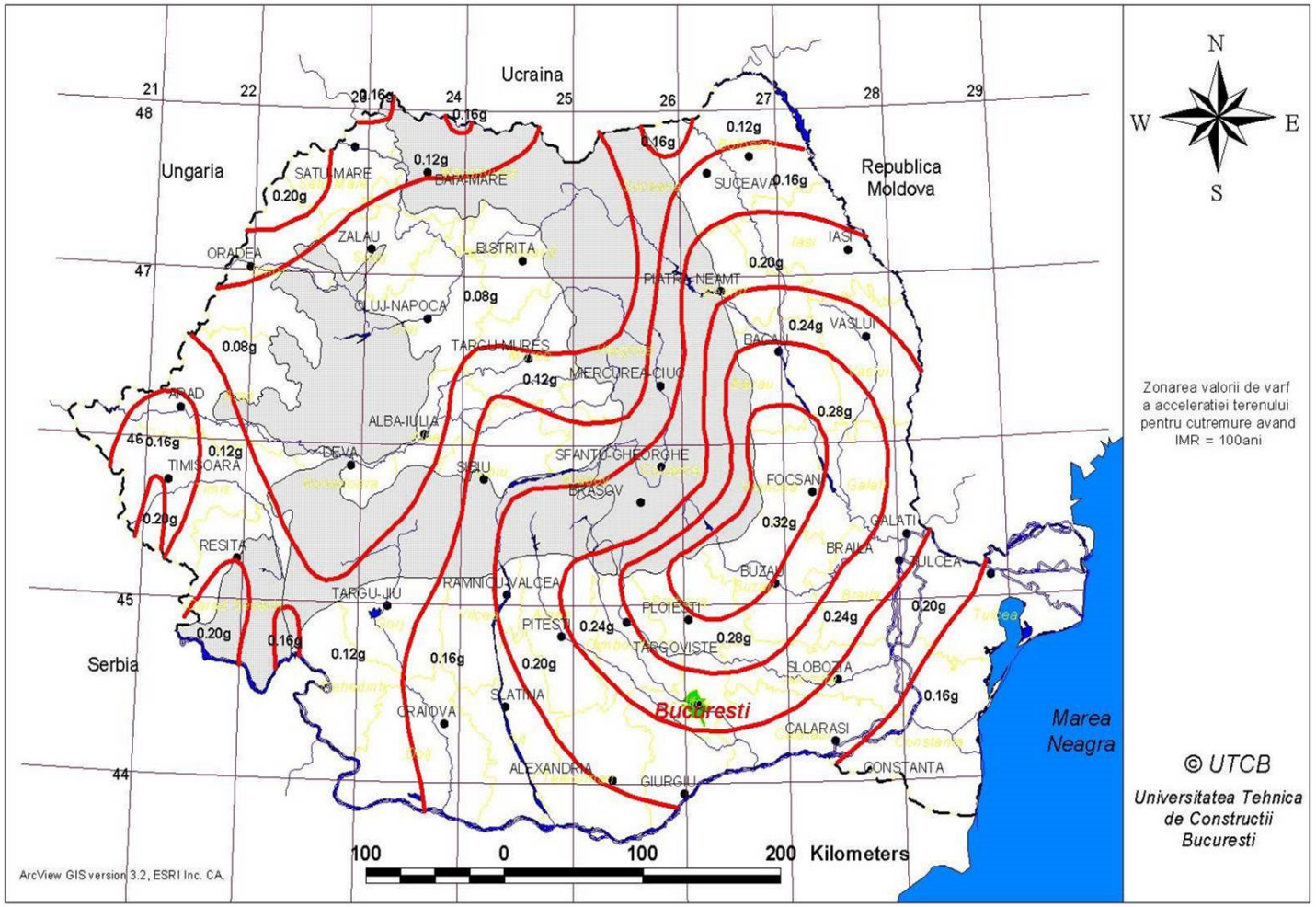
Seismic intensity map



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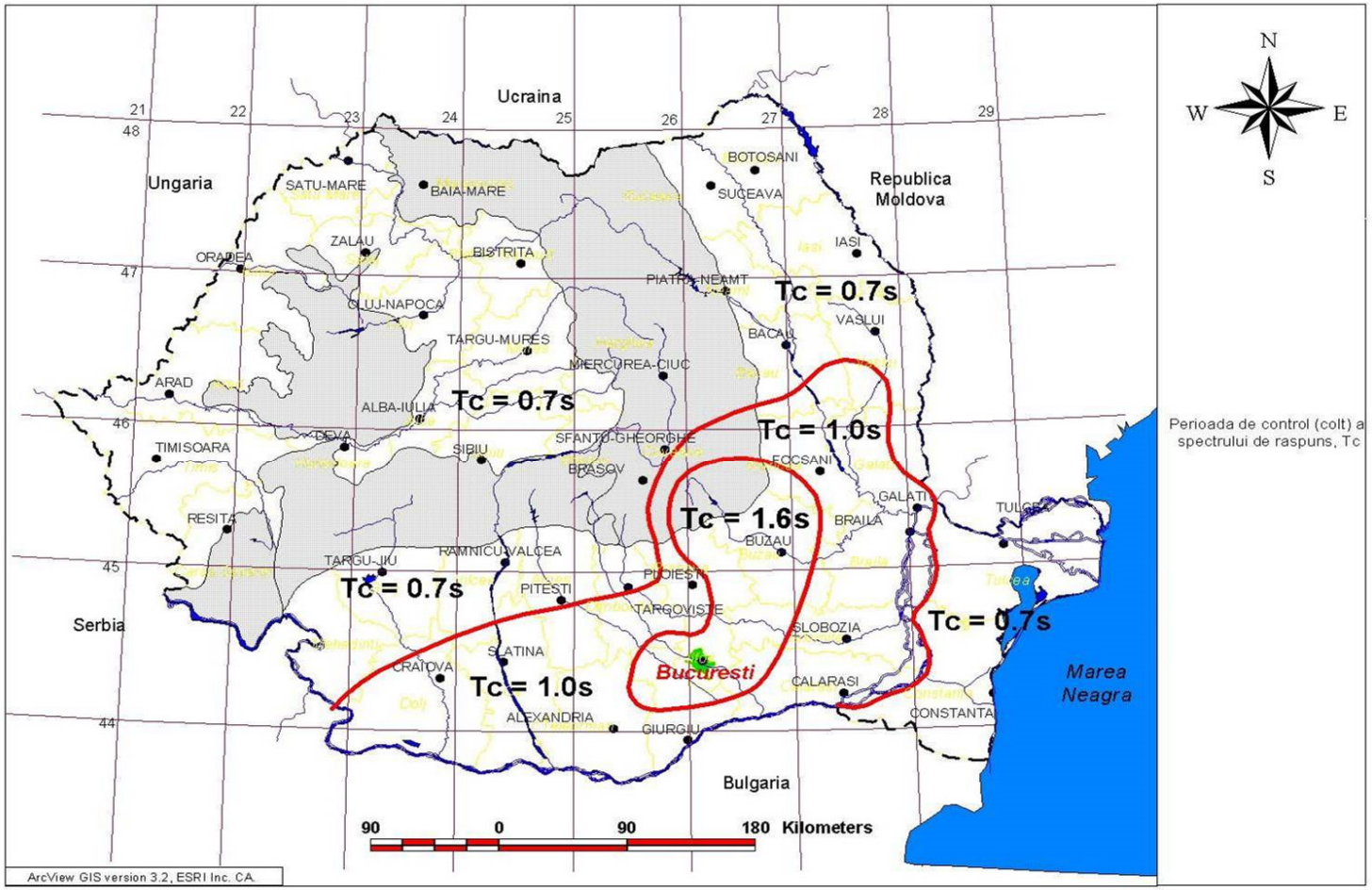
Corner period



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Behaviour factors q

Tabelul 6.3. Factorii de comportare maximi q

Tipuri de structuri	Clasa de ductilitate H	Clasa de ductilitate M
a) Cadre necontravântuite - Structuri parter	3,0	2,5
$\frac{\alpha_u}{\alpha_1} = 1,1$; $\frac{\alpha_u}{\alpha_1} = 1,0$	$5\frac{\alpha_u}{\alpha_1}$	4
- Structuri etajate $\frac{\alpha_u}{\alpha_1} = 1,2$; $\frac{\alpha_u}{\alpha_1} = 1,3$ - Zone disipative in grinzi si la baza stâlpilor	$5\frac{\alpha_u}{\alpha_1}$	4
b) Cadre contravântuite centric Contravântuiri cu diagonale intinse	4	4
Zonele disipative - numai diagonalele întinse		
Contravântuiri cu diagonale in V - Zone disipative diagonale întinse şi comprimate	2,5	2
c) Cadre contravântuite excentric $\frac{\alpha_u}{\alpha_1} = 1,2$ - Zone disipative în barele disipative încovoiate sau forfecate	$5\frac{\alpha_u}{\alpha_1}$	4

Tabelul 6.3 (continuare) Factorii de comportare maximi q

Tipuri de structuri	Clasa de ductilitate H	Clasa de ductilitate M
d) Pendul inversat $\frac{\alpha_u}{\alpha_1} = 1$; $\frac{\alpha_u}{\alpha_1} = 1,1$ - Zone disipative la baza stâlpilor Zone disipative in stâlpi $N_{Sd}/N_{pl\,Rd} > 0,3$	$2\frac{\alpha_u}{\alpha_1}$	2
e) Structuri cu nuclee sau pereţi de beton	vezi cap. 5	
f) Cadre duale (cadre necontravântuite asociate cu cadre contravântuite în X şi alternante) $\frac{\alpha_u}{\alpha_1} = 1,2$ - Zone disipative in cadrele necontravântuite şi in diagonalele întinse	$4\frac{\alpha_u}{\alpha_1}$	4
Cadre duale (cadre necontavântuite asociate cu cadre contavântuite excentric) $\frac{\alpha_u}{\alpha_1} = 1,2$ - Zone disipative in cadrele necontravântuite şi in barele disipative	$5\frac{\alpha_u}{\alpha_1}$	4

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Rules for non dissipative members

Tabelul 6.4. Relaţia dintre clasa de secţiune şi factorul de comportare q al structurii.

Clasa de ductilitate	Factorul de comportare q	Clasa de secţiune
H	$q > 4,0$	clasa 1
M	$2,0 < q \leq 4,0$	clasa 2 sau 1
L	$q = 1,0$	clasa 3, 2 sau 1

(1) Şuruburile de ancoraj vor fi proiectate la efortul maxim de întindere rezultat din combinaţia de încărcări care include acţiunea seismică. Efectele acţiunii E_{Fd} (eforturile de la baza stâlpului) se determină cu relaţia:

$$E_{Fd} = E_{F,G} + 1,1\,\gamma_{ov}\Omega E_{F,E}$$

Semnificaţiile termenilor $E_{F,G}$, $E_{F,E}$ sunt cele de la 4.6.2.4.

Raportul Ω se calculează funcţie de tipul structurii cu relaţiile de la 6.6.3 pentru cadre necontravântuite, cu relaţiile de la 6.7.4 pentru cadre contravântuite centric şi cu relaţiile de la 6.8.3 pentru cadre contravântuite excentric.

Raportul Ω va fi limitat astfel încât să fie îndeplinită condiţia $1,1\,\gamma_{ov}\Omega < q$ (q - factorul de comportare al structurii - tabel 6.3)

În cazul unui calcul simplificat se pot adopta valorile produsului $1,1\,\gamma_{ov}\Omega$ din Anexa F.

4.6.2.4. Rezistenţa fundaţiilor

(1) Sistemul fundaţiilor va fi verificat în acord cu prevederile "Normativului pentru proiectarea structurilor de fundare directă", NP 112- 04.

(2) La dimensionarea fundaţiilor, acţiunea suprastructurii în combinaţia de încărcări care include acţiunea seismică corespunde mecanismului de plastificare asociat tipului de structură utilizat, considerând efectele suprarezistenţei elementelor structurale.

(3) În cazul fundaţiilor elementelor verticale individuale (stâlpi, pereţi), condiţia de la paragraful (1) se poate considera satisfăcută dacă efectele acţiunii E_{Fd} asupra fundaţiei se determină după cum urmează:

$$E_{Fd} = E_{F,G} + \gamma_{Rd}\,\Omega\,E_{F,E} \qquad (4.23)$$

în care:

$E_{F,G}$ - efectul acţiunii (efortul secţional) din încărcările neseismice, incluse în combinaţia de acţiuni considerate în calculul la cutremur

$E_{F,E}$ - efectul acţiunii (efortul secţional) din încărcările seismice de proiectare

γ_{Rd} - factorul de suprarezistenţă, egal cu 1 pentru $q \leq 3$, şi 1,15 în celelalte cazuri

Ω - valoarea $(R_{di}/E_{di}) \leq q$ în zona disipativă a elementului i a structurii care are influenţa cea mai mare asupra efortului E_F considerat, iar

R_{di} - rezistenţa (efortul capabil) elementului i

E_{di} - valoarea de proiectare a efortului în elementul i corespunzătoare acţiunii seismice de proiectare

(4) Raportul Ω se calculează astfel:

- în cazul fundaţiilor de stâlpi şi pereţi, Ω se determină ca valoare a raportului momentelor M_{Rd}/M_{Ed} în secţiunea transversală de la baza zonei plastice

- în cazul fundaţiilor stâlpilor cadrelor cu contravântuiri centrice Ω este valoarea minimă a raportului forţelor axiale N_{Rd}/N_{Ed}, determinate pentru toate diagonalele întinse

- în cazul fundaţiilor stâlpilor cu contravântuiri excentrice, Ω este valoarea minimă a raportului forţelor tăietoare V_{Rd}/V_{Ed} determinată pentru toate zonele disipative forfecate sau a raportului momentelor încovoietoare M_{Rd}/M_{Ed} determinate în toate zonele disipative.

(5) Pentru fundaţii comune pentru mai multe elemente verticale (grinzi de fundaţii, radiere sau infrastructuri tip cutie) relaţia ($E_{Fd} = E_{F,G} + \gamma_{Rd}\,\Omega\,E_{F,E}$) se aplică pentru baza tuturor elementelor verticale, considerând $\gamma_{Rd} = 1$.

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Combination of actions (CR 0-2005)

Combination of actions (CR 0-2005)

- In the static and dynamic analysis, the following action were considered:
  - Dead load (P)
  - Snow (Z)
  - Wind transversal (VT)
  - Wind longitudinal (VL)
  - Crane vertical (PodV)
  - Crane transversal (PodT)
  - Crane longitudinal (PodL)
  - Seismic action (S)

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Combination of actions (CR 0-2005)

- Combination of actions for the two limit states were as follows:

A) Fundamental combinations

- **Ultimate Limit State**

$$1.35\sum_{j=1}^{n} G_{k,j} + 1.5Q_{k,1} + \sum_{i=2}^{m} 1.5\psi_{0,i}Q_{k,i}$$

- **Serviceability Limit State**

$$\sum_{j=1}^{n} G_{k,j} + Q_{k,1} + \sum_{i=2}^{m} \psi_{0,i}Q_{k,i}$$

B) Special Combinations

- **Ultimate Limit State**

$$\sum_{j=1}^{n} G_{k,j} + \gamma_I A_{E,k} + \sum_{i=2}^{m} \psi_{2,i}Q_{k,i}$$

- **Serviceability Limit State**

$$\sum_{j=1}^{n} G_{k,j} + \gamma_I A_{E,k} + \sum_{i=2}^{m} \psi_{2,i}Q_{k,i}$$

ν = 0.5 for SLS (this value affects the $A_{E,k}$)

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Special combinations - Ultimate Limit State

1	P + Z + Seism	1.0	0.4	1.0	
2	P + Z + $1.1\gamma_{ov}\Omega$ Seism	1.0	0.4	$1.1\times1.25\times\Omega$	Calcul stalpi
3	P + Z + $\gamma_{Rd}\,\Omega$ Seism	1.0	0.4	$1.15\times\Omega$ - ptr. $q\leq3$ $1.0\times\Omega$ – ptr. $Q>3$	Calcul ancoraje in fundatii

Special combinations - Serviceability Limit State

		P	Z	Seism
		1.0	0.4	0.5
1	P + Z + Seism			

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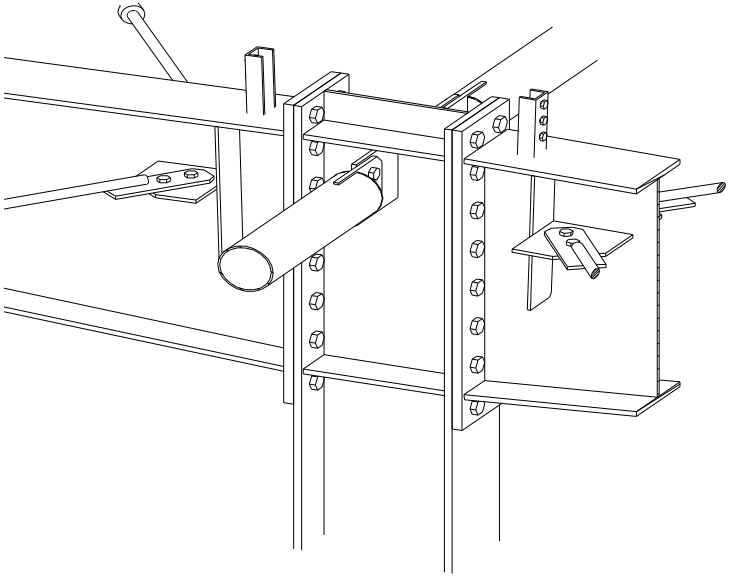
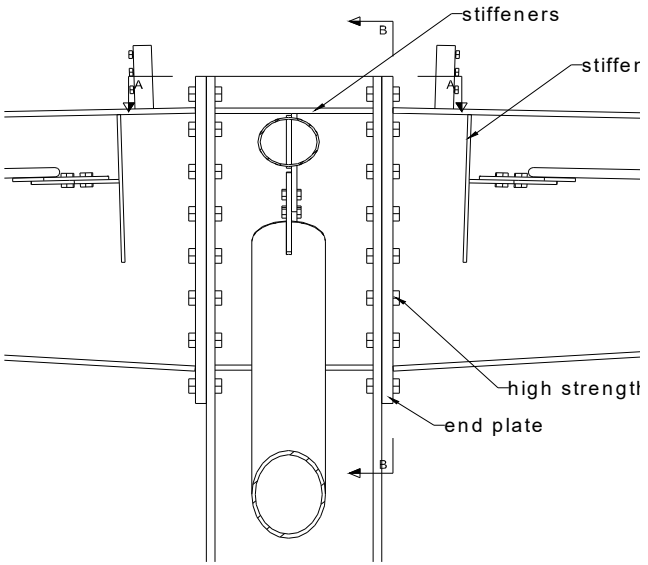
Detailing

- Beam to column joints
  - Extended end plate bolted connection
  - M24, 10.9 grade
  - End plate t_p = 30 mm

Perspective View



Basic View



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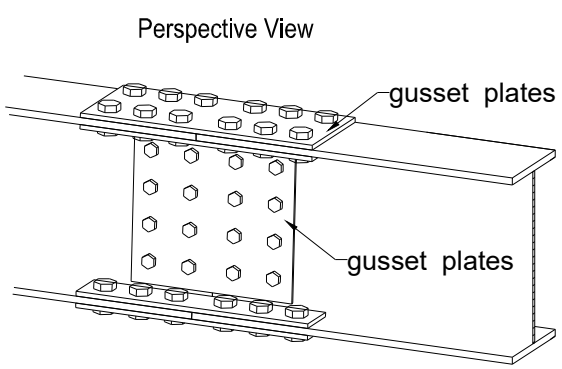
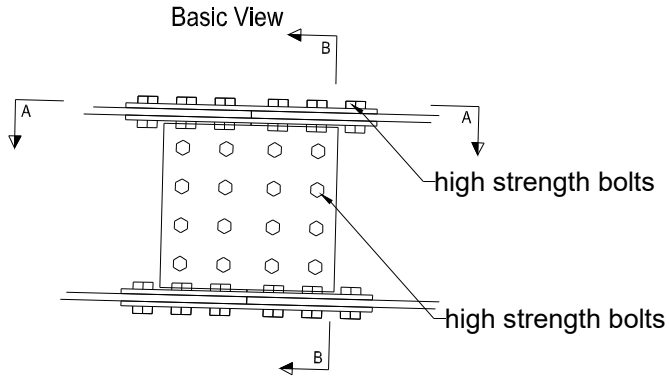
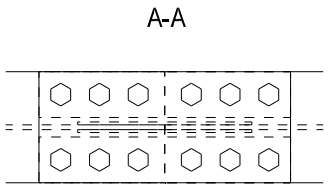
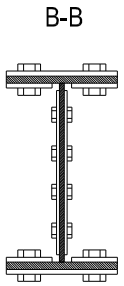
Detailing

- Beam splice
  - M16, 10.9 grade – web
    - M20, 10.9 grade – flanges
    - Web cleat t_{cl} = 8mm
    - Flange cleat t_{cl} = 12 mm









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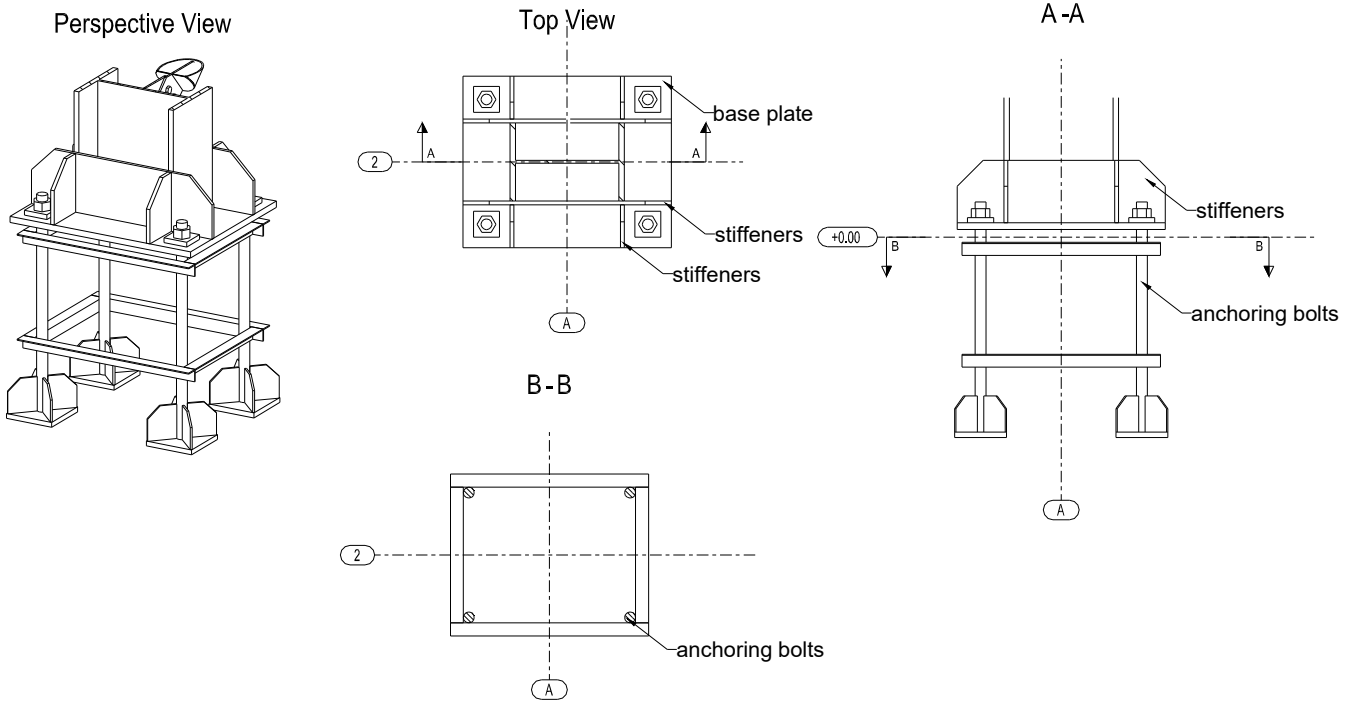
Detailing

- Column base
  - M48, 6.6. grade
  - Anchoring length = 980 mm
  - Base plate t_p = 35mm
  - Stiffeners t_s = 12 mm



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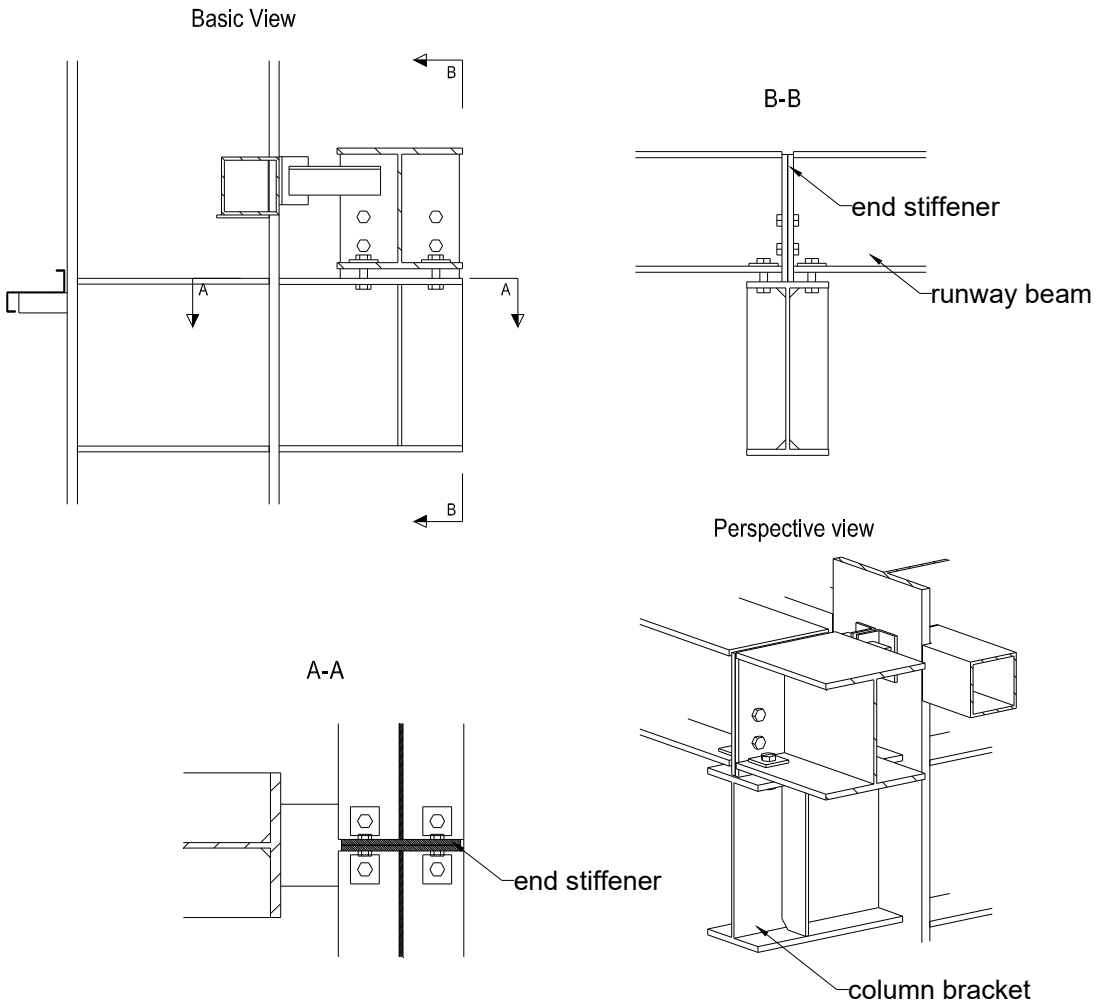
Runway beam to column connection



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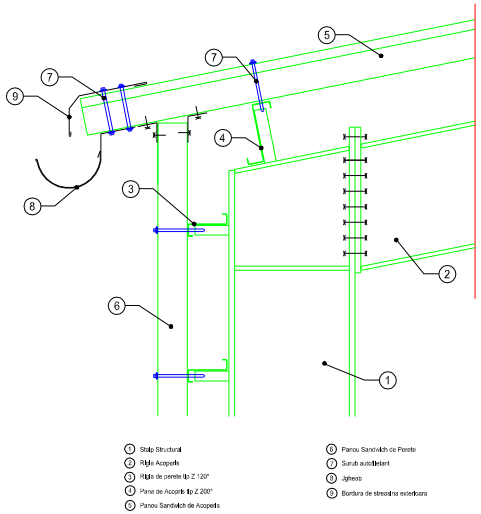
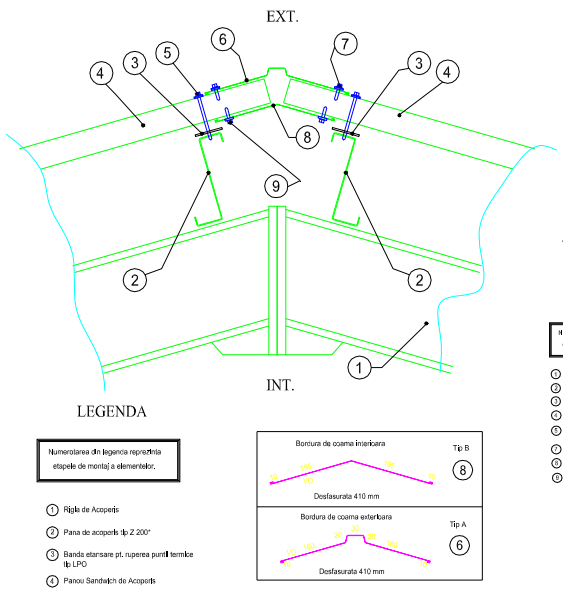
Secondary system

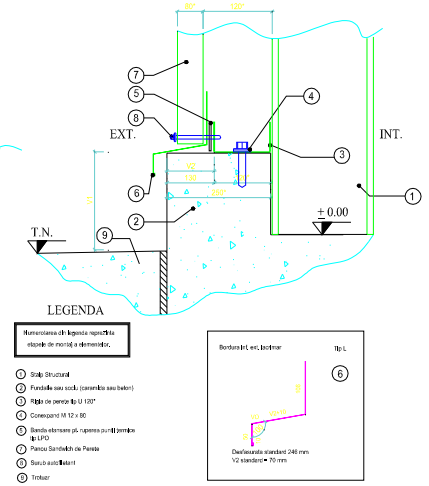
Detaliu prindere elemente de inchideri - Streasina



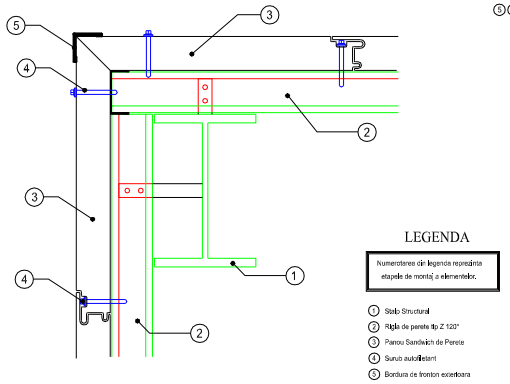
Detaliu prindere elemente de inchideri - Coama



Detaliu prindere elemente de inchideri - Soclu



Detaliu prindere elemente de inchideri - Fronton



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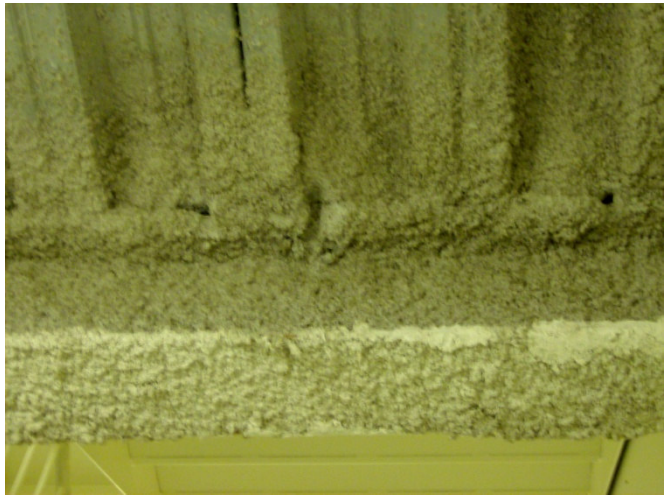
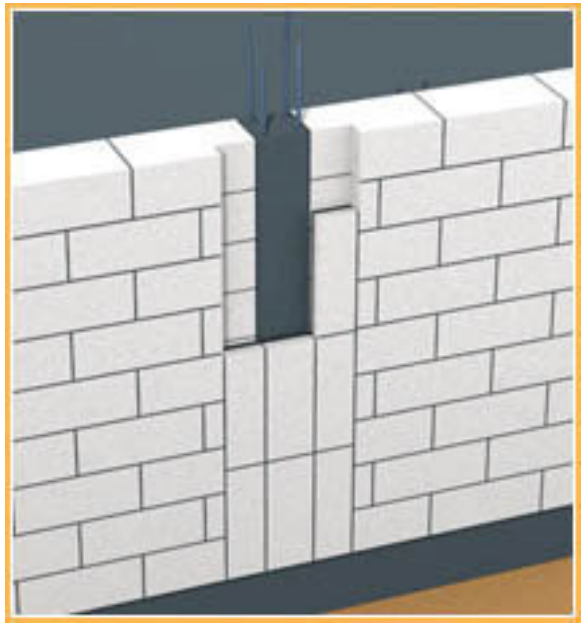
Fire protection

• Columns, beams: sprayed-on mineral fibers



• Wall: Autoclaved aerated concrete



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Drawings

- Drawing are the means used by designer to show how each individual structural member or component can be fabricated and/or constructed. Then, it is the contractor's responsibility to ensure that all structural items are fabricated or constructed to the correct dimensions, to use the correct materials.
- There are two main types of drawings:

– Fabrication drawings: These show the details of structural members and give all information regarding materials, sizes, welding, drilling, etc. for fabrication. The mark number of the detail refers to the number on the marking plan. There are two types of drawings here:

- *Assembly drawings:* are typically workshop drawings showing fabrication information for one assembly. In most cases, an assembly consists of single parts that are either bolted or welded to a main part. The following scales are recommended: 1:25; 1:20, 1:10, 1:5.
- *Part drawings:* Single-part drawings are workshop drawings which show the fabrication information for one part (usually without welds). They might be also included in the assembly drawings but generally are produced separately. The following scales are recommended: 1:5 - 1:1.

– General arrangements drawings: these consist of plans, elevations and sections for the steel building showing the location and mark numbers for all steel members. Connections between steel members must be specified. The following scales are recommended: 1:200, 1:100, 1:50. Create general arrangement (GA) drawings when you need:

- Several views on one drawing, including the entire model or a part of it
- Erection, marking, or anchor bolt plans
- Information from model views, including 3D views

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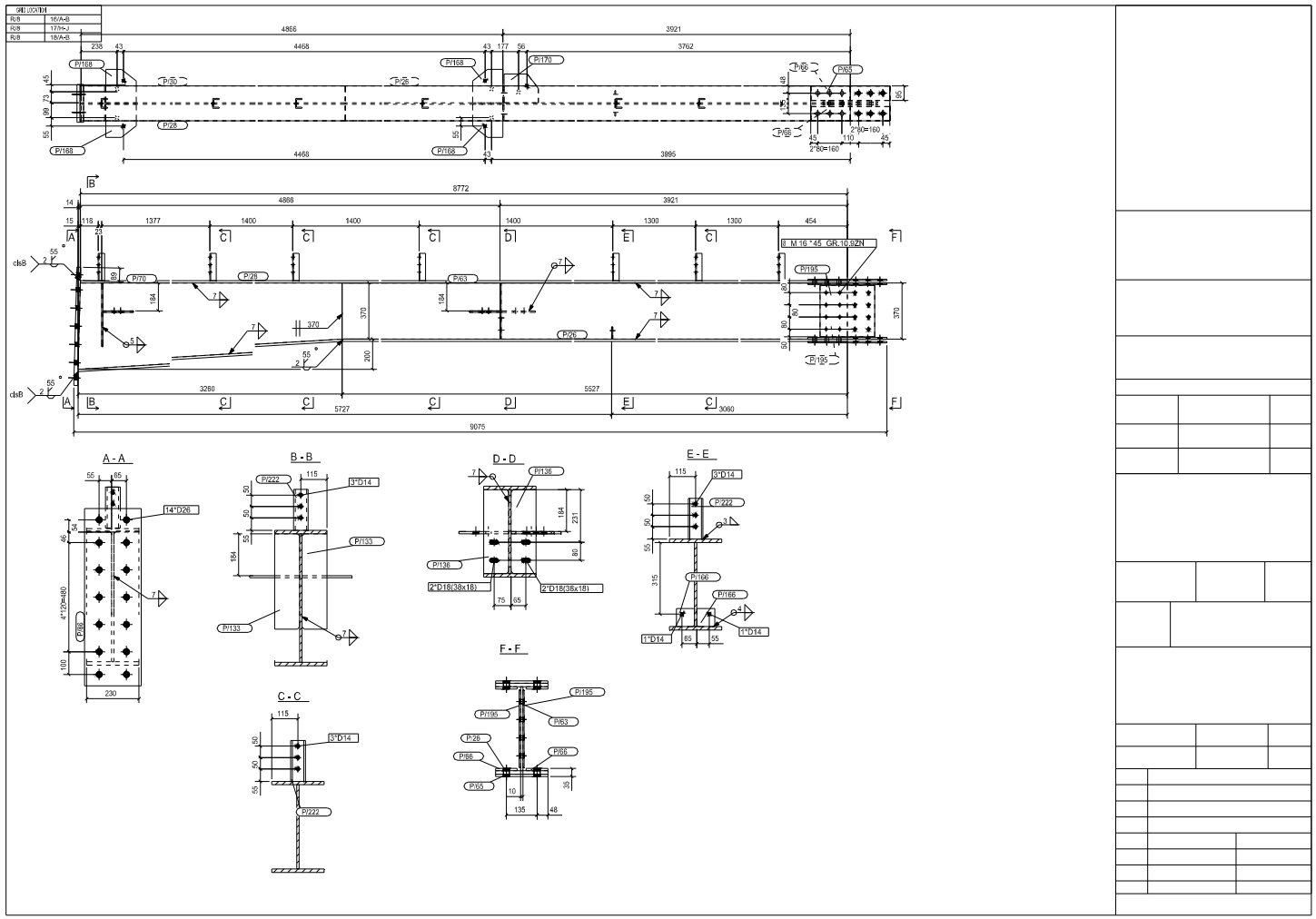
Fabrication drawings: Assembly drawings



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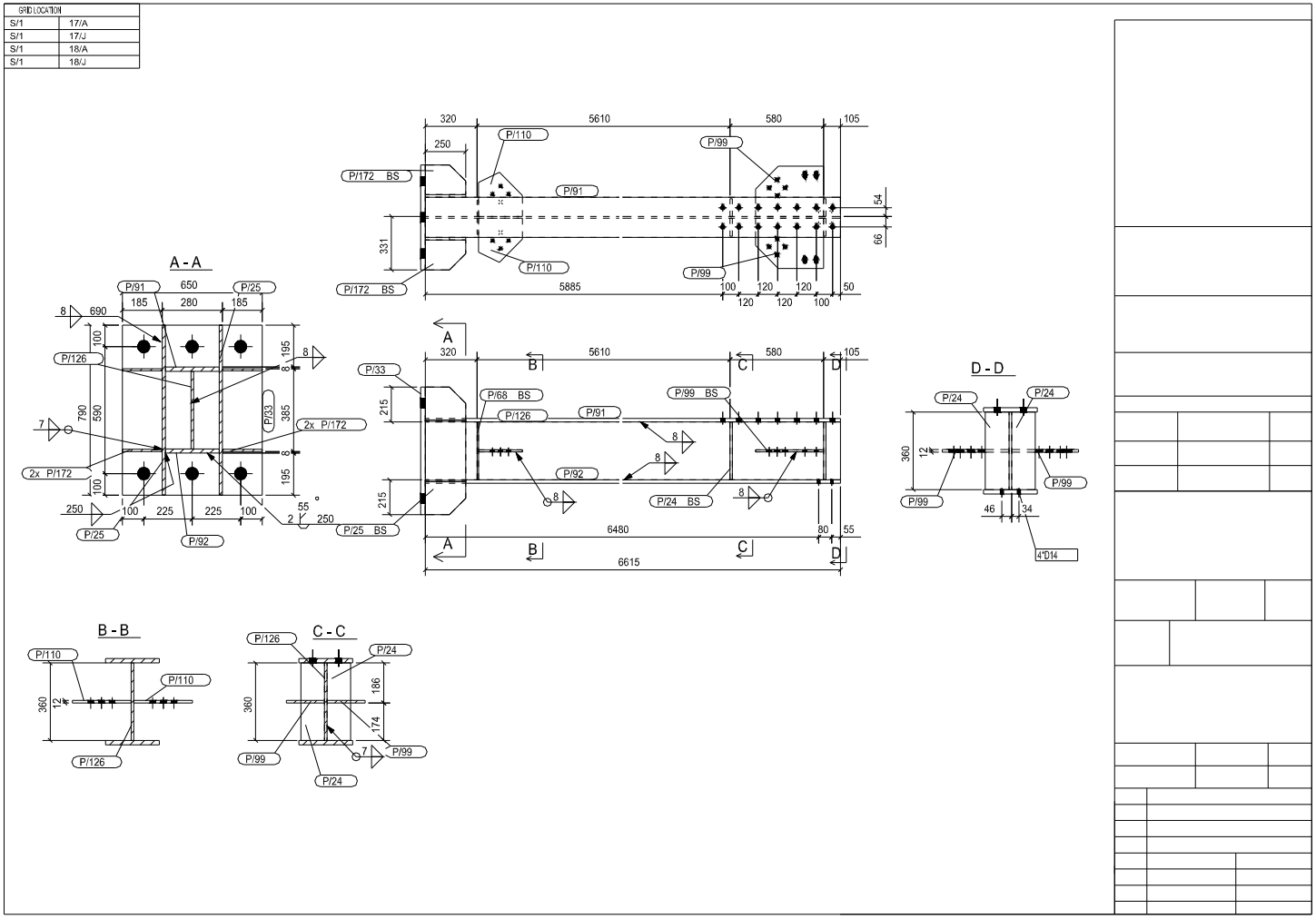
Fabrication drawings: *Assembly drawings*



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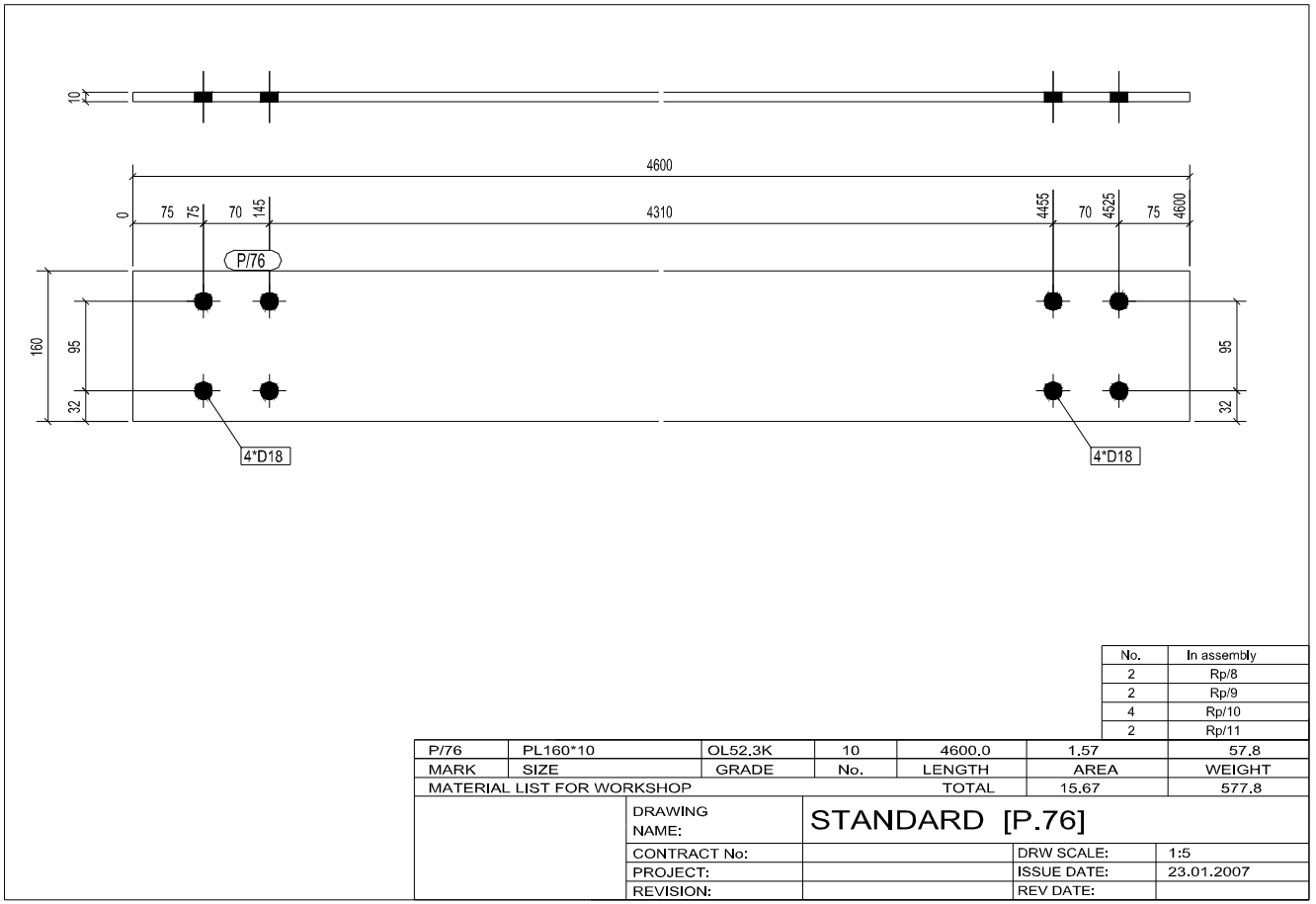
Fabrication drawings: *Part drawings*



No.	In assembly
2	Rp/8
2	Rp/9
4	Rp/10
2	Rp/11

P/76	PL160*10	OL52.3K	10	4600.0	1.57	57.8
MARK	SIZE	GRADE	No.	LENGTH	AREA	WEIGHT
MATERIAL LIST FOR WORKSHOP				TOTAL	15.67	577.8

DRAWING NAME:	STANDARD [P.76]			
CONTRACT No:		DRW SCALE:	1:5	
PROJECT:		ISSUE DATE:	23.01.2007	
REVISION:		REV DATE:		

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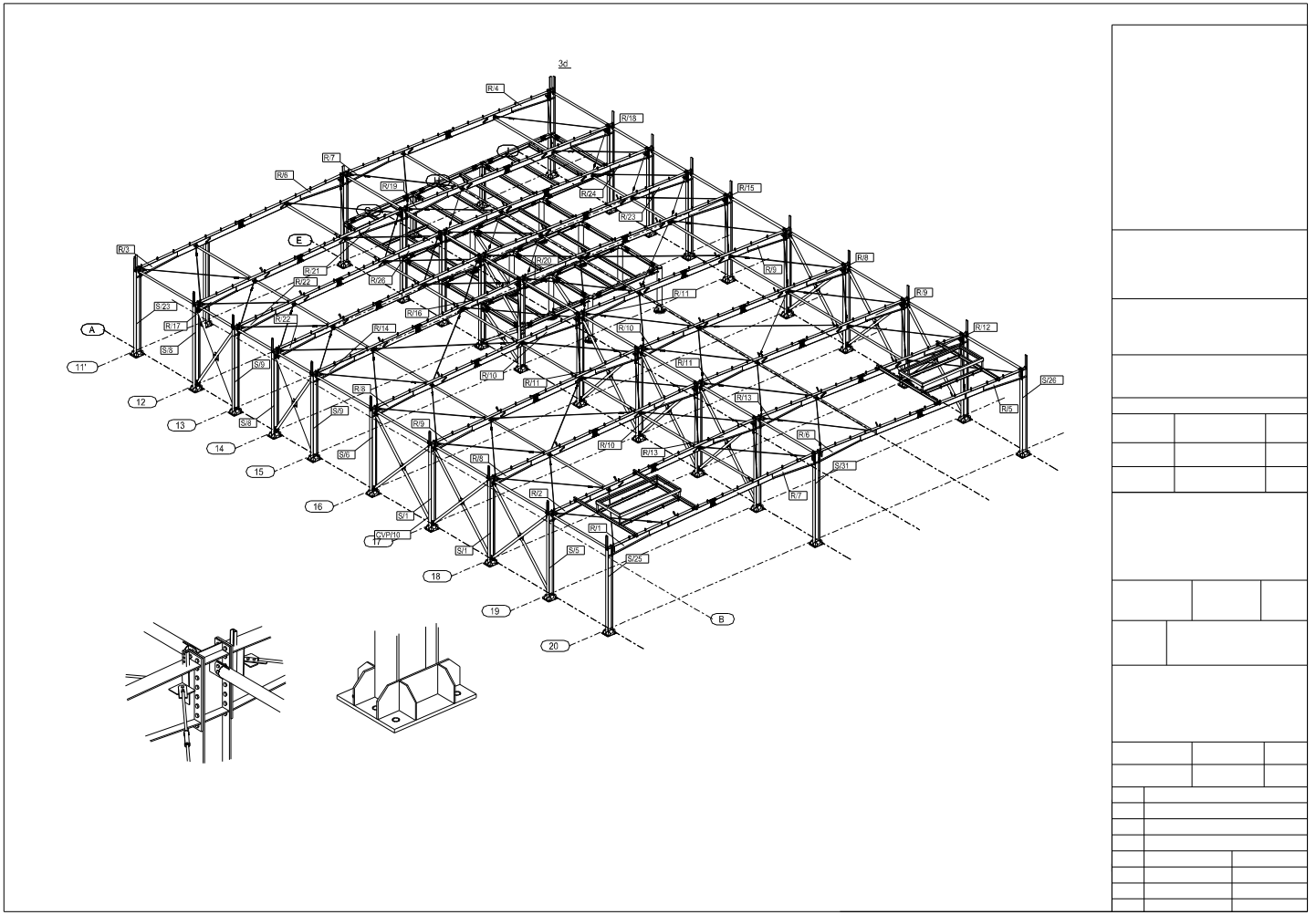
General arrangements drawings



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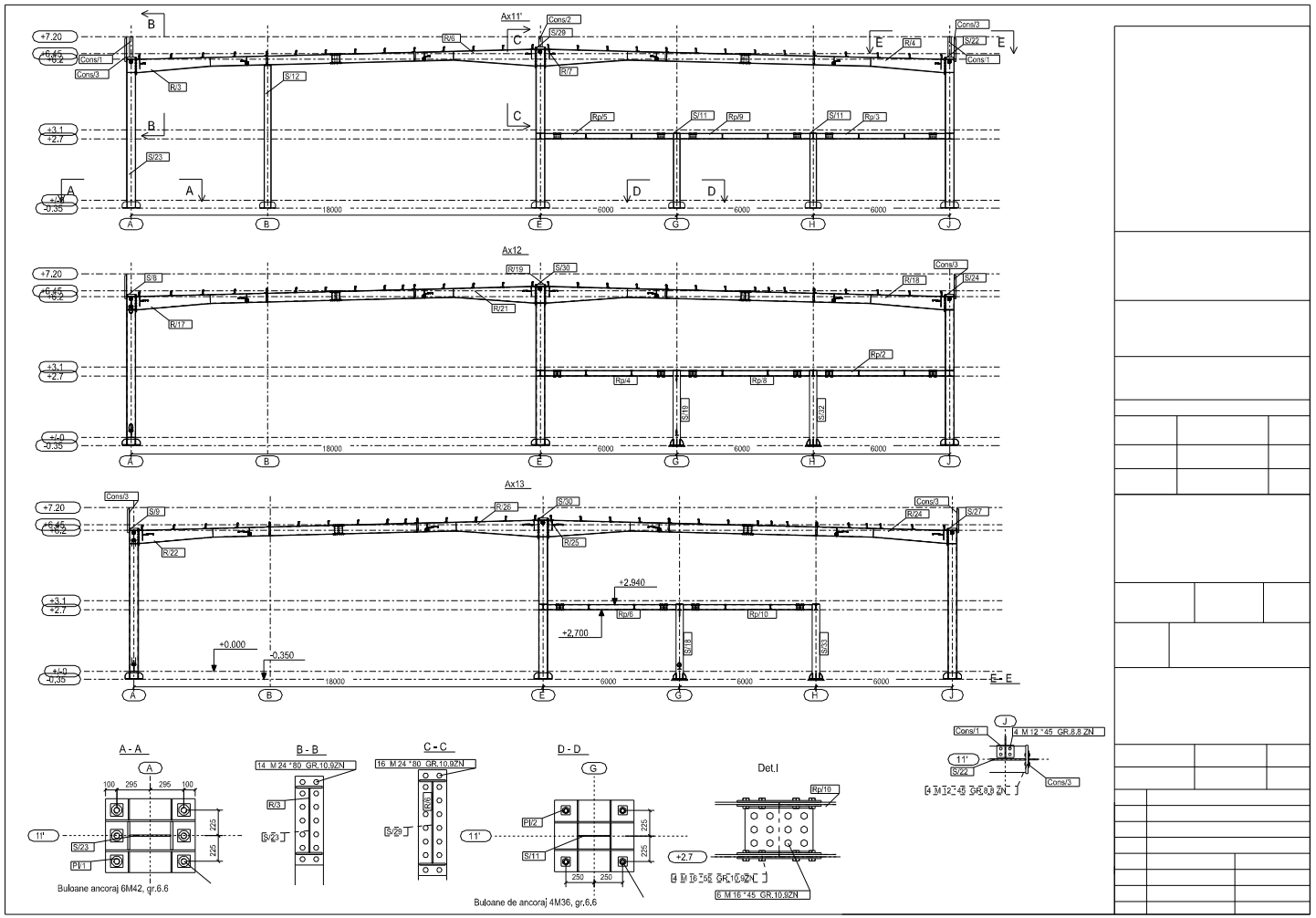

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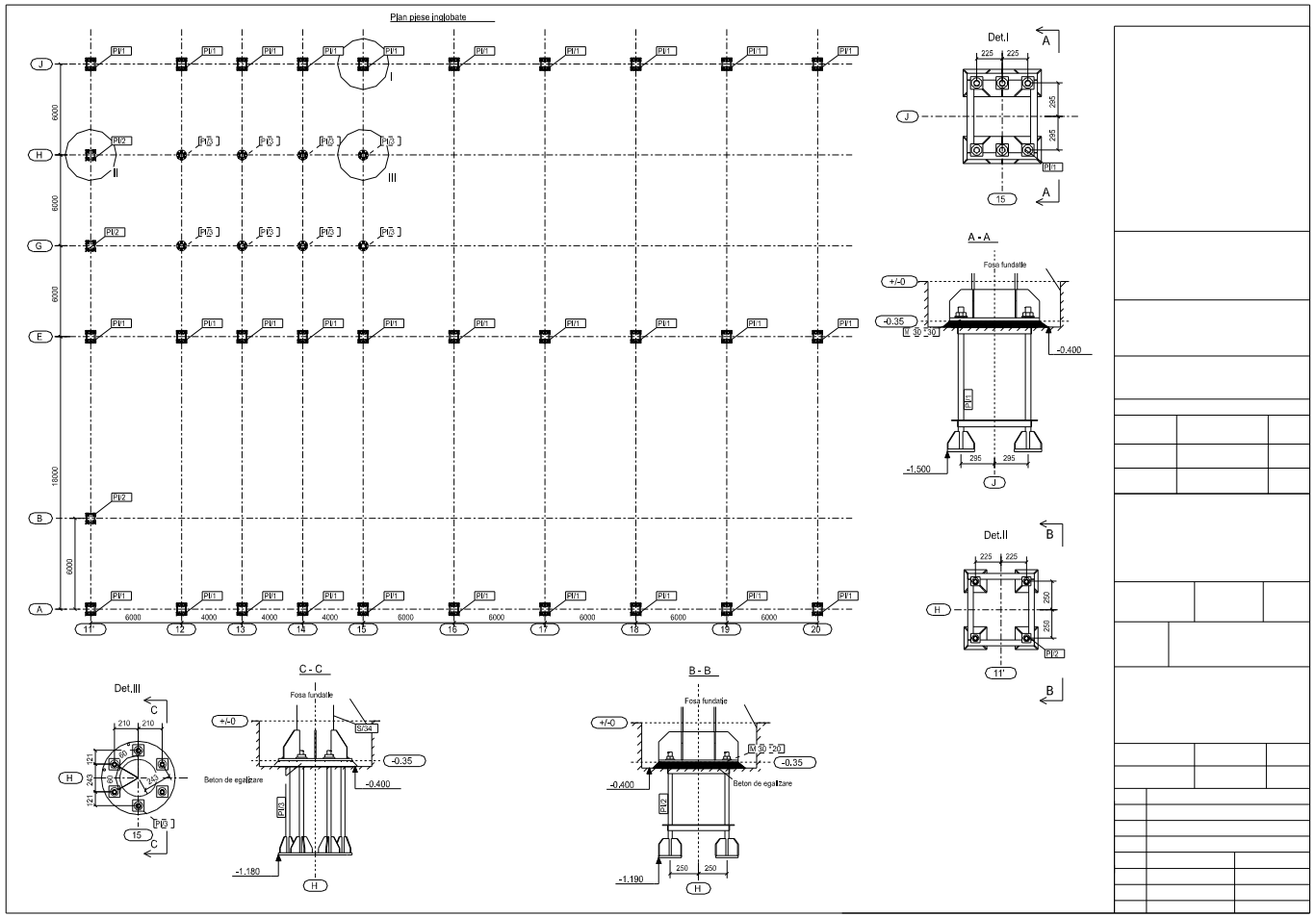

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Material reports

- Designer should also provide the bill of materials (including part list, bolt list, washer list, nut list and more). Quality of material should be specified (steel grade, bolt grade, etc)
- Attention should be paid to this issue, as the quantities in excess or in minus may be very costly and time consuming
- CAD/CAM packages offer a large variety of material reports: material list, bolt list, washer list, nut list, and more.

Material list

Project name:										
Drawing number:										
Part no.	Part name	No. of pieces		Dimensions		Length	Weight (kg)			Material
		for 1 elem.	total	b (mm)	t (mm)	l (mm)	for 1m	for 1 piece	total	
Column S1 (3 pcs)										
101	Tube 120X4	1	3			3256	11.5	37.4	37.44	OL 37-3k
102	Plate	1	3	600	25.0	670	117.8	78.9	78.89	OL 52-3k
106	Plate	4	12	180	12.0	8213	17.0	139.3	557.04	OL 52-3k
107	Plate	6	18	196	6.0	8213	9.2	75.8	454.91	OL 52-3k
111	Channel 60x40x4	7	21			125	3.9	0.5	3.41	OL 37-3k
134	Channel 100x50X4	1	3			90	4.5	0.4	0.41	OL 37-3k
135	Tube 80X80X4	1	3			80	9.3	0.7	0.74	OL 37-3k
142	Bolt M12x35	2	6			35		0.1	0.11	gr 4.6 ZN
						Weight for 1 element: 1429.3				kg/piece
						Total weight: ***4287.8***				***Kg***

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Photos during construction

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Case study: design of a tall building

Plan	**25,5m x 41,5m**
Span	**7,5m**
Bay	**7,5m**
Storey height	**4,0m**
Total height	**106,3m - tallest building in Bucharest**
Nr. stories	**3B + 26S**
Foundations	**Mat foundations + piles, "Top-down" method**
Main structure	**Partially encased steel columns, steel beams and bracings, composite slabs**
Location	**The new building neighbors the city centre of Bucharest**
Construction	**March - November 2006**
Completion	**2008**

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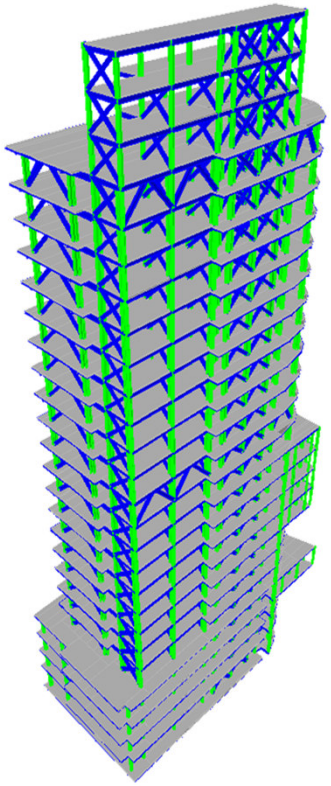
The structural system

- Dual frame structure: rigid frames + centrically brace frames + outrigger and truss systems at top and midheight
- S355 steel
- Cruciform cross sections columns from hot rolled profiles
- Hot rolled profiles for regular beams
- Centric X braces made of hot rolled profiles



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Facade

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Analysis and design of structure

- Static and dynamic 3D analysis
  - Design of steel structure: Romanian codes + European codes: EN 1993, EN1994, EN 1998
  - Wind tunnel test
  - Seismic design
- Performance based seismic evaluation
- Fire resistance design with advanced nonlinear analysis
- Column loss scenario

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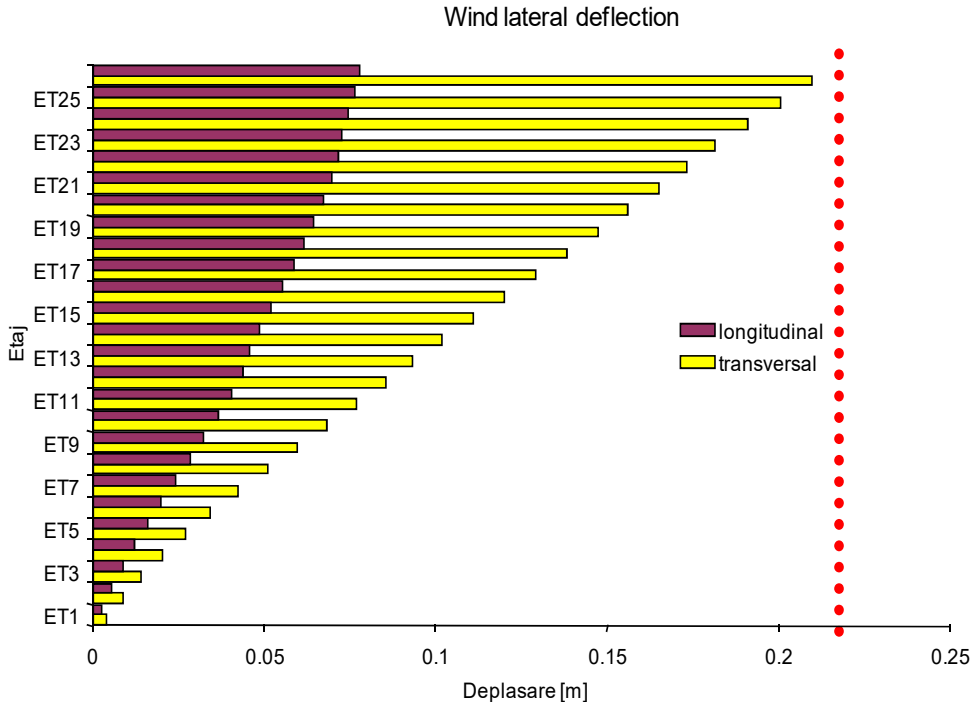
Wind design

- High demands from wind load
- Top lateral deflection - H/500
- Concentrically braced frames (in X and V)



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Wind tunnel tests

- Wind tunnel tests
  - Rigid model to evaluate the distribution of pressure coefficients
  - Aeroelastic model to evaluate the aerodynamic coefficients
  - Boundary layer wind tunnel
- Motivation
  - Irregular shape
  - Complex Surroundings (dense constructed area)
  - To optimize safety and cost of project

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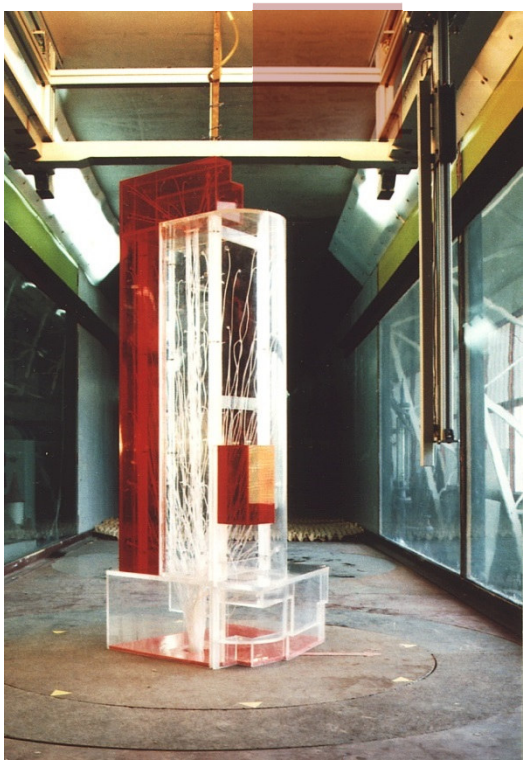
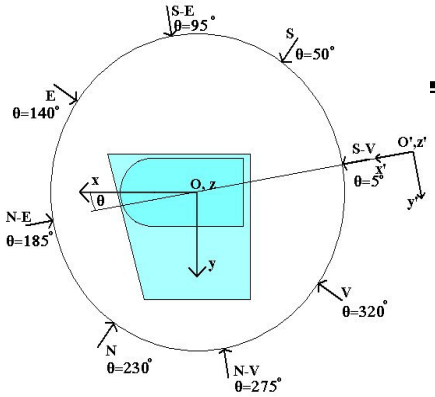
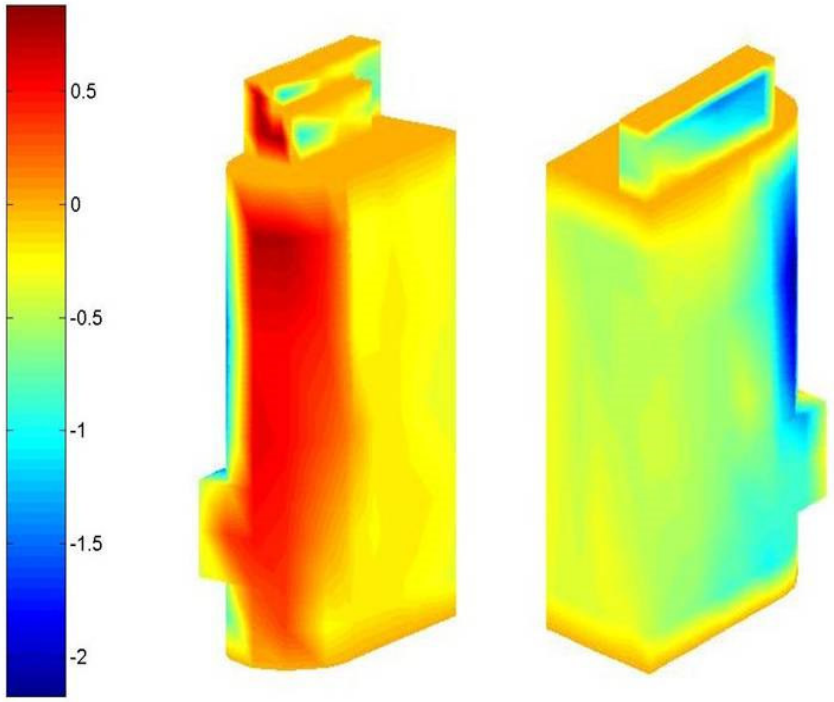
Rigid model







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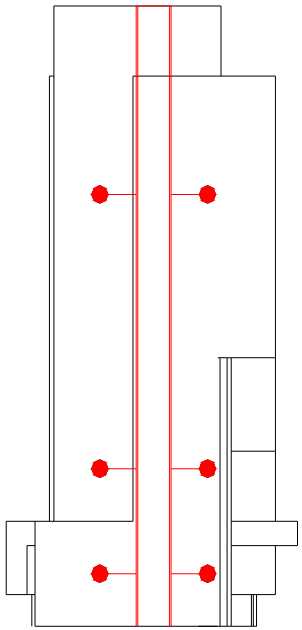
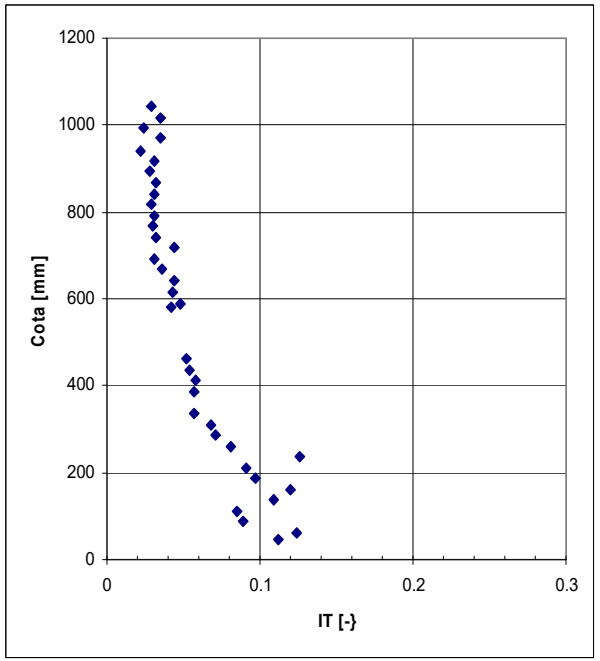
Aeroelastic model





Turbulence intensity profile

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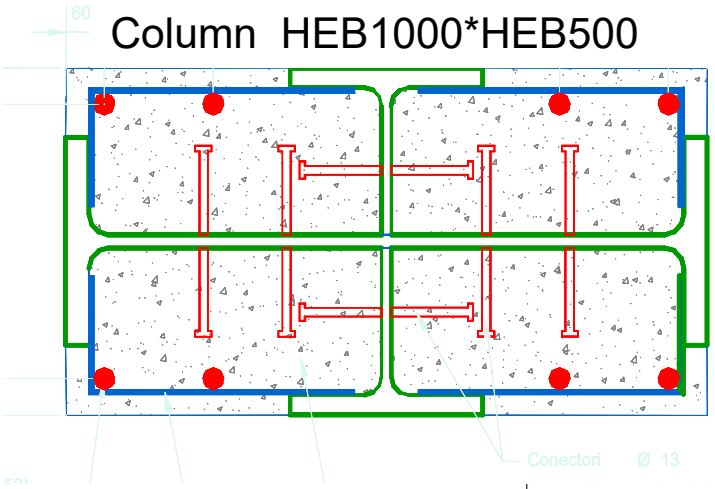
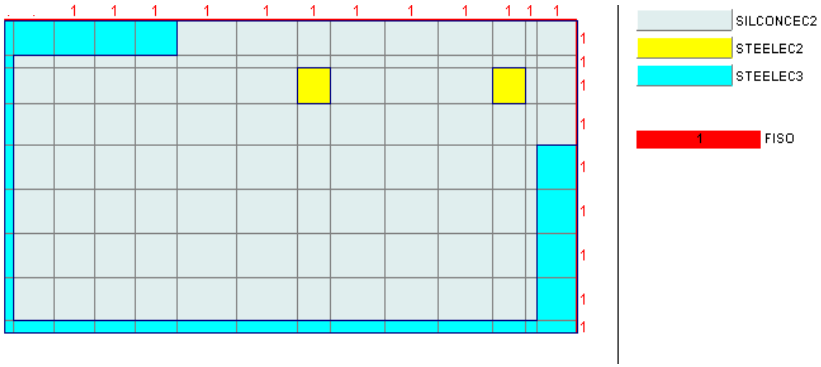
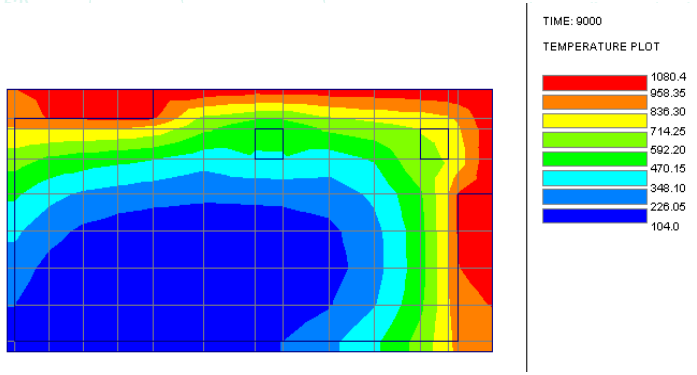
Fire analysis

Column HEB1000*HEB500

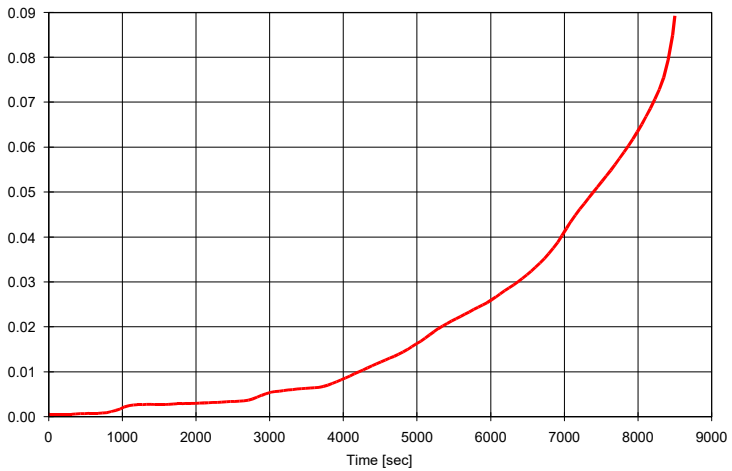


Model of column cross section





Temperature distribution on the cross section after 21/2 hrs. (ISO fire)



Time-deflection at column mid-height, first storey-column

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Robustness

- Buildings should have enough robustness to avoid progressive collapse which may be generalized due to localized failures in case of extreme events
- A robust structures is the one characterized by:
  - high redundancy, which can be achieved by a proper design, e.g. conception and detailing
  - a good balance between stiffness, overstrength and ductility of its components
  - multiple routes for force transfer
  - plastic capacity in structural members
  - sufficient strength for structural members that cannot be allowed to collapse

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Progressive collapse

- Progressive collapse - A chain reaction failure of building members to an extent disproportionate to the original localized damage
- Exaples of progressive collapse:
  - Collapse of the Ronan Point apartment building in the United Kingdom (UK) in 1968
  - World Trade Center (WTC) towers during the terrorist attacks on September 11, 2011 (9-11)\
  - The Murrah Federal Building in Oklahoma City, 1995
  - The U.S. embassies in Africa

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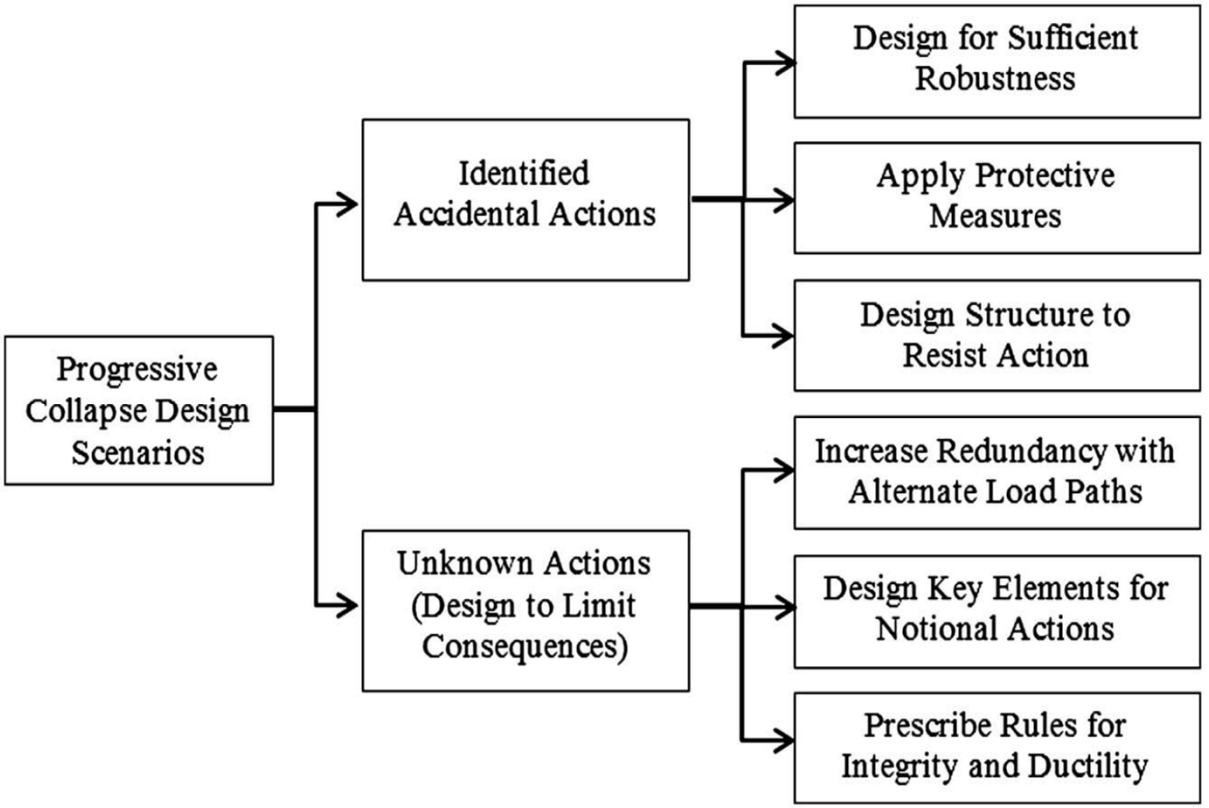
Design strategies for preventing disproportionate collapse



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Ronan Point building
1968 collapse

The Murrah Federal Building,
Oklahoma City 1995

World Trade Center 1 and 2
11 September 2001

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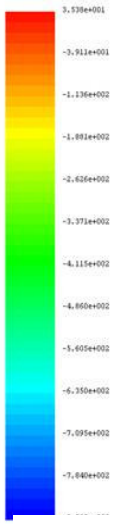
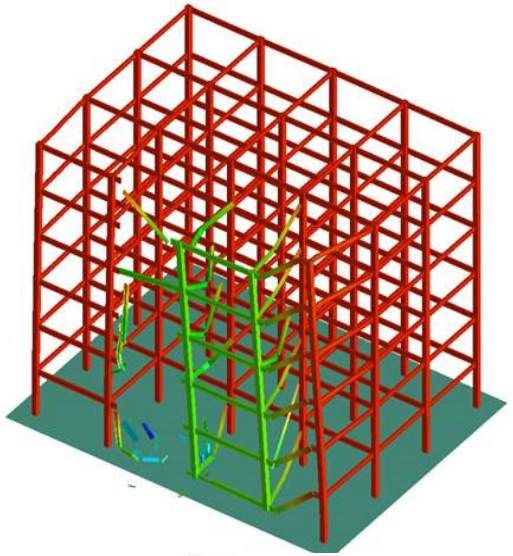
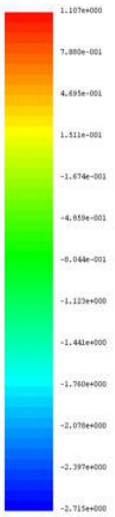
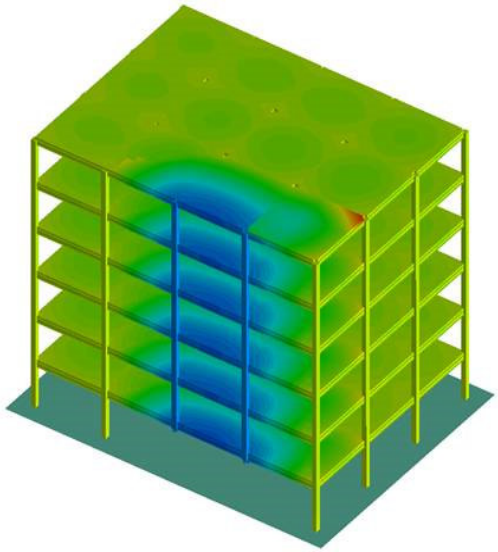
Collapse control design

- Alternate load path method is the preferred approach for providing the collapse control design of multi-story buildings
- The method can be used for assessing the level of damage and the possibility of collapse due to different hazard levels
- The mthod can be incorporated in a Performance Based Collapse Control Design









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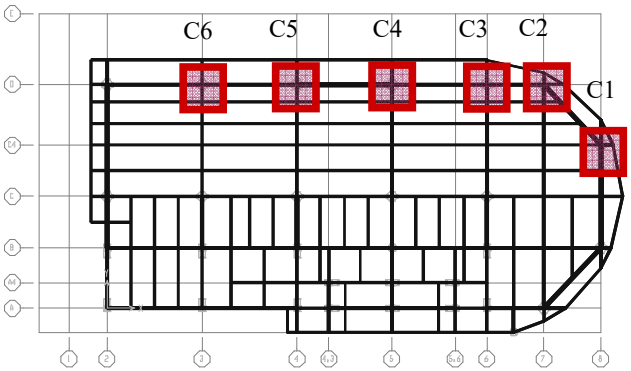
Robustness in case of column loss

- Investigation of collapse
  - high-rise buildings can be subjected to exceptional situations
  - As the external loads generated by impact and blast cannot be clearly assumed, the loss of vertical members was assumed.
  - Different hazard scenarios:
- Loss of interior columns
- Loss of corner columns
- Location: 1st storey, 13th storey
  - Columns are removed one by one
  - 3d dynamic nonlinear analysis



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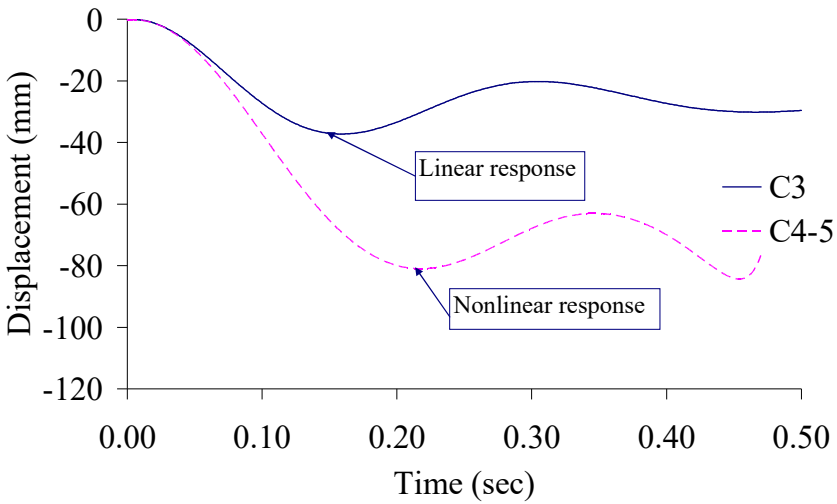
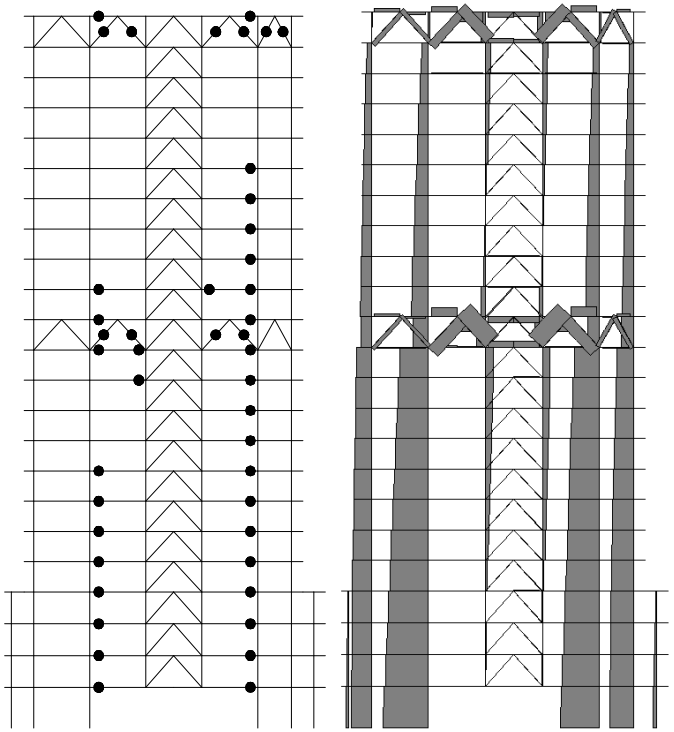
Robustness in case of column loss





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Views during construction

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Building completed

- **This lecture was prepared for the Edition 1 of SUSCOS (2012/14) by RUI SIMÕES (UC) and FLOREA DINU (UPT).**

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